



LABORATÓRIO DE SÍNTESE EM
BIODIVERSIDADE
Universidade Federal de Mato Grosso do Sul



FUNDAÇÃO
UNIVERSIDADE
FEDERAL DE
MATO GROSSO DO SUL

Eco-evolutionary dynamics in metacommunities: Integrating space, phenotypes, and environment to understand multi-scale patterns

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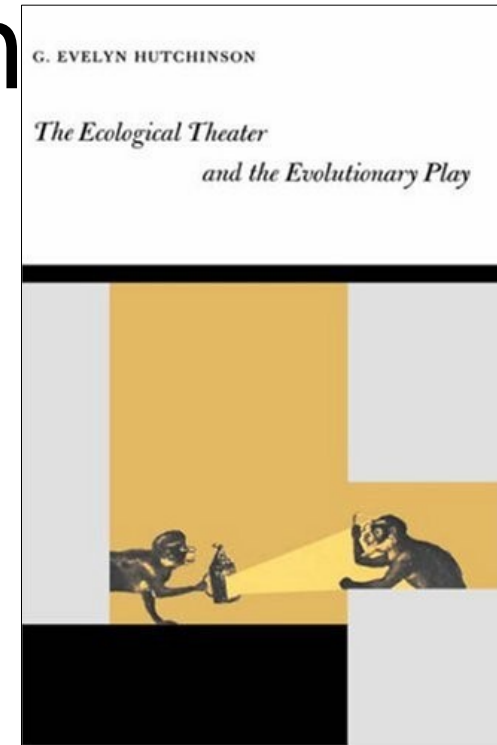
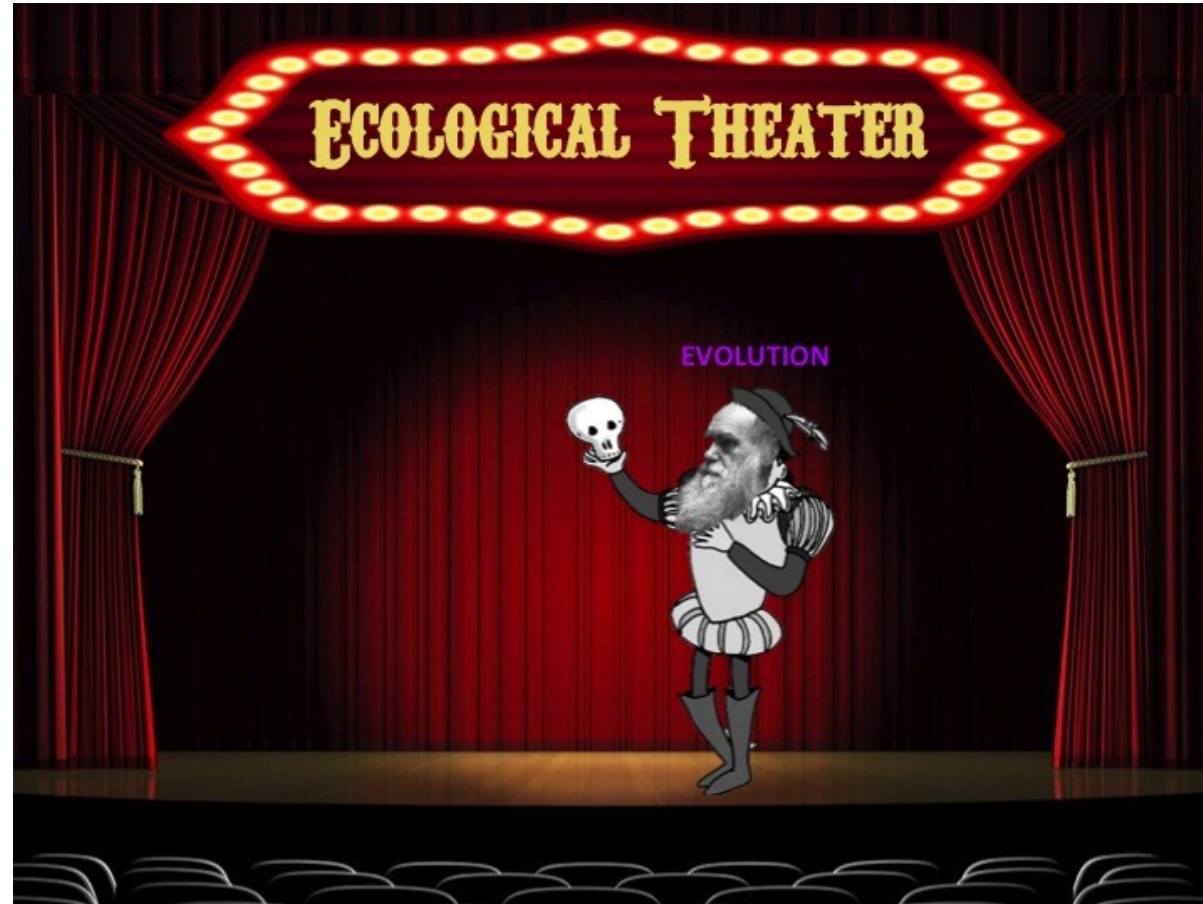
Outline

- Why we can't just ignore evolution
- Evolution can occur in very short time scales... And influence ecological dynamics
- Recap of population genetics
- Selection
- Examples of interaction between processes
- What are eco-evolutionary dynamics and what they are not
 - Evolving metacommunities approach
- EvoluCITY: my research project

Traditional view of the relationship between ecology and evolution

Evolution is something that happens at long time scales and takes too many generations for its effects to manifest

Ecology could just ignore evolution





SÓ QUE NON!

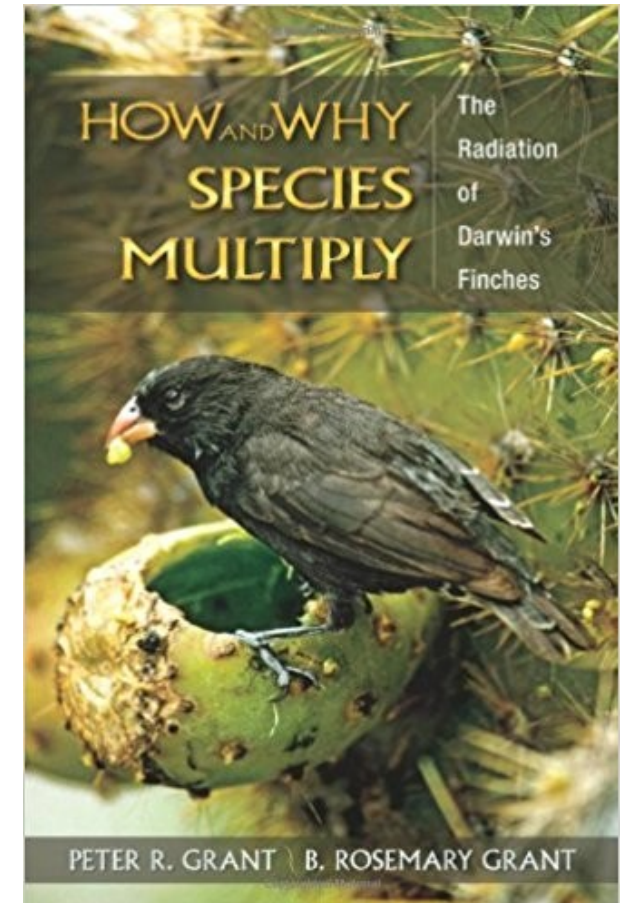
“from so simple a beginning, endless forms most beautiful and most wonderful have been ***and are being evolved.***”

Charles Darwin (1859)
Origin of species by means
of natural selection



“ “Nothing in evolutionary biology makes sense except in the light of ecology”
Grant and Grant (2008, p. 167)

” ”



2017

Review

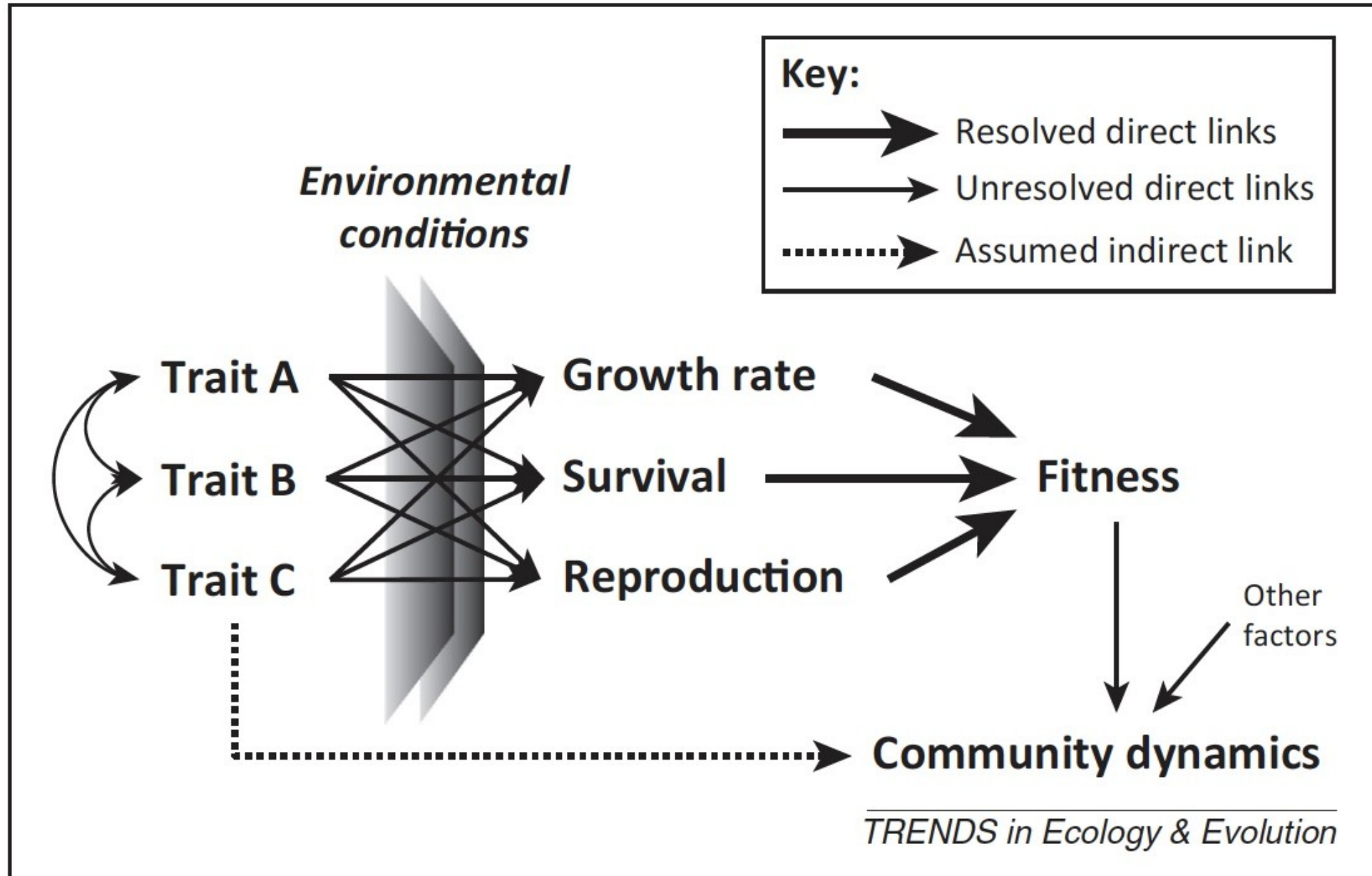
Evolution in a Community Context: On Integrating Ecological Interactions and Macroevolution

Marjorie G. Weber,^{1,*} Catherine E. Wagner,²
Rebecca J. Best,^{3,4} Luke J. Harmon,^{3,5} and Blake Matthews³

VOL. 191, NO. 3 THE AMERICAN NATURALIST MARCH 2018

Evolution in a Community Context: Trait Responses to Multiple Species Interactions*

Casey P. terHorst,^{1,†} Peter C. Zee,¹ Katy D. Heath,² Thomas E. Miller,³ Abigail I. Pastore,³
Swati Patel,⁴ Sebastian J. Schreiber,⁵ Michael J. Wade,⁶ and Matthew R. Walsh⁷



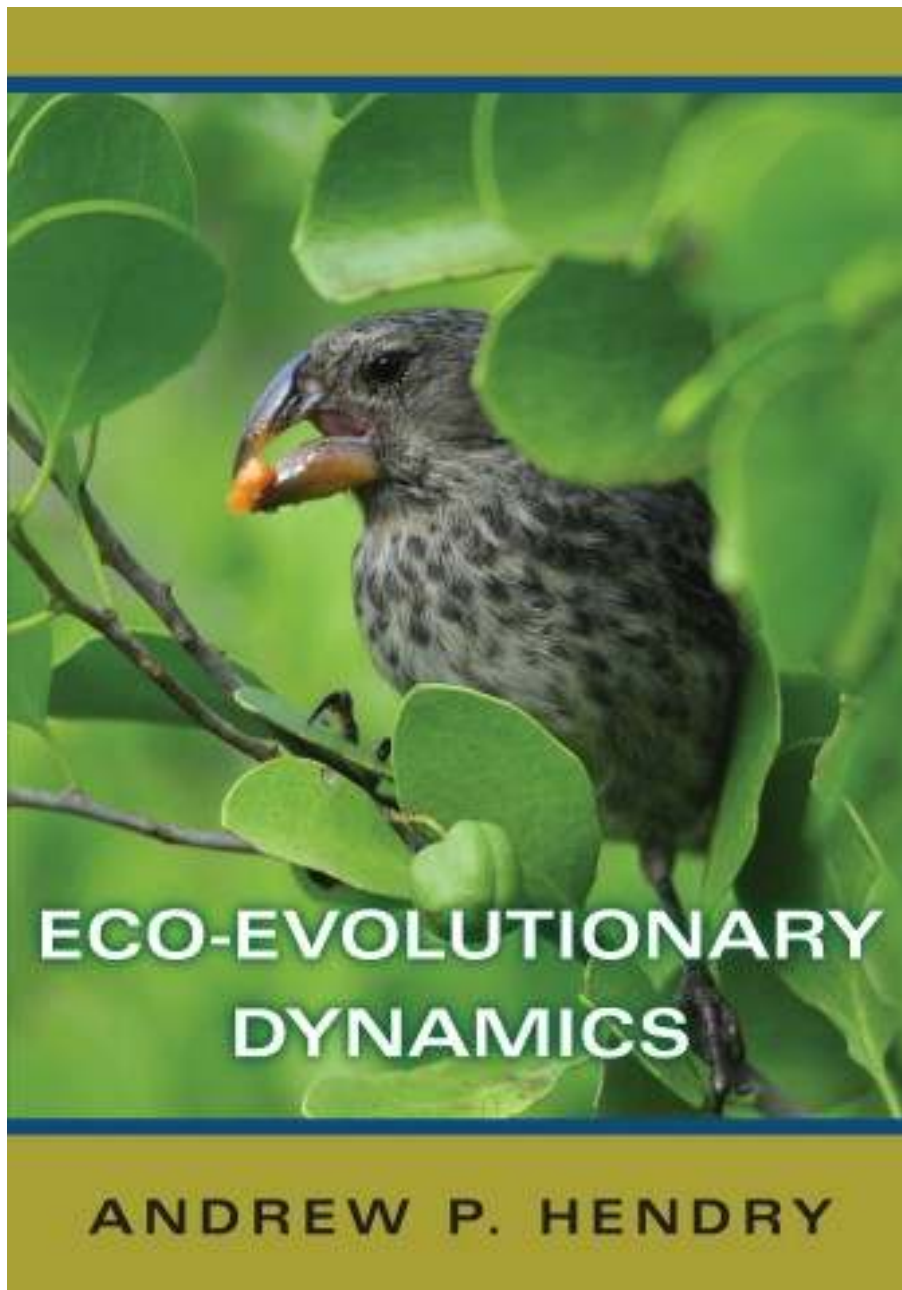
The ecological theater re-
conceptualized

EDITORIAL

Evolution on ecological time-scales

S. P. CARROLL,^{*†} A. P. HENDRY,[‡] D. N. REZNICK[§] and C. W. FOX[¶]

Thus, while it has long been recognized that adaptive evolution occurs in an ecological context, it is now clear that evolutionary change feeds back directly and indirectly to demographic and community processes, and that ecological studies should consider this reciprocal interplay between ecology and evolution. Because we are now in a period of accelerated environmental change, evolutionary biologists have an unprecedented opportunity to study evolutionary processes and their influence on ecological processes. Likewise, evolutionary biology has an unprecedented opportunity



2017

The Newest Synthesis: Understanding the Interplay of Evolutionary and Ecological Dynamics

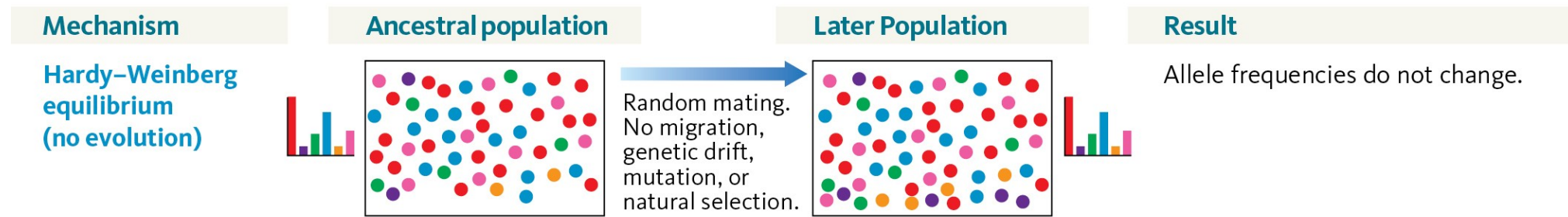
Thomas W. Schoener

28 JANUARY 2011 VOL 331 **SCIENCE** www.sciencemag.org



Evolution can happen very fast and
its effects can change ecological
dynamics

A brief stop-over to remember a few
concepts of population genetics



Null model of population
genetics and evolution



Predicts allelic frequency in a
population in the absence of
selection, migration, mutation, and
drift

Mechanisms of Evolution: Altering Allele Frequencies

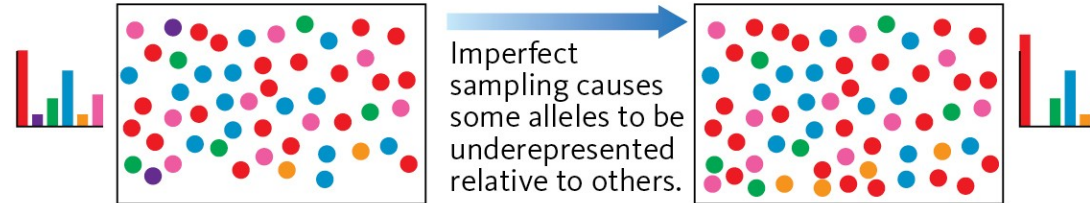
Mechanism

Ancestral population

Later Population

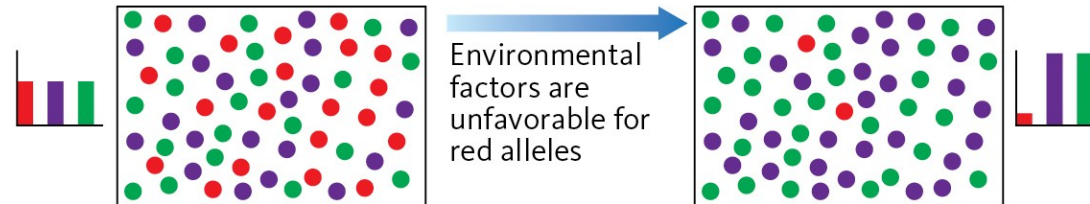
Result

Drift



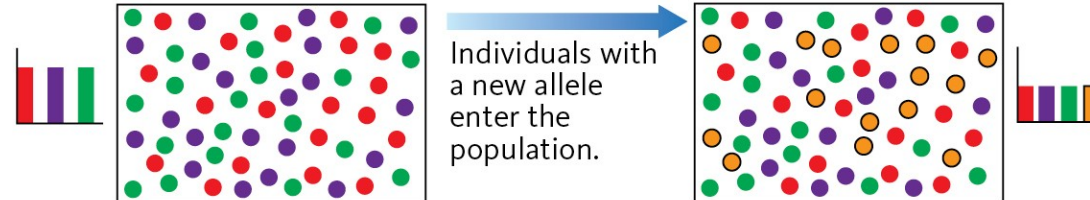
Purple allele is lost.

Natural selection



Red allele becomes less common.

Migration (gene flow)



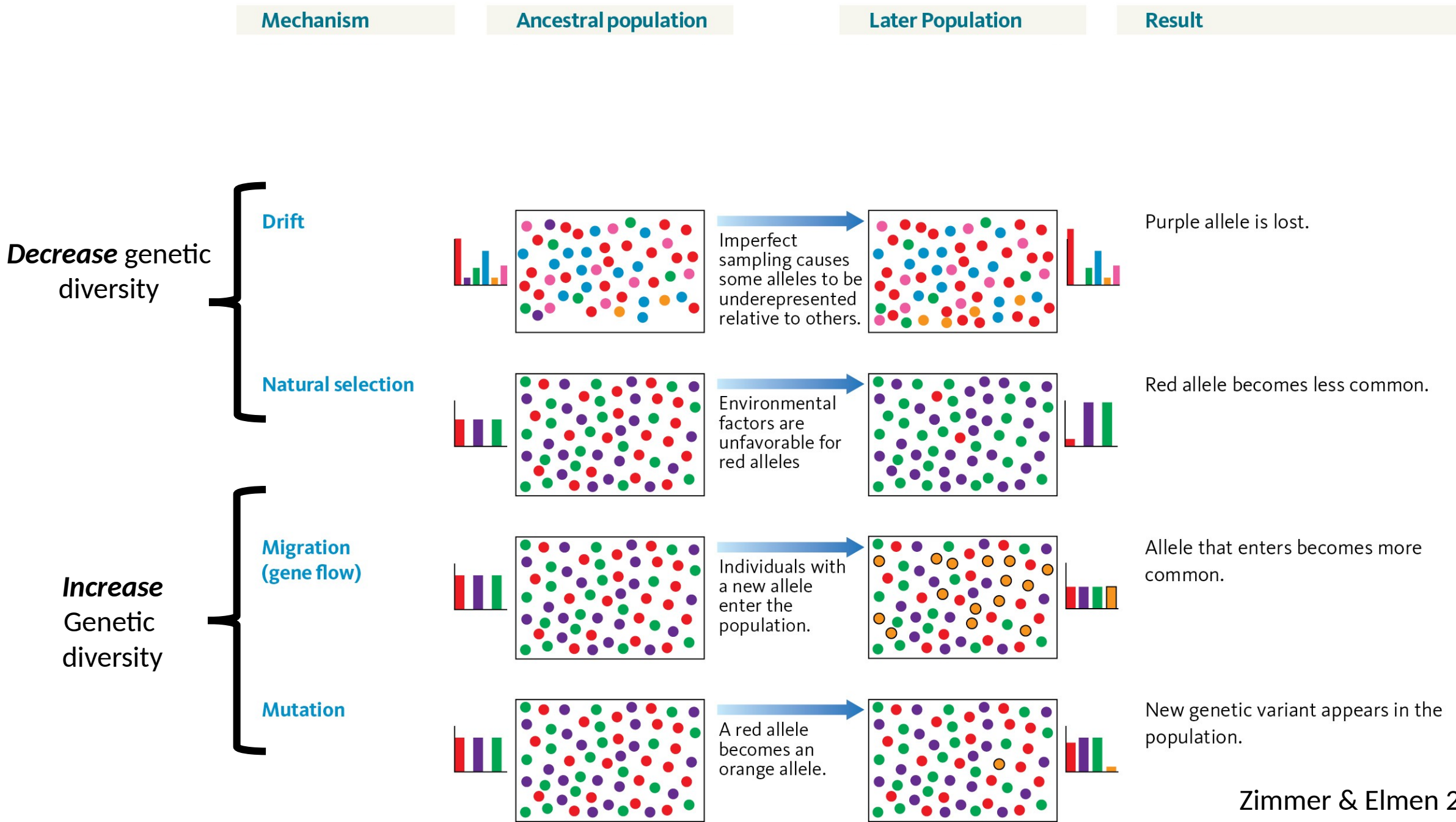
Allele that enters becomes more common.

Mutation



New genetic variant appears in the population.

Mechanisms of Evolution: Altering Allele Frequencies



Mechanisms of Evolution: Altering Allele Frequencies

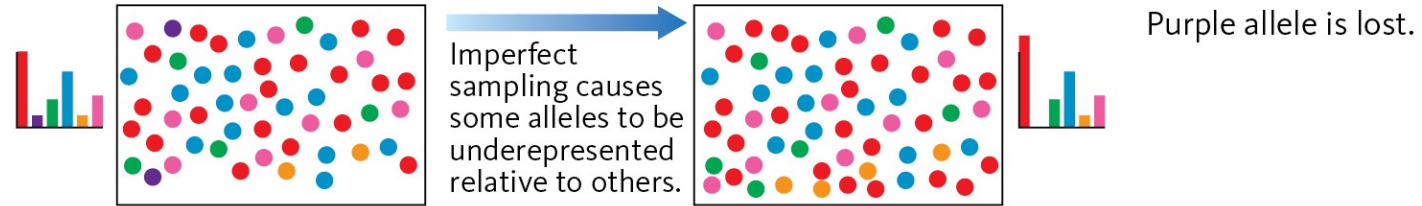
Mechanism

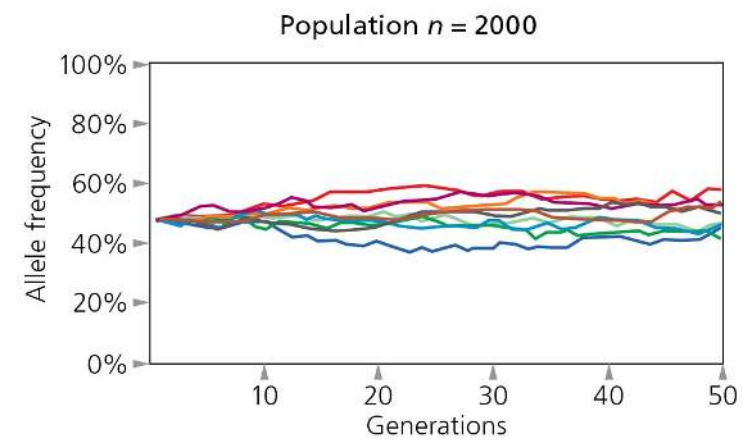
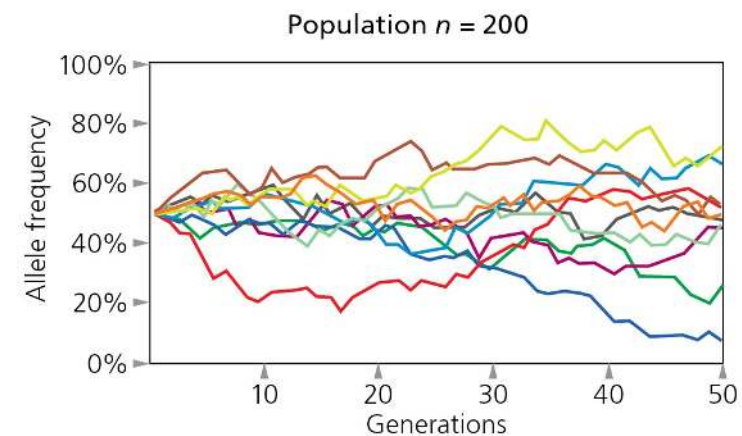
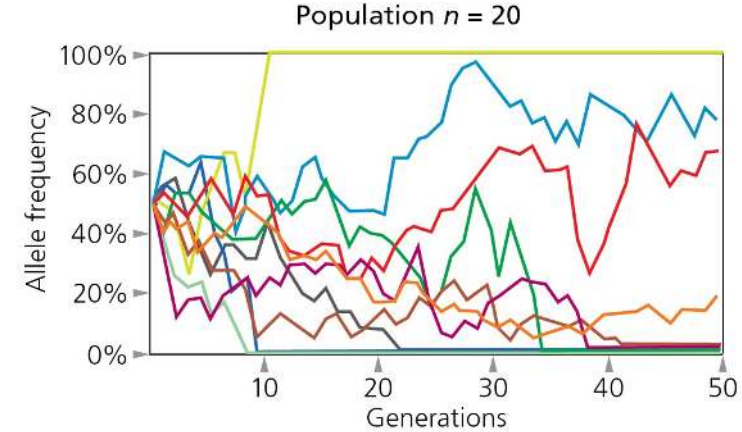
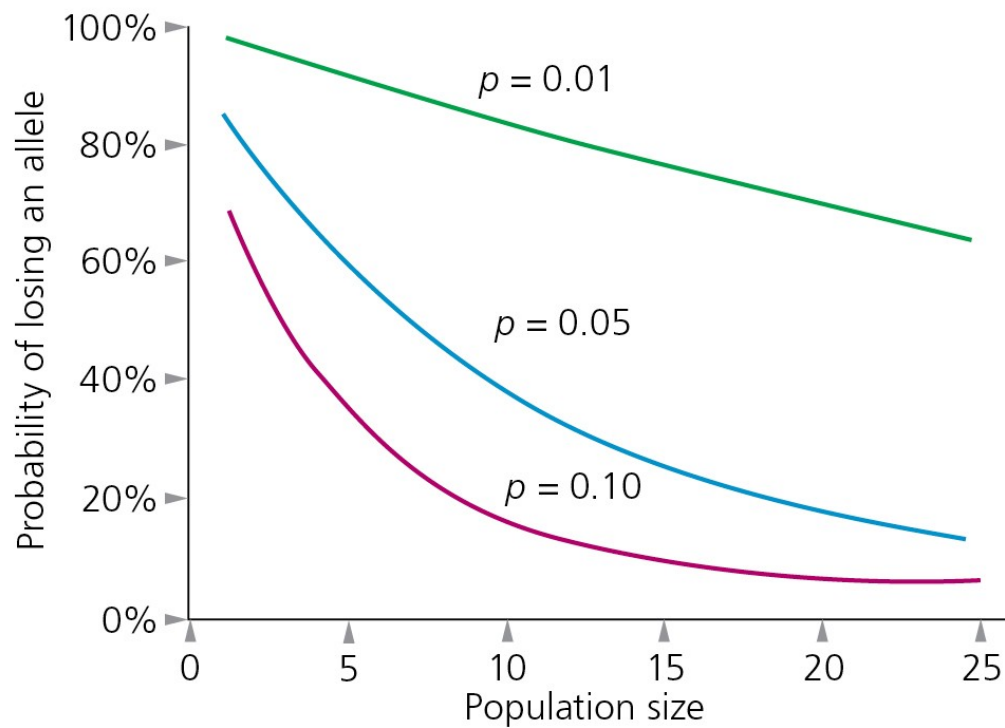
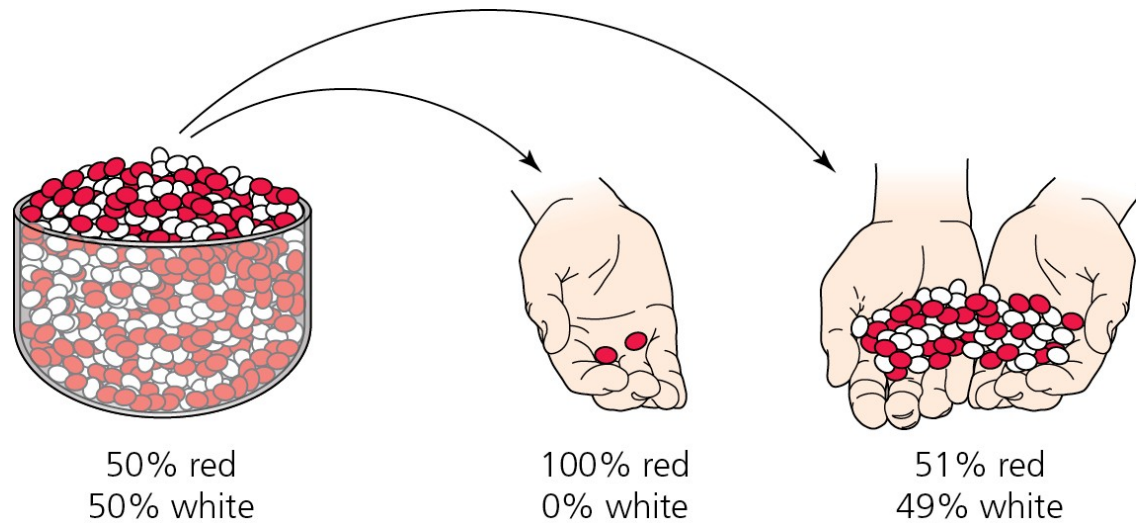
Ancestral population

Later Population

Result

Drift





Mechanisms of Evolution: Altering Allele Frequencies

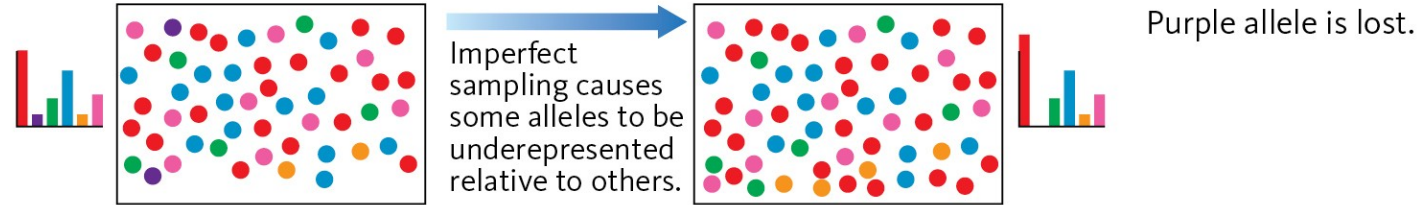
Mechanism

Ancestral population

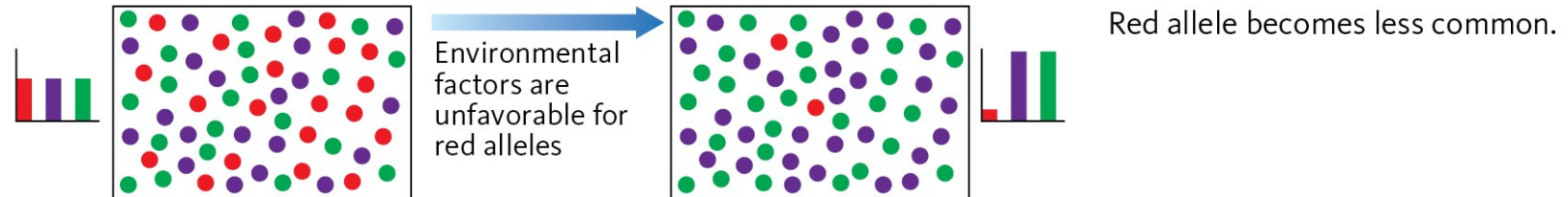
Later Population

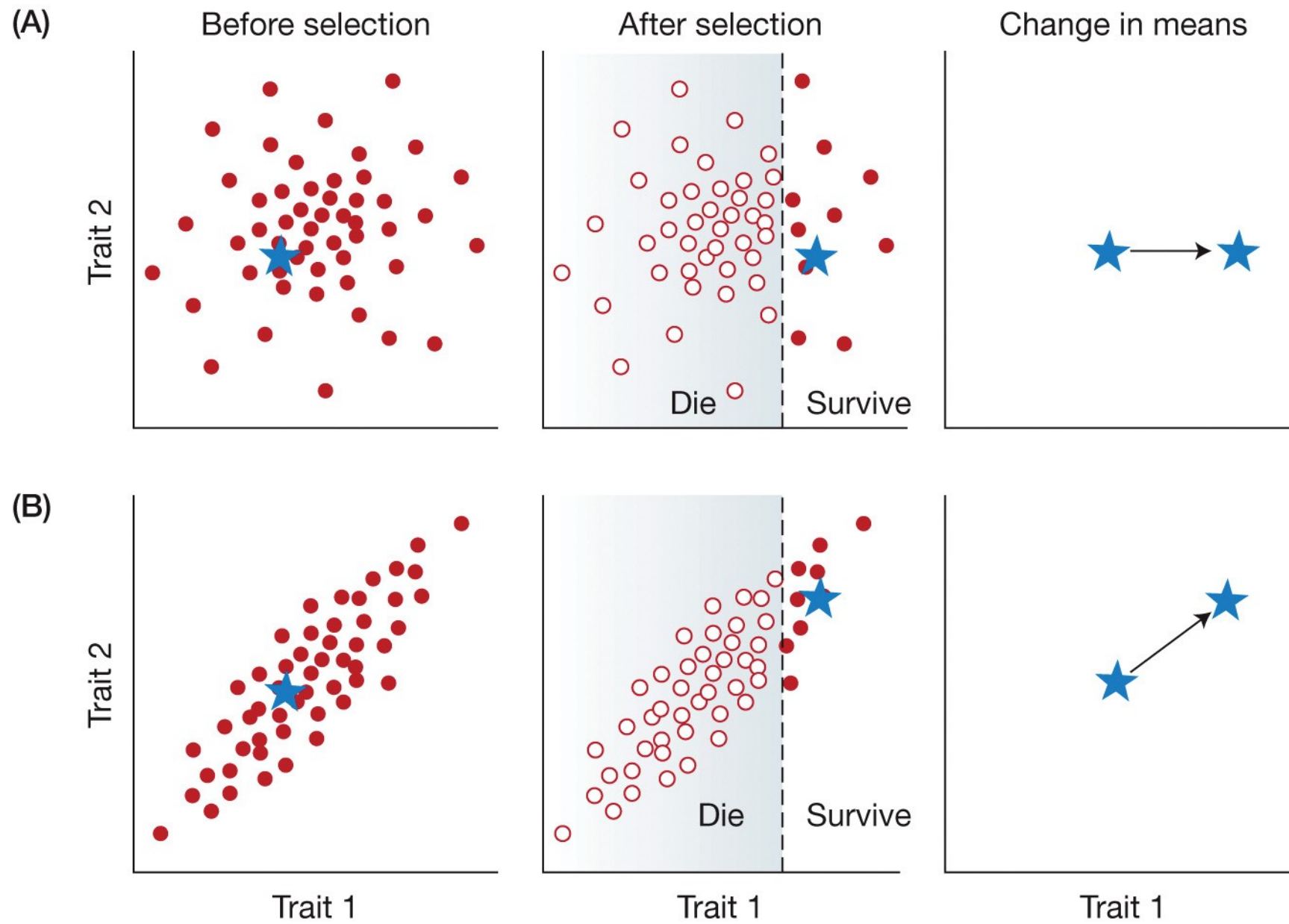
Result

Drift



Natural selection





Mechanisms of Evolution: Altering Allele Frequencies

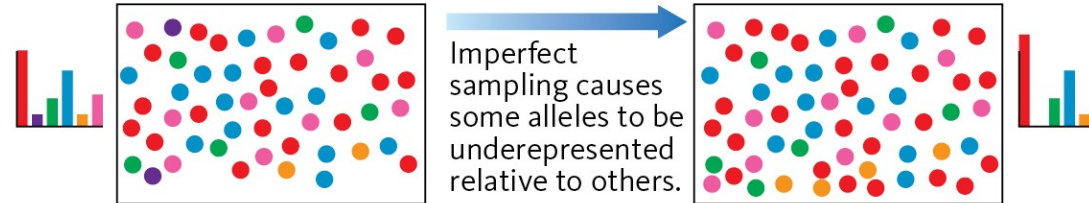
Mechanism

Ancestral population

Later Population

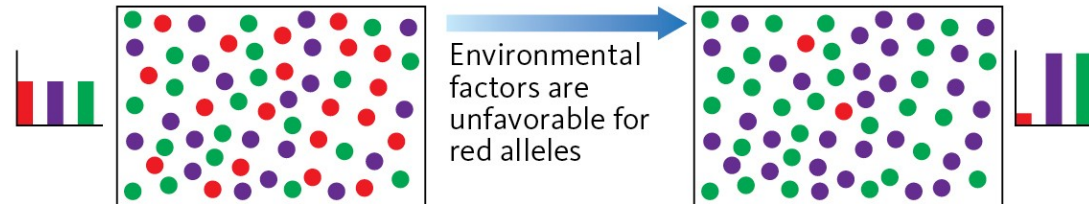
Result

Drift



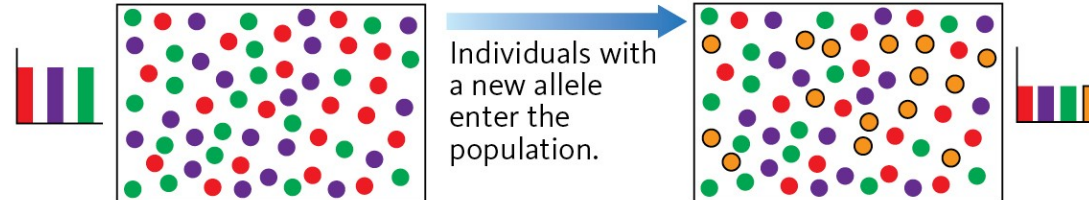
Purple allele is lost.

Natural selection

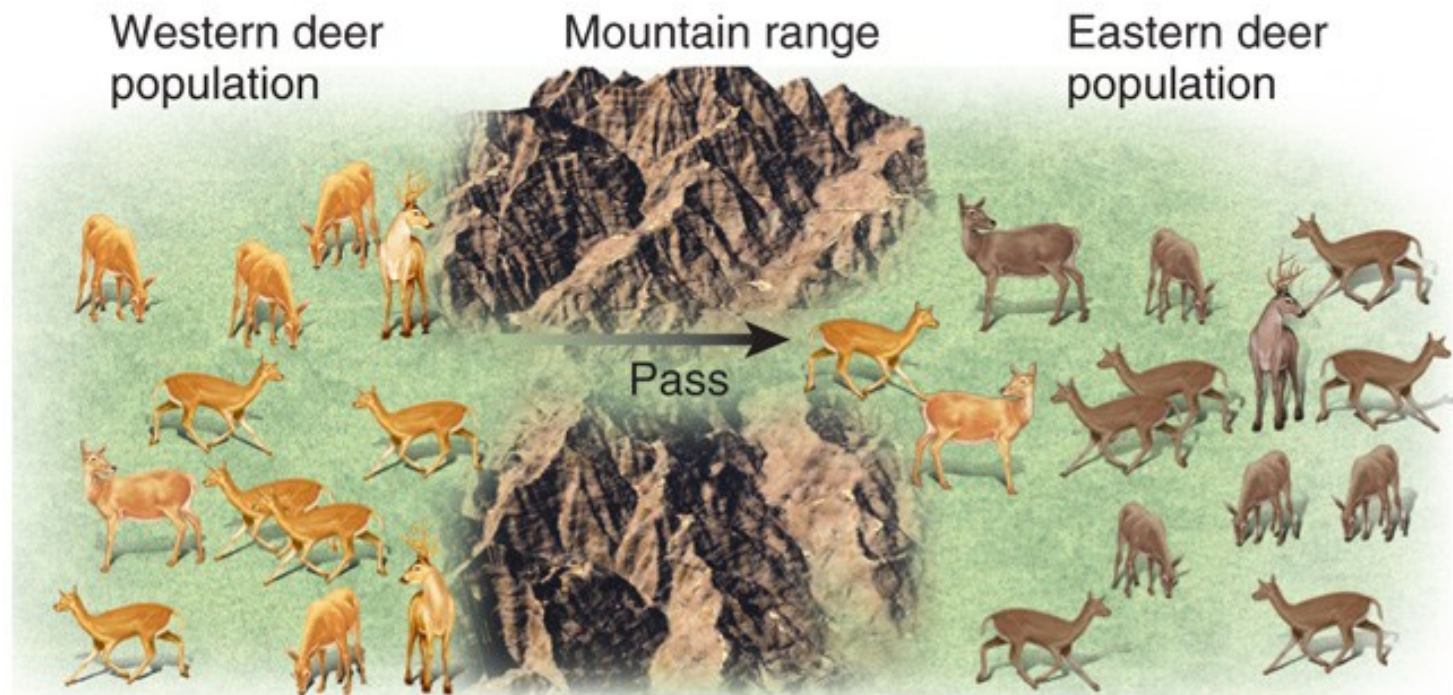
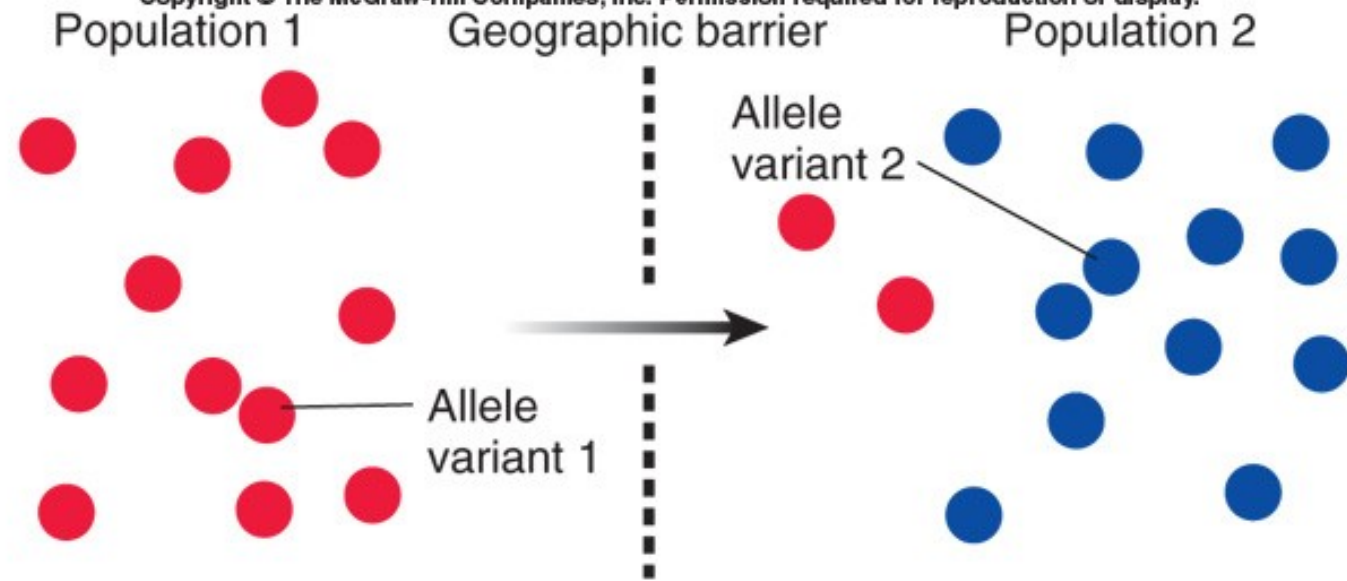


Red allele becomes less common.

Migration (gene flow)



Allele that enters becomes more common.



Mechanisms of Evolution: Altering Allele Frequencies

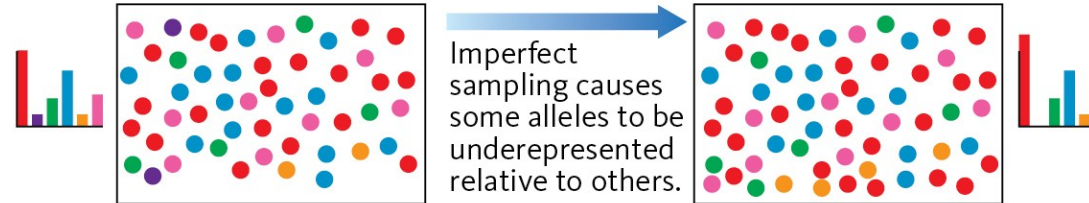
Mechanism

Ancestral population

Later Population

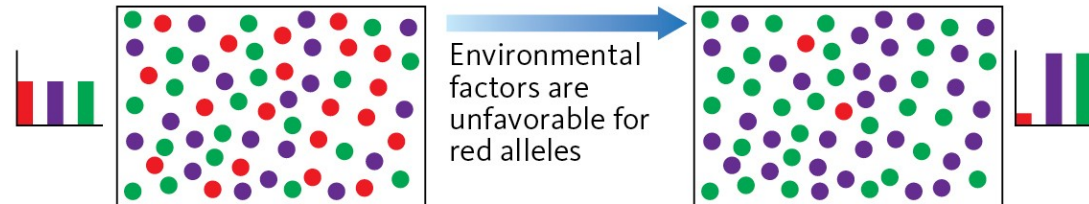
Result

Drift



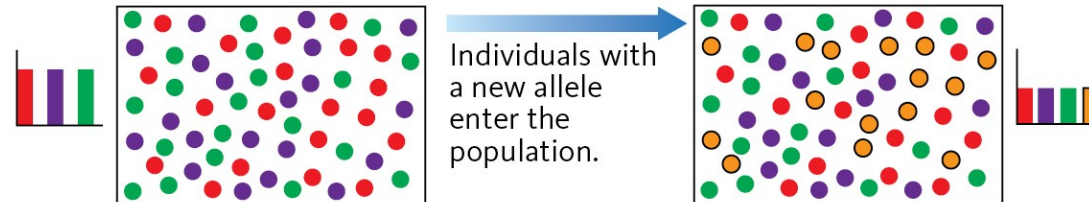
Purple allele is lost.

Natural selection



Red allele becomes less common.

Migration (gene flow)



Allele that enters becomes more common.

Mutation

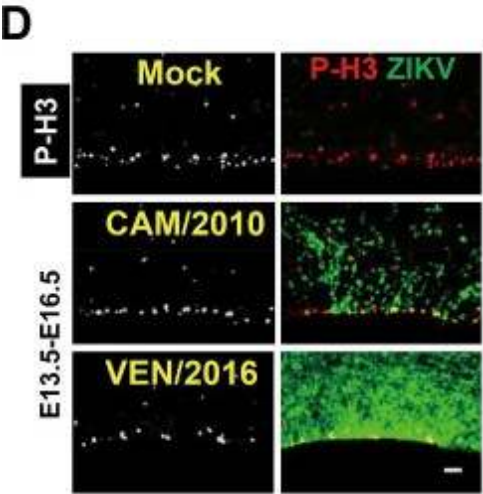
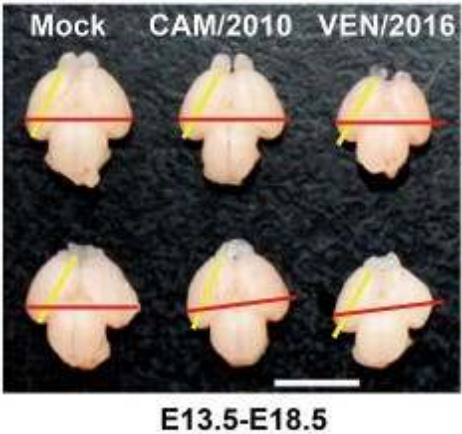


New genetic variant appears in the population.

Cite as: L. Yuan *et al.*, *Science*
10.1126/science.aam7120 (2017).

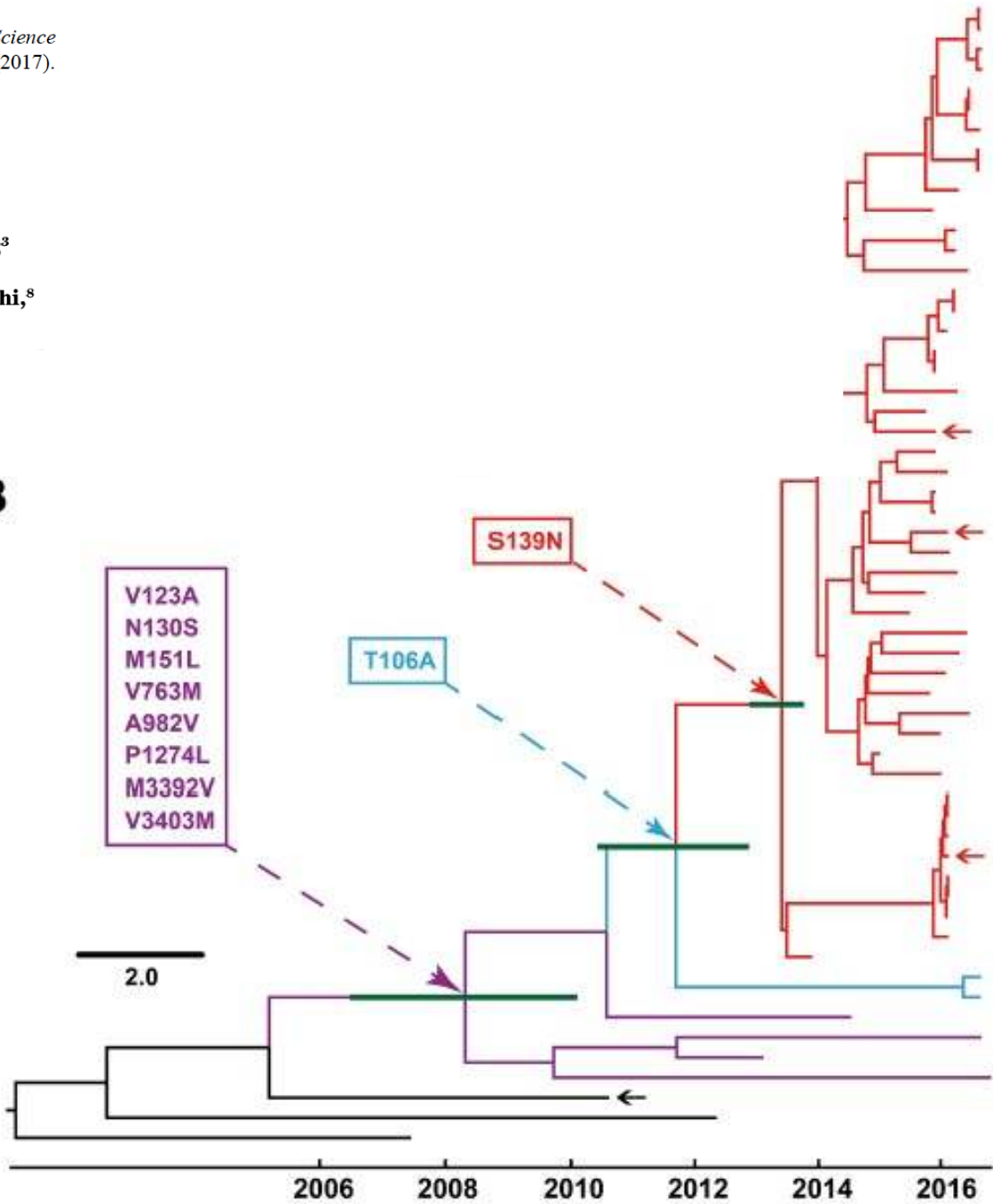
A single mutation in the prM protein of Zika virus contributes to fetal microcephaly

Ling Yuan,^{1,2*} Xing-Yao Huang,^{3*} Zhong-Yu Liu,^{3*} Feng Zhang,^{1,2*} Xing-Liang Zhu,^{1,2*} Jiu-Yang Yu,^{3*} Xue Ji,³ Yan-Peng Xu,³ Guanghui Li,^{1,2} Cui Li,^{1,2} Hong-Jiang Wang,³ Yong-Qiang Deng,³ Menghua Wu,⁴ Meng-Li Cheng,^{3,5} Qing Ye,³ Dong-Yang Xie,^{3,5} Xiao-Feng Li,³ Xiangxi Wang,⁶ Weifeng Shi,⁷ Baoyang Hu,⁴ Pei-Yong Shi,⁸ Zhiheng Xu,^{1,2,9†} Cheng-Feng Qin^{3†}



Ancestral strain
(Camboja)
Virulent strain
(atual)

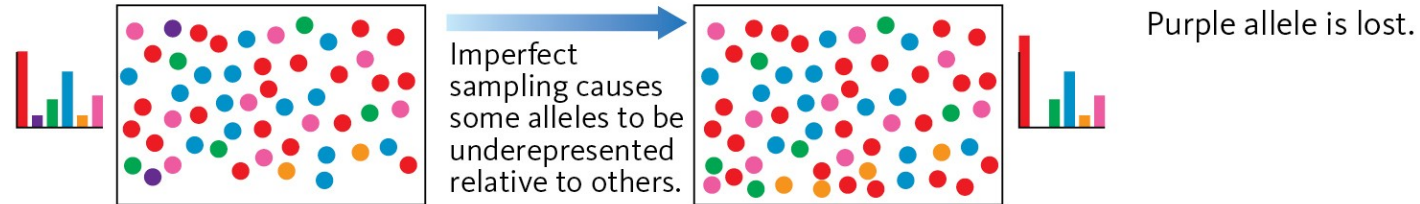
B



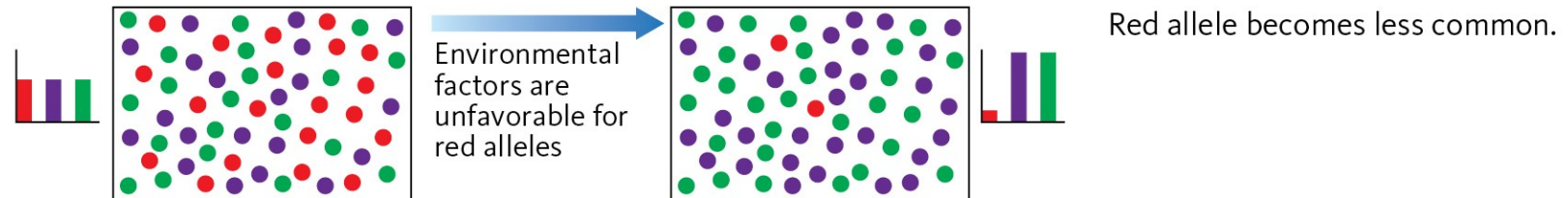
Mechanisms of Evolution: Altering Allele Frequencies

Mechanism	Ancestral population	Later Population	Result
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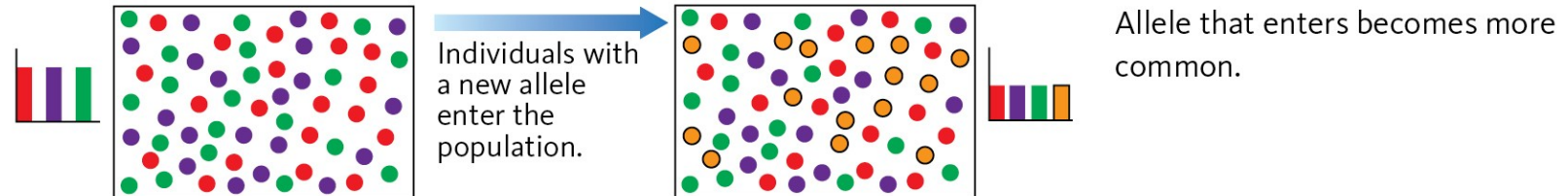
Drift



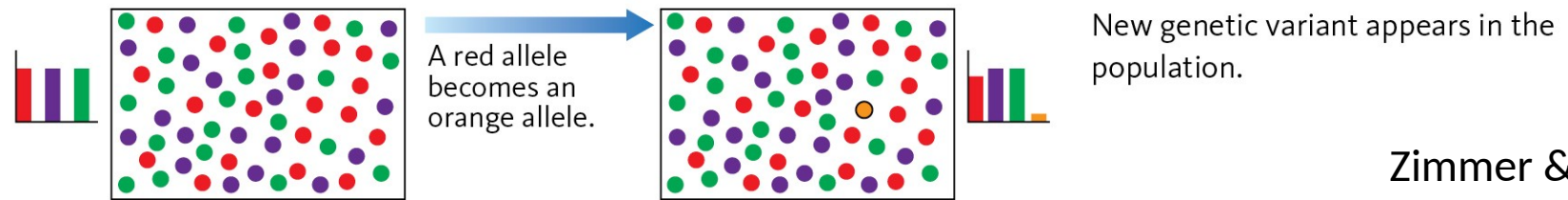
Natural selection



Migration (gene flow)



Mutation



Continuous traits

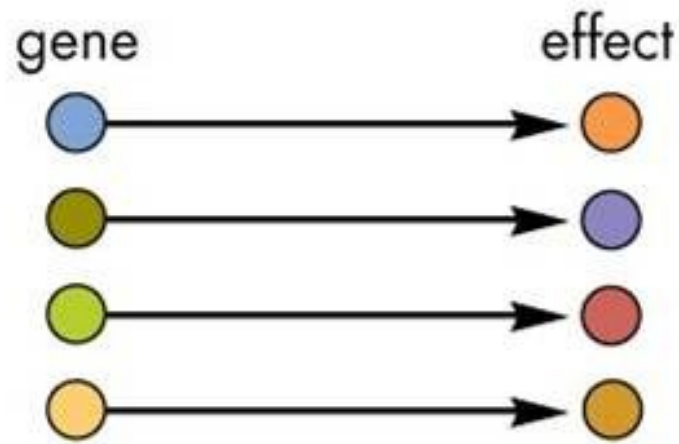
Distribution in a population

Genetic architecture and evolutionary adaptation

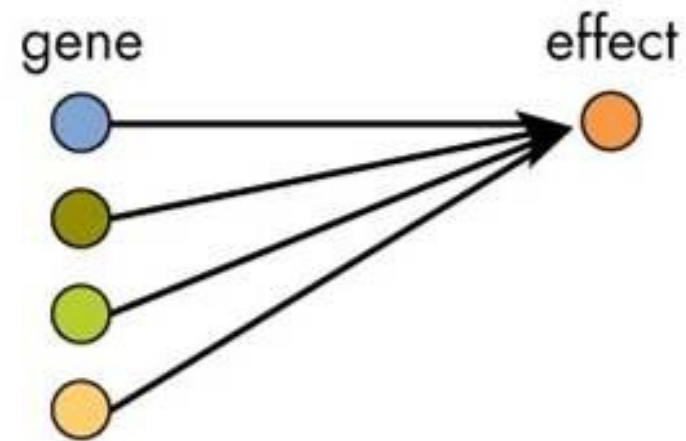
or how four letters created all diversity we see

Modes of genetic inheritance

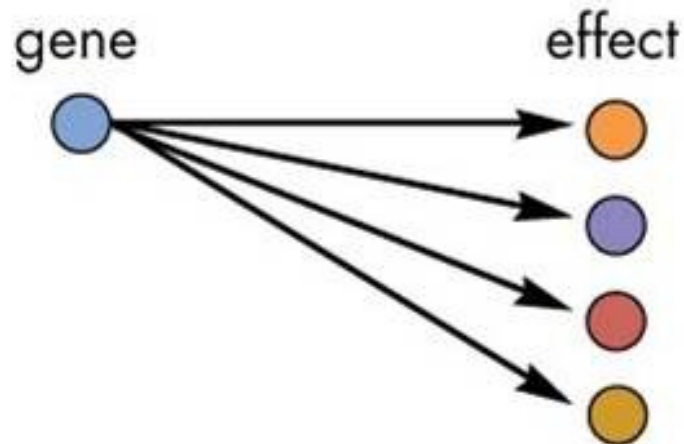
- Simple dominance
 - Autosomal chromosome
 - Sexual chromosome
- Epistasis => effect of one locus depends on one or more “modifier” genes
- Pleiotropy => when one allele influences two or more non-related phenotypic traits
- Polygenic inheritance => trait is the result of several genes
 - Height, eye color, skin color



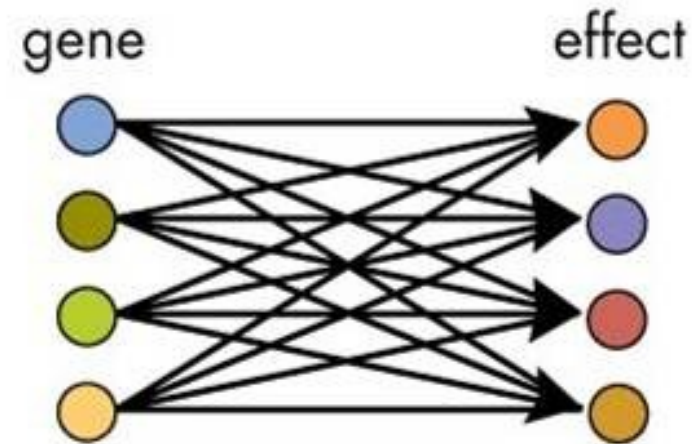
Each gene has a distinct biological effect.



Polygenic trait: Many genes contribute to a single effect.



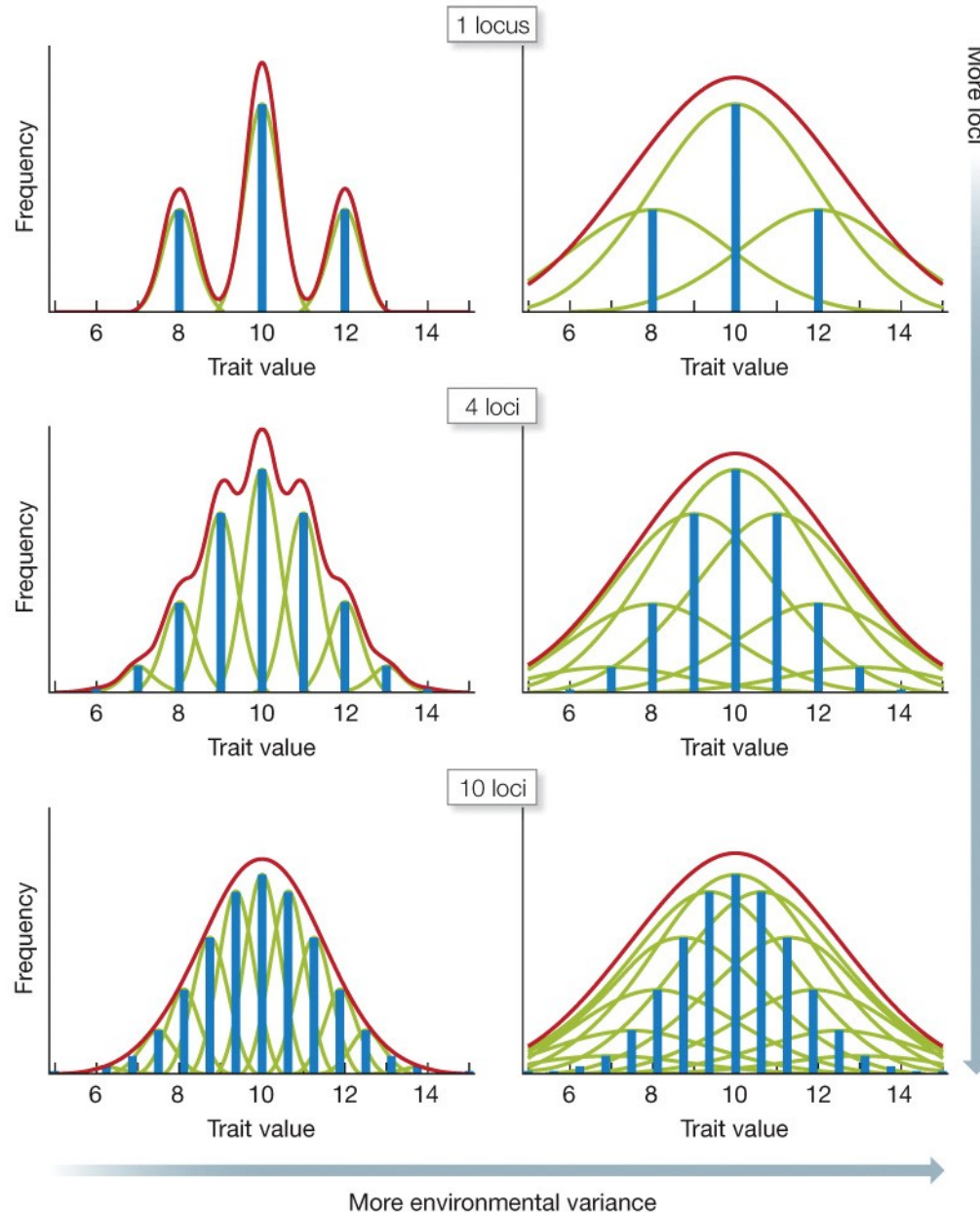
Pleiotropy: A gene has multiple effects.



Polygenic traits and pleiotropy



EVOLUTION 4e, Figure 6.2
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EVOLUTION 4e, Figure 6.3
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(399)

XV.—The Correlation between Relatives on the Supposition of Mendelian Inheritance. By R. A. Fisher, B.A. Communicated by Professor J. ARTHUR THOMSON. (With Four Figures in Text.)

(MS. received June 15, 1918. Read July 8, 1918. Issued separately October 1, 1918.)

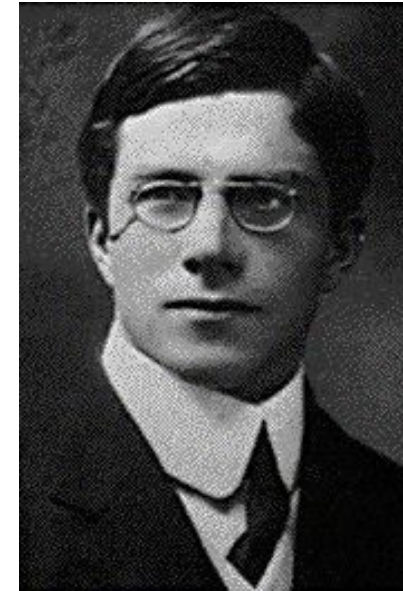
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Several attempts have already been made to interpret the well-established results of biometry in accordance with the Mendelian scheme of inheritance. It is here attempted to ascertain the biometrical properties of a population of a more general type than has hitherto been examined, inheritance in which follows this scheme. It is hoped that in this way it will be possible to make a more exact analysis of the causes of human variability. The great body of available statistics show us that the deviations of a human measurement from its mean follow very closely the Normal Law of Errors, and, therefore, that the variability may be uniformly measured by the standard deviation corresponding to the square root of the mean square error. When there are two independent causes of variability capable of producing in an otherwise uniform population distributions with standard deviations σ_1 and σ_2 , it is found that the distribution, when both causes act together, has a standard deviation $\sqrt{\sigma_1^2 + \sigma_2^2}$. It is therefore desirable in analysing the causes of variability to deal with the square of the standard deviation as the measure of variability. We shall term this quantity the Variance of the normal population to which it refers, and we may now ascribe to the constituent causes fractions or percentages of the total variance which they together produce. It is desirable on the one hand that the elementary ideas at the basis of the calculus of correlations should be clearly understood, and easily expressed in ordinary language, and on the other that loose phrases about the "percentage of causation,"

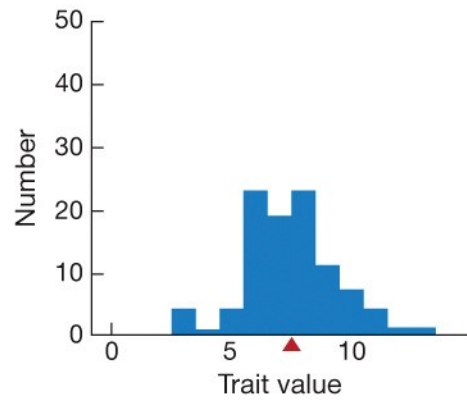
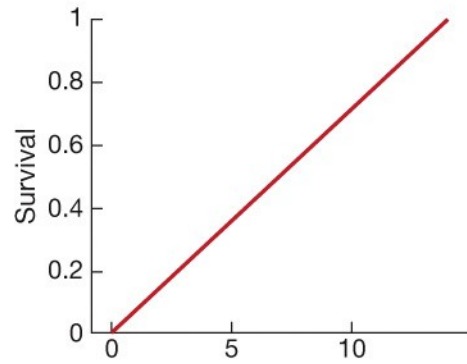
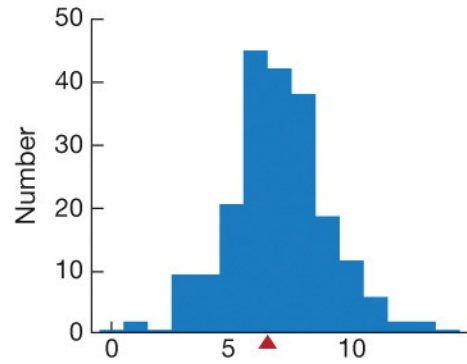
TRANS. ROY. SOC. EDIN., VOL. LII, PART II (NO. 15)

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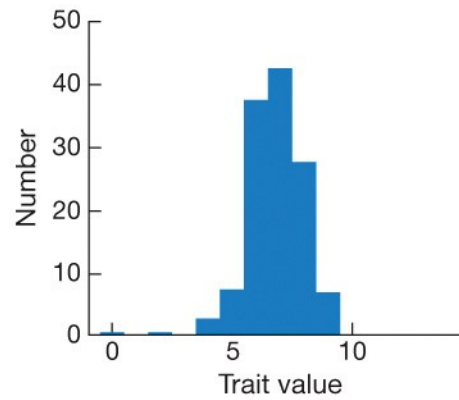
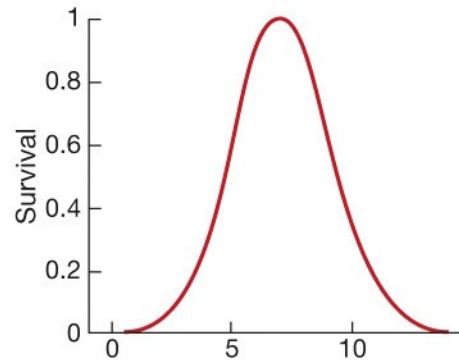
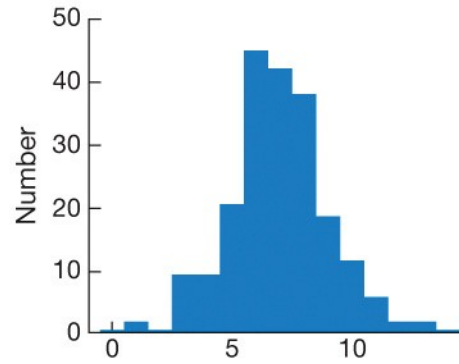


The three types of selection

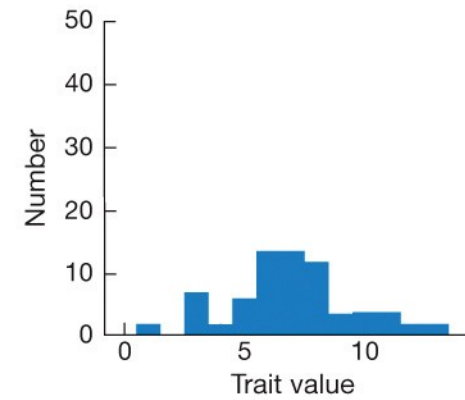
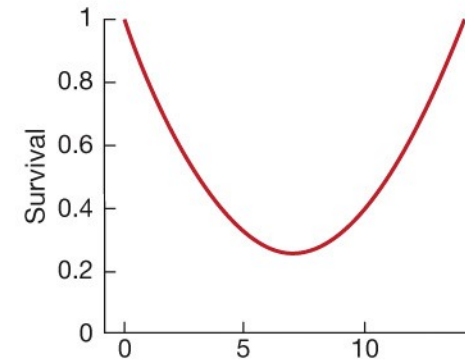
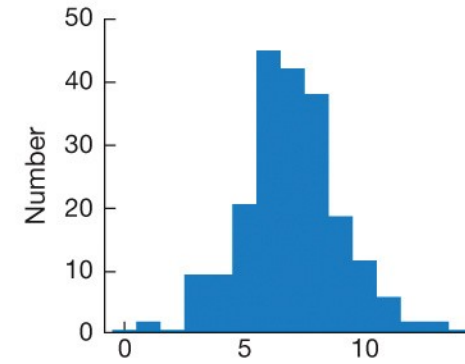
(A) Directional



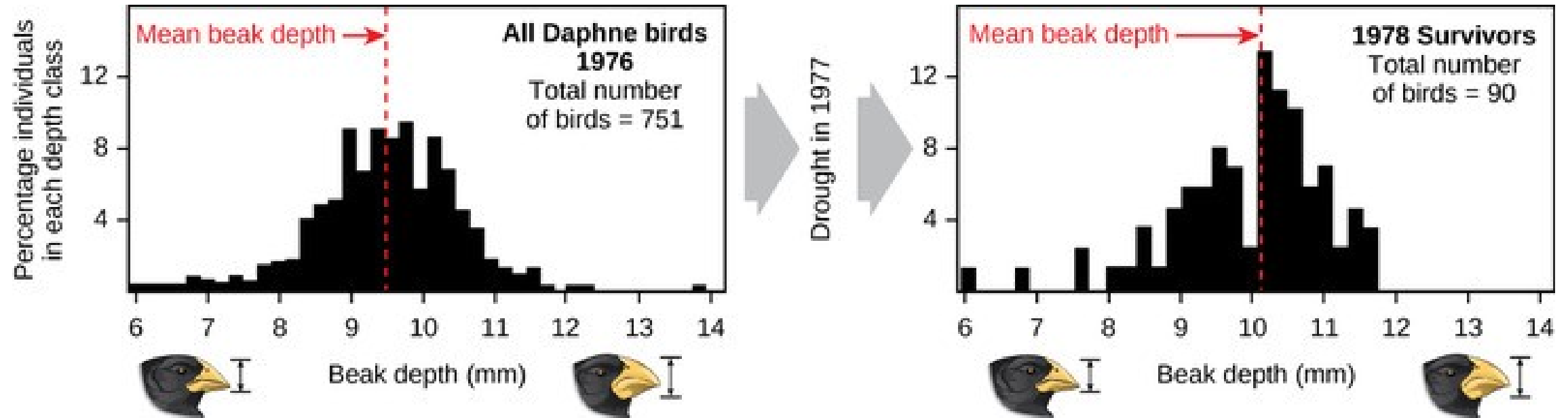
(B) Stabilizing



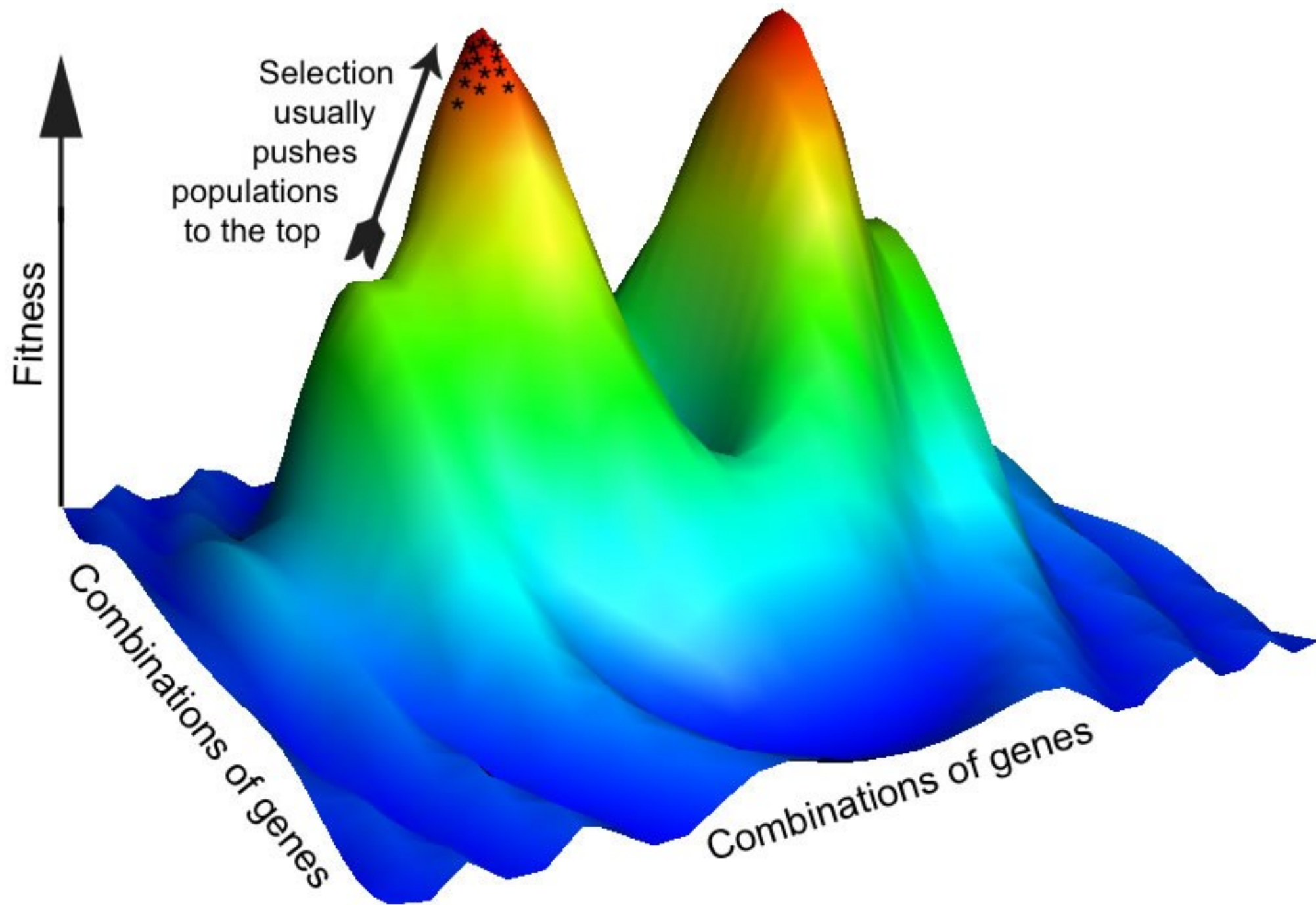
(C) Disruptive



Change in one generation in beak height of Darwin finches, due to drought and decrease in the availability of small, soft seeds

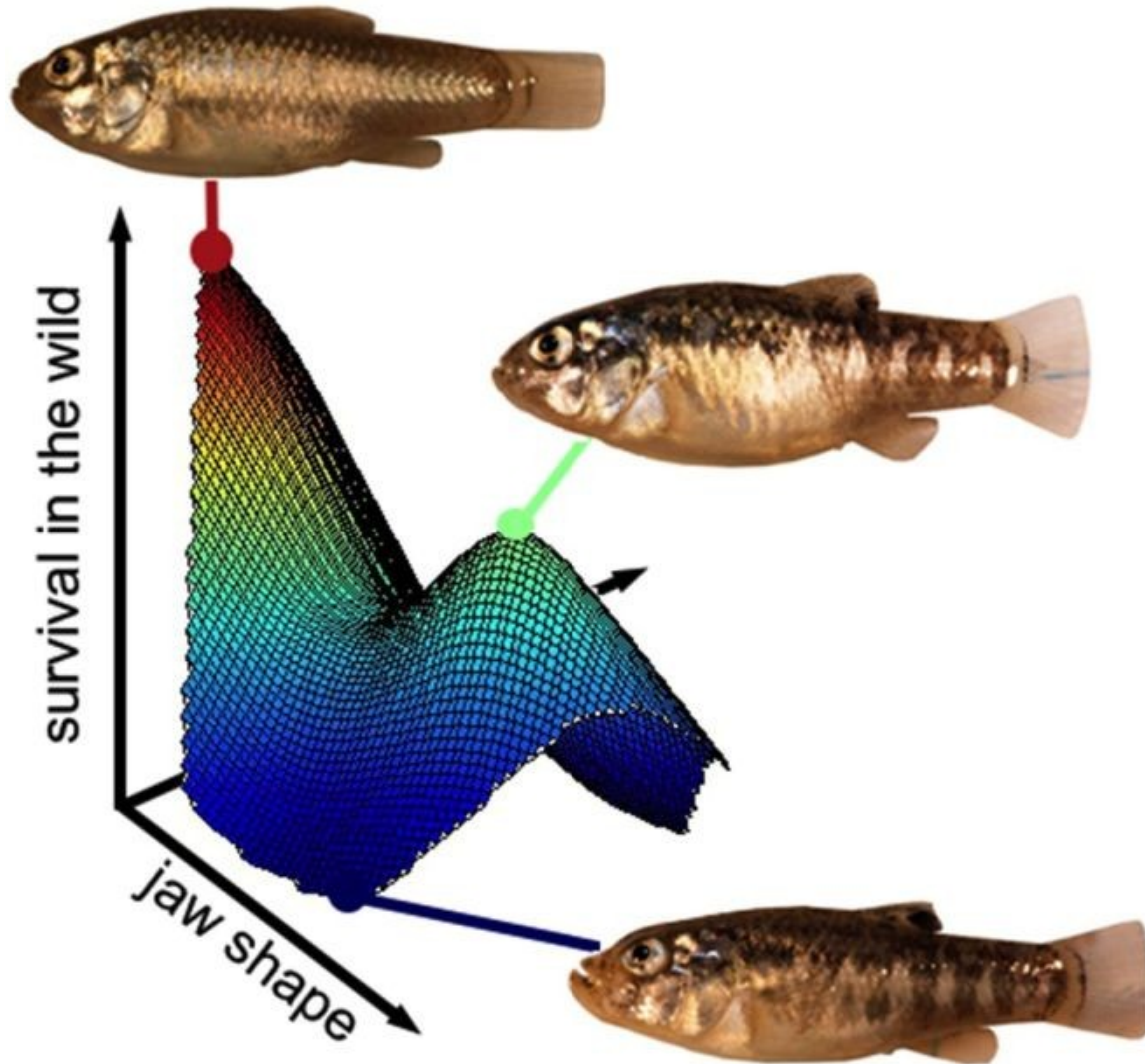


Adaptive landscapes



Conceptual bridge between micro-
and macroevolution

How to estimate adaptive landscapes in wild populations



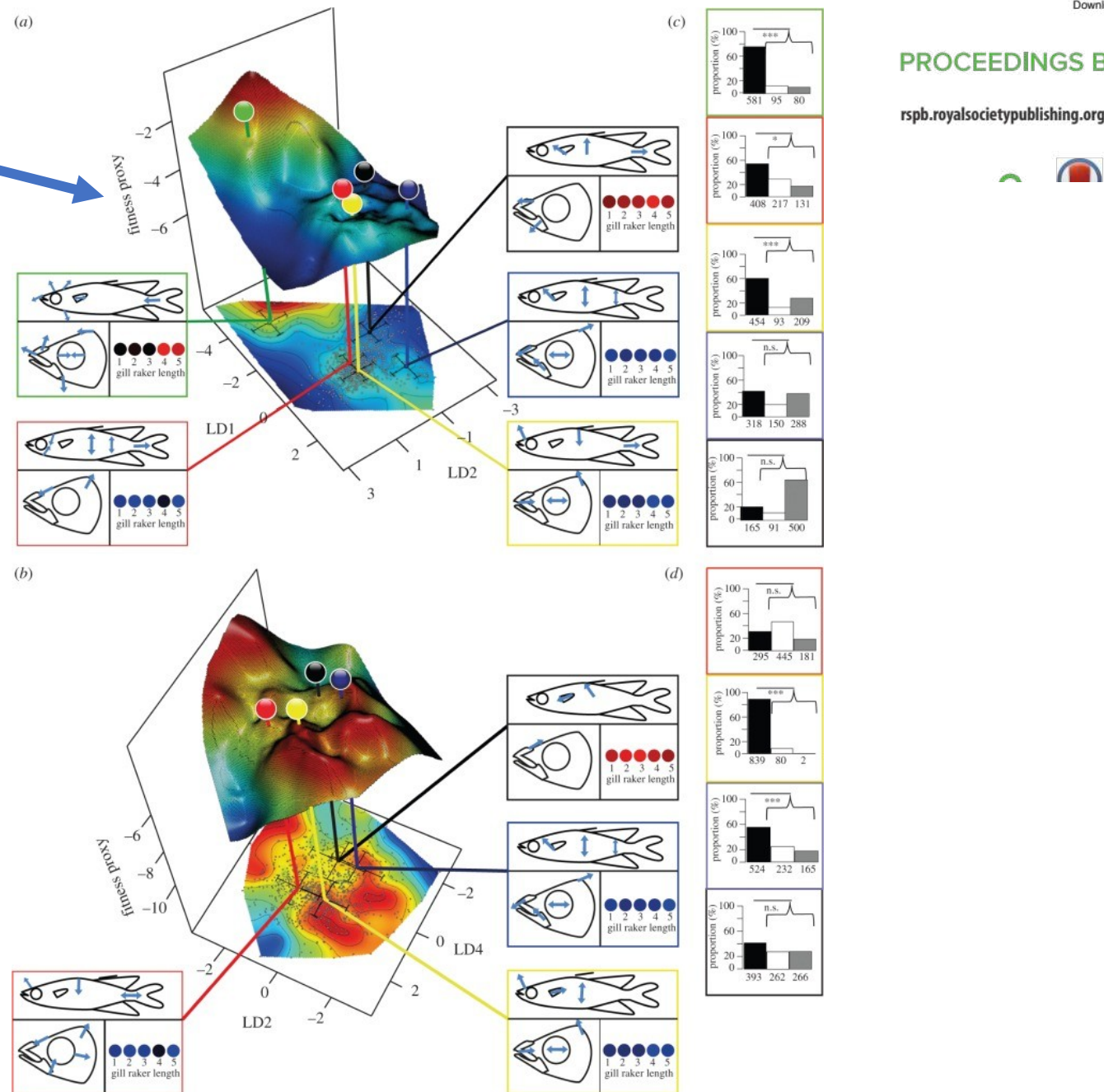
- Jaw shape
- Diet specialization
 - Specialist in snails diversify faster

REPORTS

Multiple Fitness Peaks on the Adaptive Landscape Drive Adaptive Radiation in the Wild

Christopher H. Martin* and Peter C. Wainwright

Relative liver
weight



PROCEEDINGS B

rspb.royalsocietypublishing.org

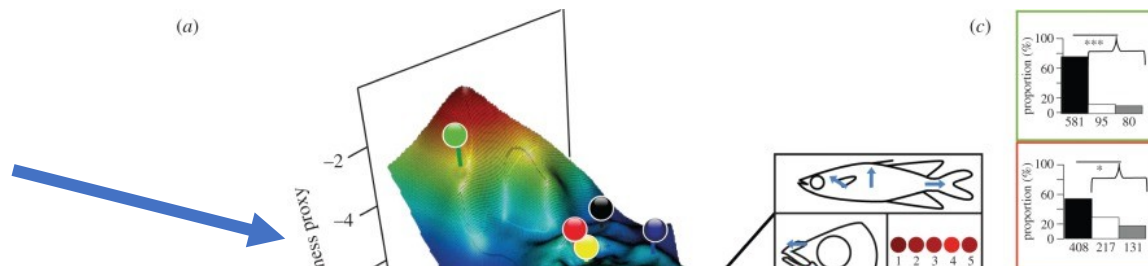


Rugged adaptive landscapes shape a complex, sympatric radiation

Jobst Pfaender^{1,2}, Renny K. Hadiaty³, Ulrich K. Schliewen⁴ and Fabian Herder¹

¹Department of Biology, University of Konstanz, 78434 Konstanz, Germany; ²Department of Biology, University of Tübingen, 72076 Tübingen, Germany; ³Department of Biology, University of Tübingen, 72076 Tübingen, Germany; ⁴Department of Biology, University of Tübingen, 72076 Tübingen, Germany

Relative liver
weight



PROCEEDINGS B

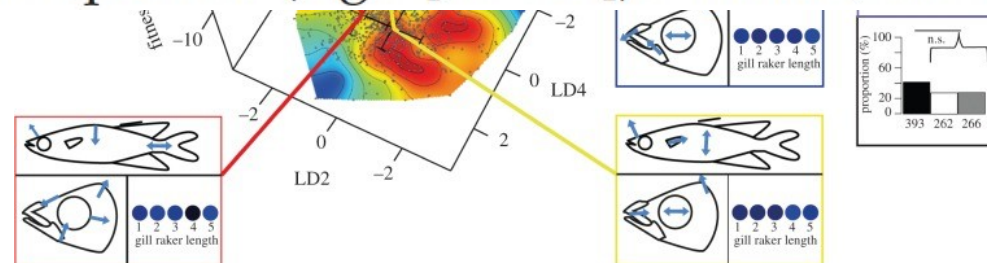
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Rugged adaptive landscapes shape
a complex, sympatric radiation

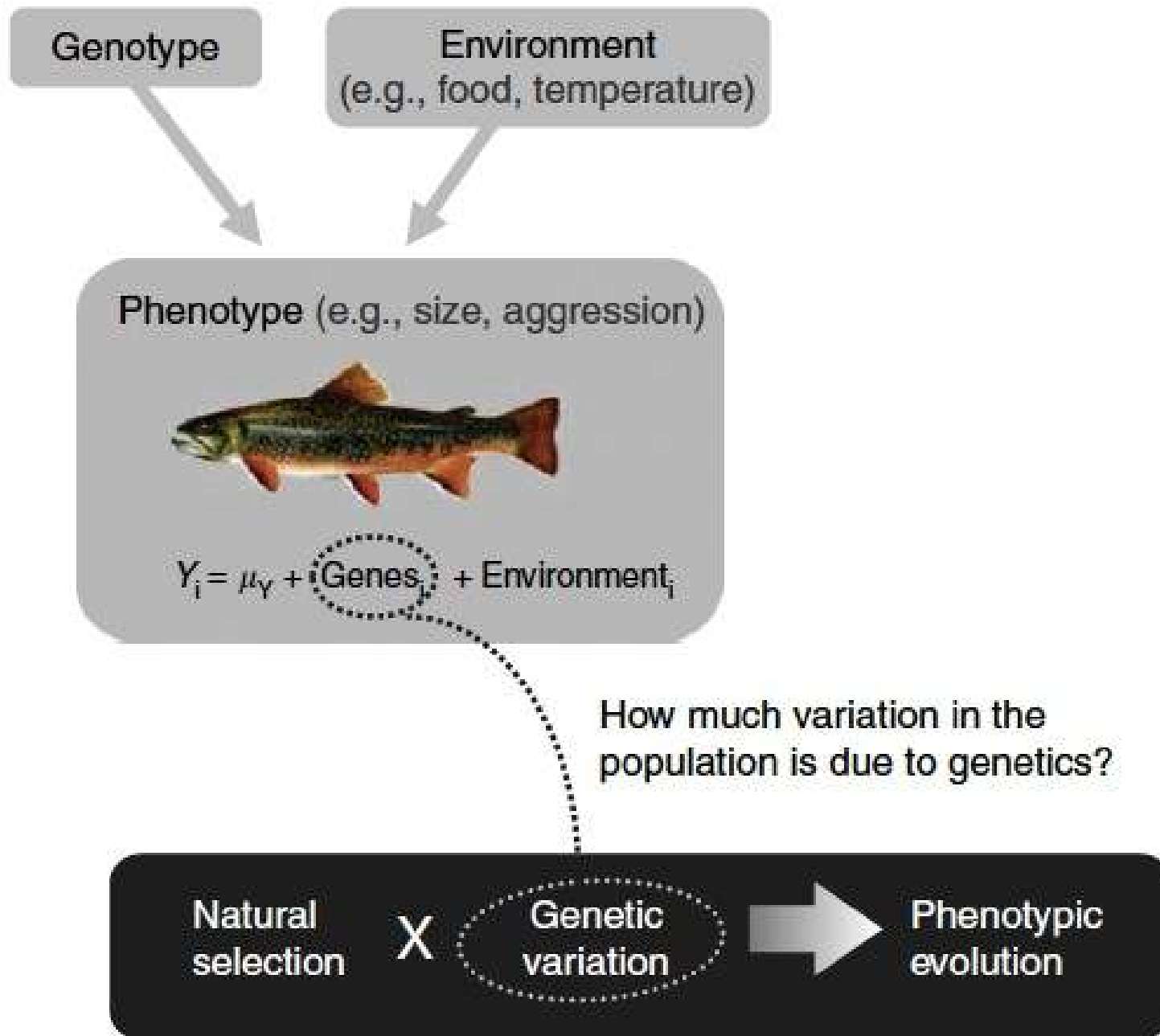
Jobst Pfaender^{1,2}, Renny K. Hadiaty³, Ulrich K. Schliwen⁴ and Fabian Herder¹

(j) Fitness proxies

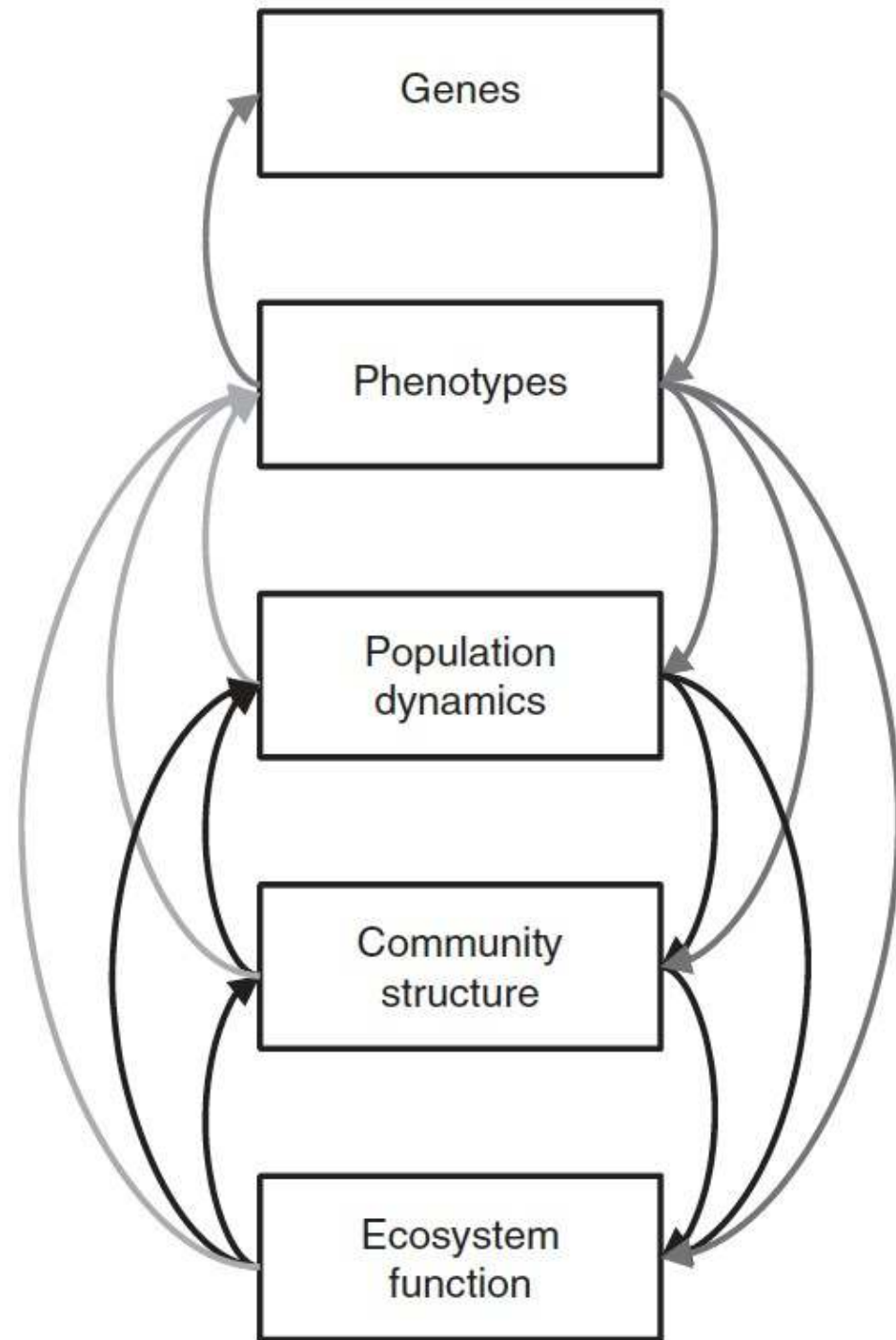
Estimating fitness is challenging, especially in large natural populations. For fishes, various proxies have been applied to quantify lifetime reproduction and survivorship [48] (see also [8,48,49]). Here, we use the relative liver weight as proxy for individual fitness, because the liver is an important fat-storage organ in fishes [50] and is frequently used as an estimator for individual condition in fishes (e.g. [51,52]). Moreover, individual condition is tightly correlated with reproductive capacity in other fish species (e.g. [53–56]). Hence, it is reasonable to



As ecologists, we need to learn (or
talk more with) quantitative
genetics more than ever!



Eco-evolutionary
dynamics



Or, how evolution
influences ecology and
vice versa

Natural selection in the wild can
happen very fast

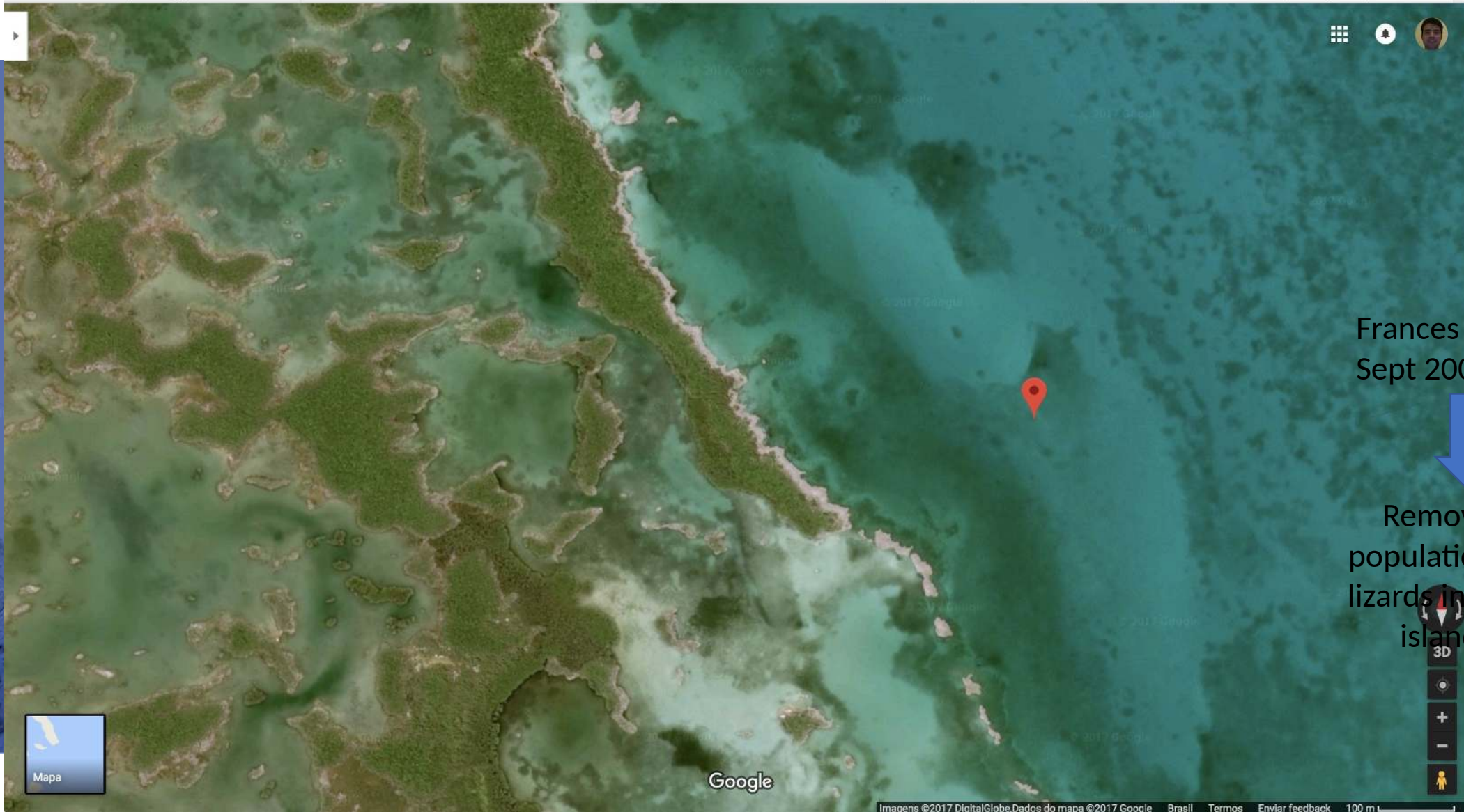
Founder Effects Persist Despite Disturbance: A Field

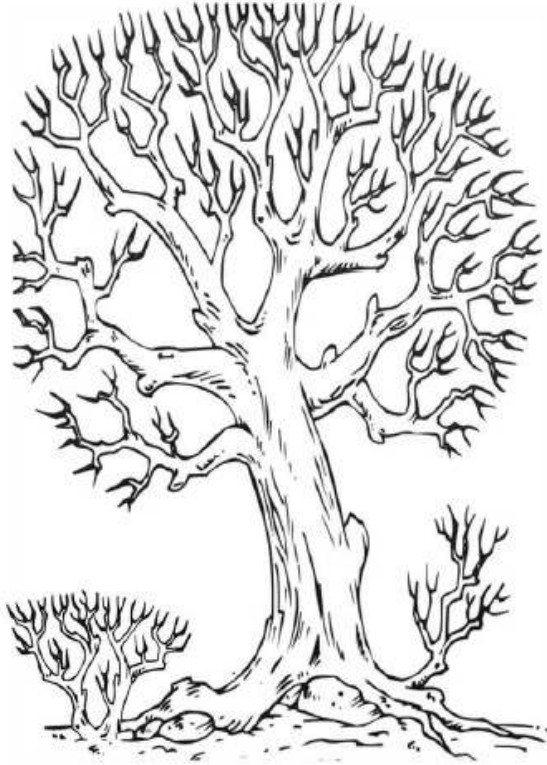
ner,³ David A. Spiller,³ Jonathan B. Losos¹

NCE www.sciencemag.org

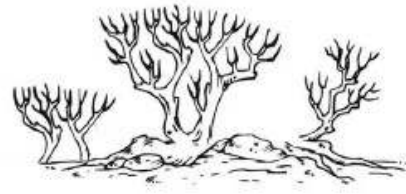
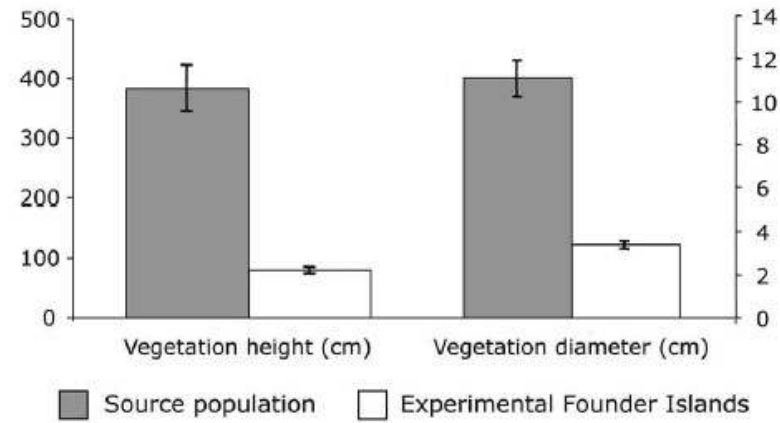
Frances hurricane
Sept 2004

Removed
populations of
lizards in small
islands



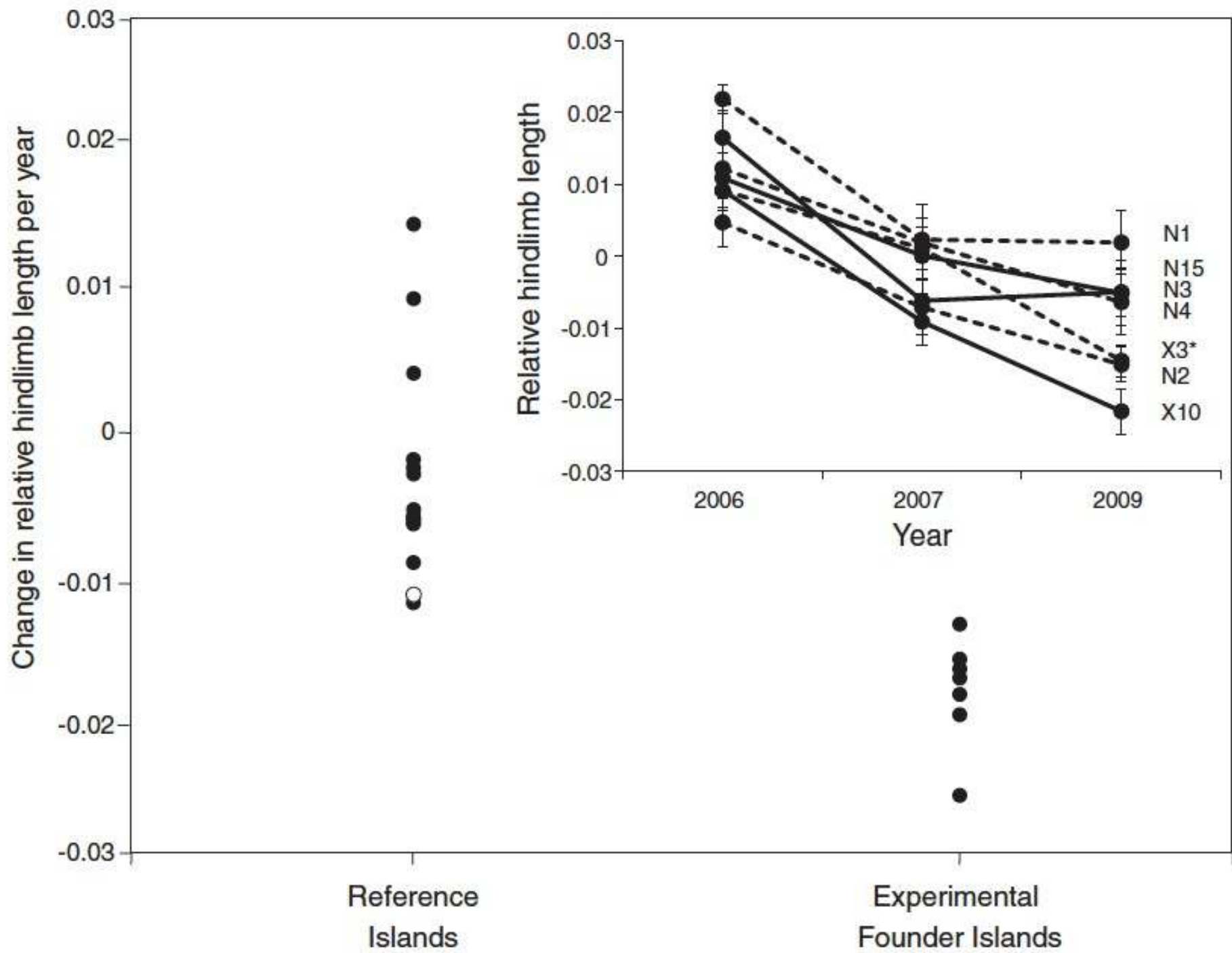


Source Population
(Iron Cay)



Experimental
Founder Islands





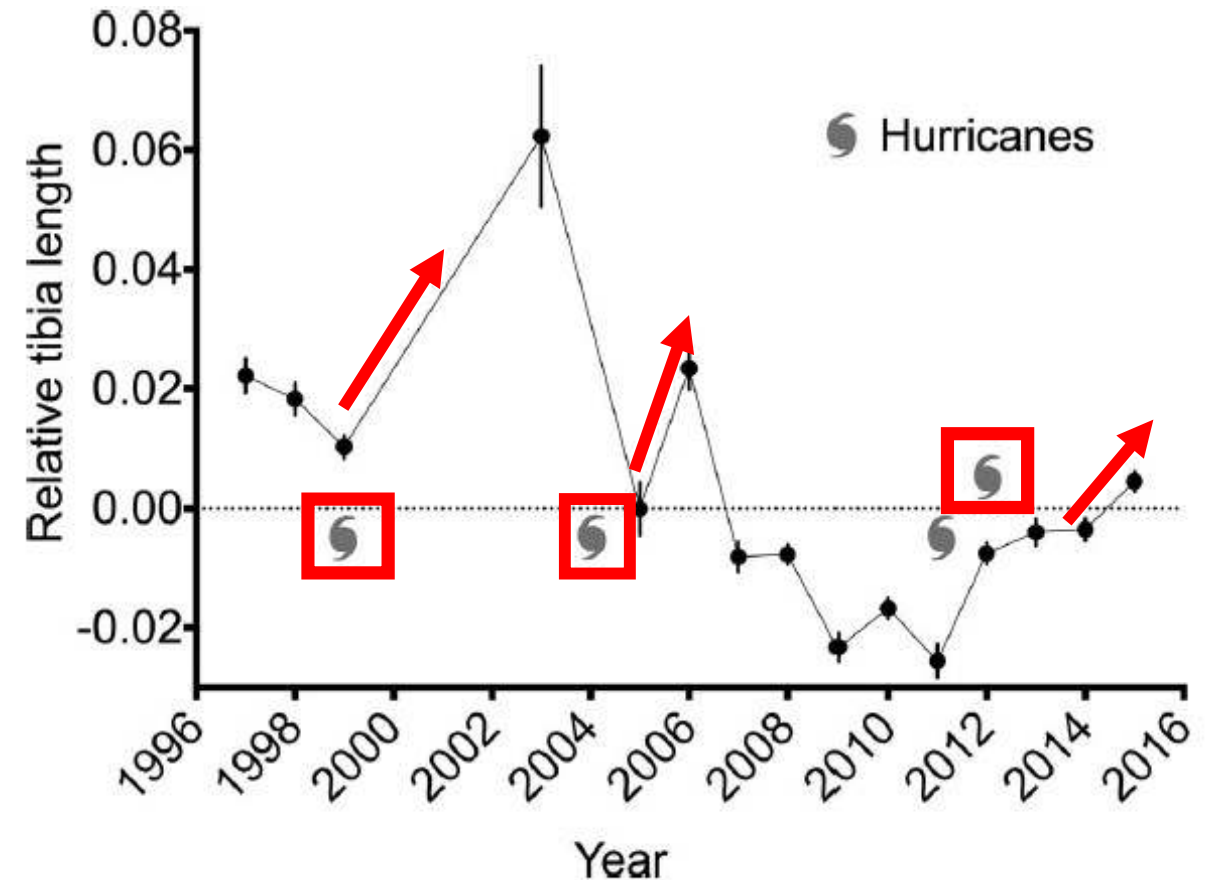
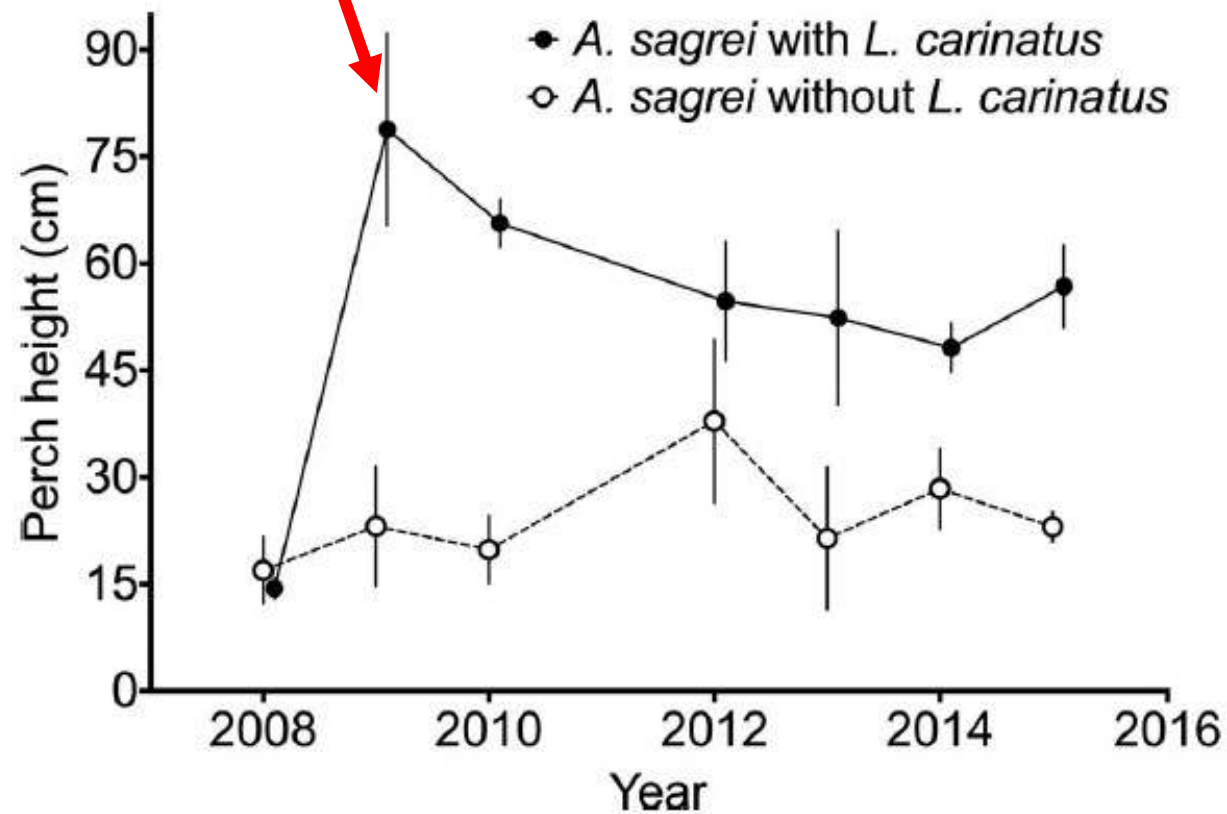
Study conclusions

- Differentiation in the source population is basically a result of adaptation to the new environment in experimental islands
- However, the signature of the founder effect is apparent, even in the face of adaptive divergence. Indeed, the current variation among experimental islands is better explained by the initial phenotypes than the current environmental conditions
- In summary: the populations differentiate faster due to adaptation to a new environmental setting, but the genetic *pool* is constrained to the founder population, i.e., it is contingent

A Multigenerational Field Experiment on Eco-evolutionary Dynamics of the Influential Lizard *Anolis sagrei*: A Mid-term Report

Thomas W. Schoener¹, Jason J. Kolbe², Manuel Leal³, Jonathan B. Losos⁴, and David A. Spiller¹

Spend less time in the ground in the presence of a predator



Partitioning the effect of ecology and
evolution on community and ecosystem
dynamics

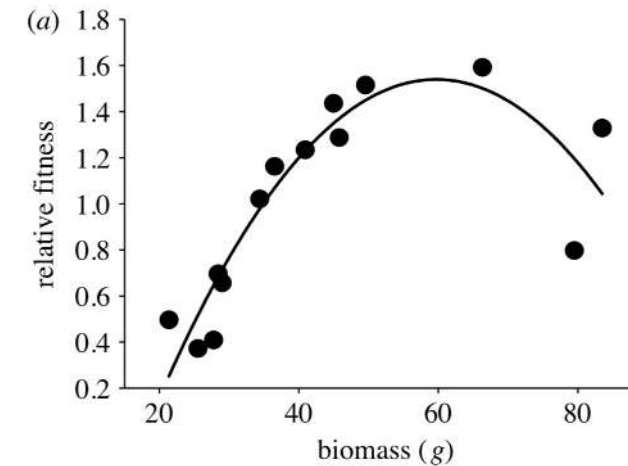
Lande's equation
(multivariate extension of the
breeder's equation)

$$\begin{bmatrix} \Delta \bar{z}_1 \\ \Delta \bar{z}_2 \\ \vdots \\ \Delta \bar{z}_i \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} & \dots & G_{1j} \\ G_{21} & G_{22} & \dots & G_{2j} \\ \vdots & \vdots & & \vdots \\ G_{i1} & G_{i2} & \dots & G_{i\ddot{j}} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_i \end{bmatrix}$$

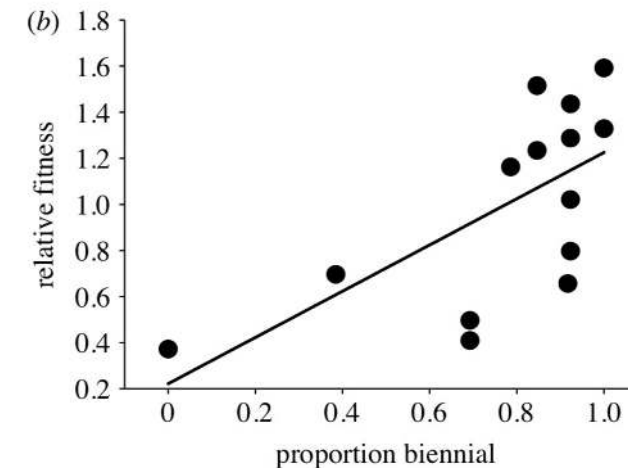
$$R = h^2 S$$

Evolution in plant populations as a driver of ecological changes in arthropod communities

Marc T. J. Johnson^{1,*}, Mark Vellend² and John R. Stinchcombe³



Stabilizing
selection



Directional
selection

Evolution in plant populations as a driver of ecological changes in arthropod communities

Marc T. J. Johnson^{1,*}, Mark Vellend² and John R. Stinchcombe³

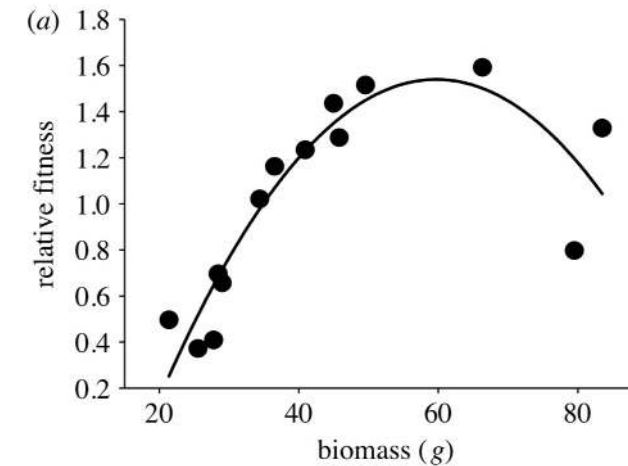
Lande's equation
(multivariate extension of the
breeder's equation)

Evolutionary
change in
the mean
value of trait

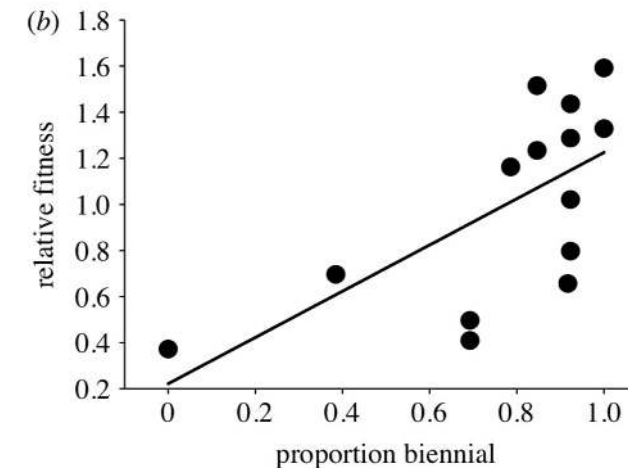
=

Genetic variance co-variance
matrix

Selection
gradient
"strength
of
selection"



Stabilizing
selection



Directional
selection

REVIEWS AND
SYNTHESES

Rapid evolution and the convergence of ecological
and evolutionary time

Nelson G. Hairston Jr,* Stephen P.
Ellner, Monica A. Geber, Takehito
Yoshida and Jennifer A. Fox

Abstract
Recent studies have documented rates of evolution of ecologically important phenotypes sufficiently fast that they have the potential to impact the outcome of ecological interactions while they are underway. Observations of this type go against accepted

Table 1 An attribute of interest X is affected by simultaneous discrete changes in an ecological variable k and an evolutionary variable z (diagonal terms $X_{t,t}$ vs. $X_{t+1,t+1}$)

	z_t	z_{t+1}
k_t	$X_{t,t} = X(z_t, k_t)$	$X_{t+1,t} = X(z_{t+1}, k_t)$
k_{t+1}	$X_{t,t+1} = X(z_t, k_{t+1})$	$X_{t+1,t+1} = X(z_{t+1}, k_{t+1})$

Table 2 Results of applying the decomposition in Table 1 to changes in mean realized fitness of *Onychodaptomus sanguineus* in Bullhead Pond, RI

Time interval	Ecological change rate	Evolutionary change rate
1979–1981	–2.5	–1.1
1981–1982	3.7	1.3
1982–1983	2.7	0.6
1983–1984	0.5	–0.2
1984–1985	–1.4	–0.9
1985–1986	–4.5	1.3
1986–1987	6.3	–0.5
Mean absolute rate	3.1	0.8

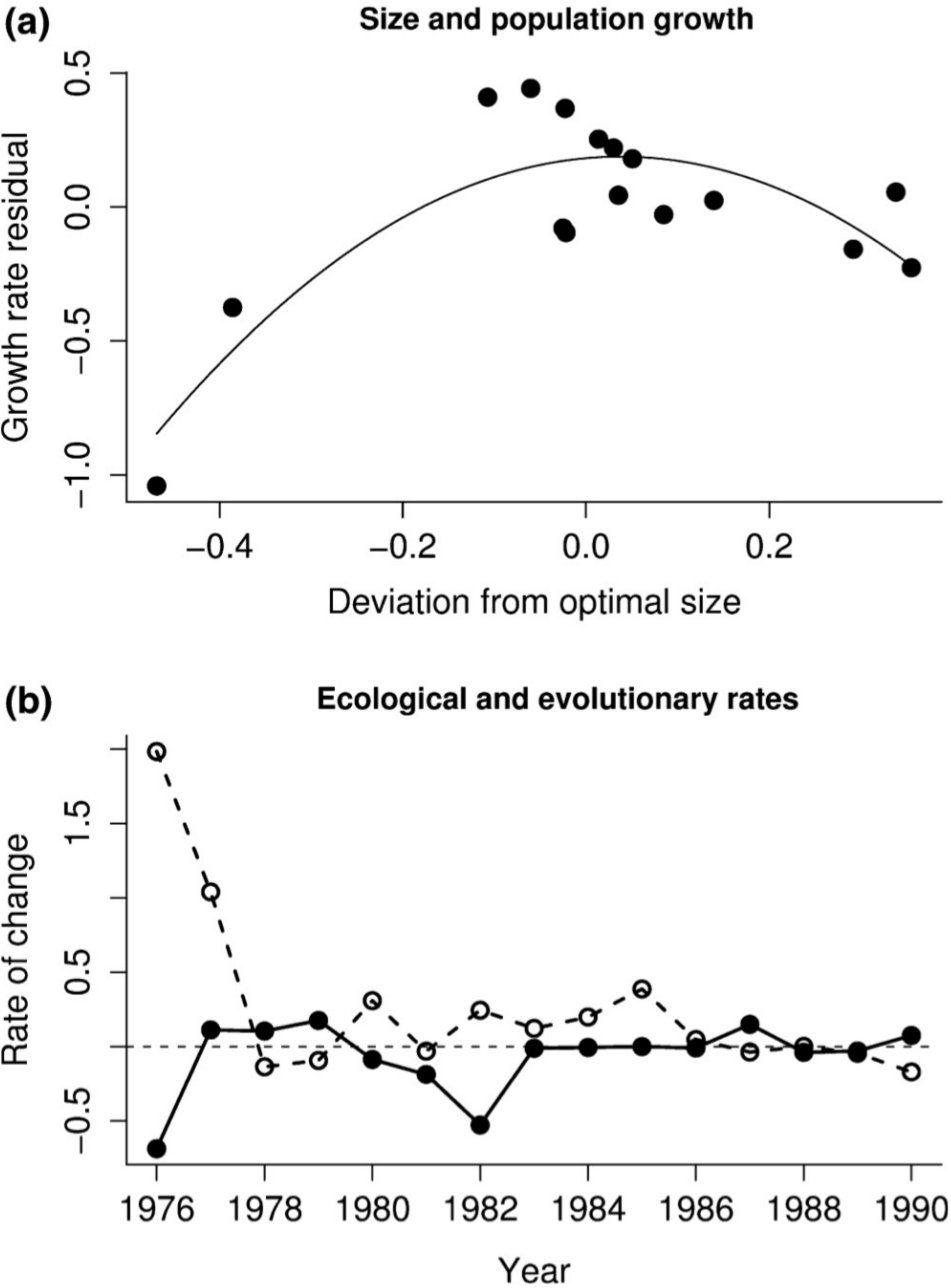


Figure 5 Ecological and evolutionary rates in the medium ground finch *Geospiza fortis* on Daphne Major island. (a) An estimated

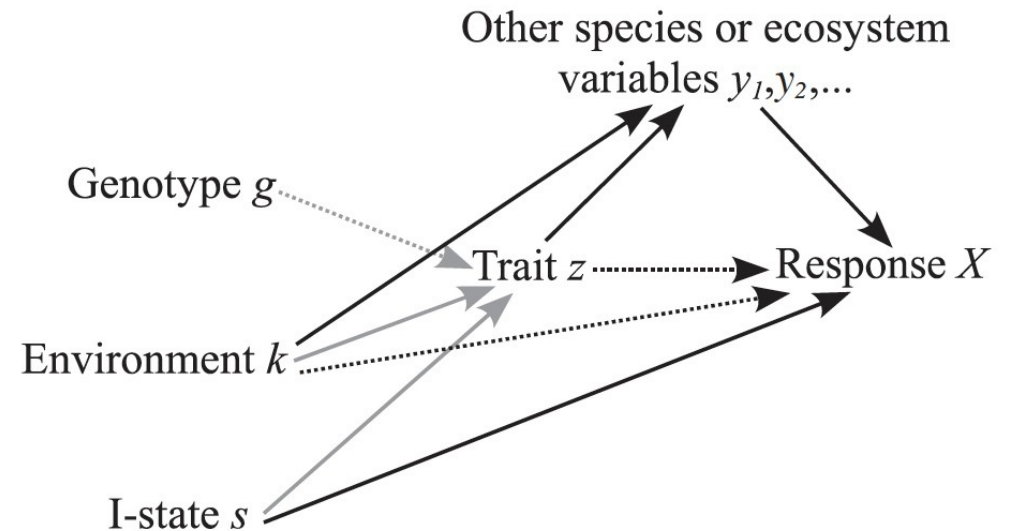
REVIEW AND SYNTHESIS

Does rapid evolution matter? Measuring the rate of contemporary evolution and its impacts on ecological dynamics

Stephen P. Ellner,* Monica A. Geber and Nelson G. Hairston Jr
Department of Ecology and

Abstract

Rapid contemporary evolution due to natural selection is common in the wild, but it remains uncertain whether its effects are an essential component of community and ecosystem structure and function. Previously we showed how to partition change in a population, community or ecosystem property into contributions from



Do we still need CWM etc?



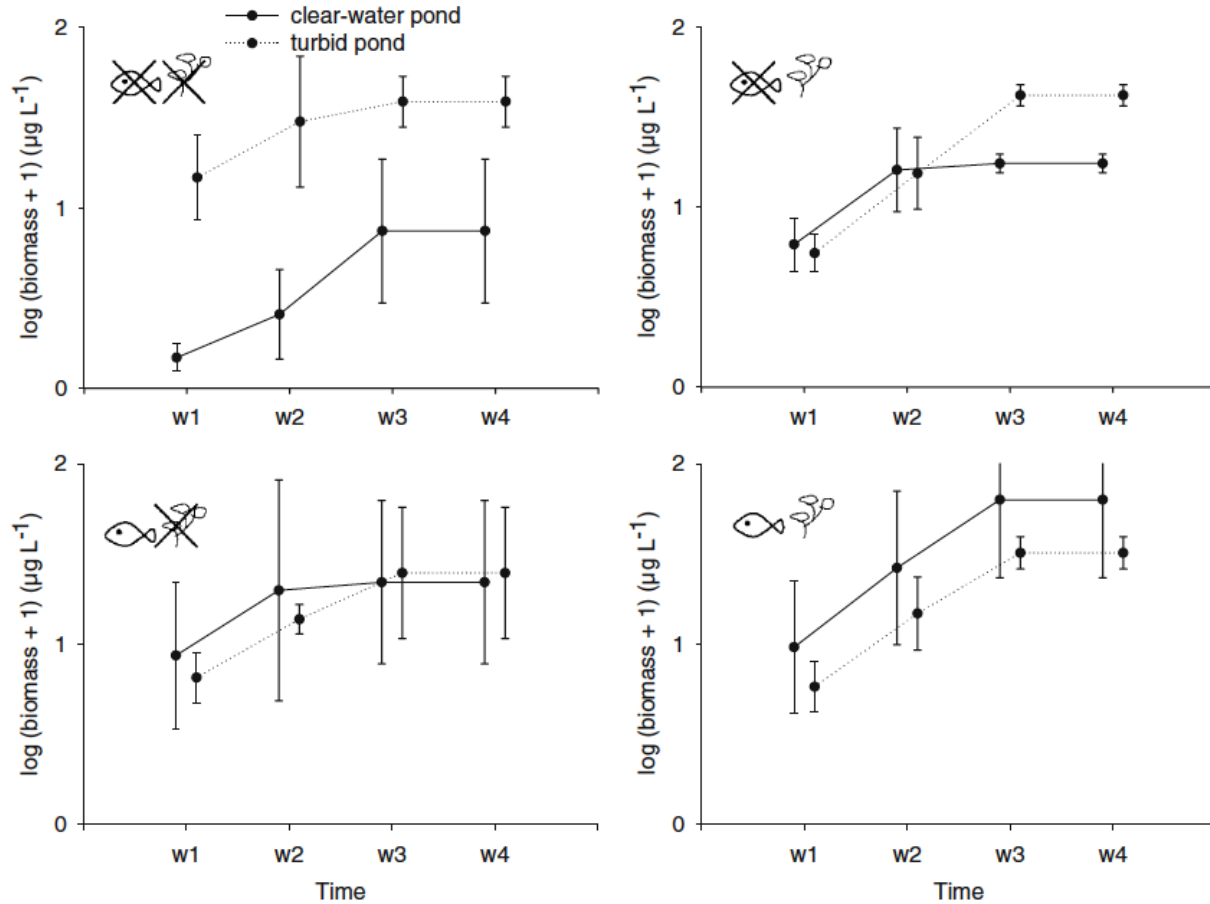
Eco-evolutionary dynamics in metacommunities

Genetic composition of resident populations influences establishment success of immigrant species

Luc De Meester · Gerald Louette · Cathy Duvivier ·
Célien Van Damme · Erik Michels

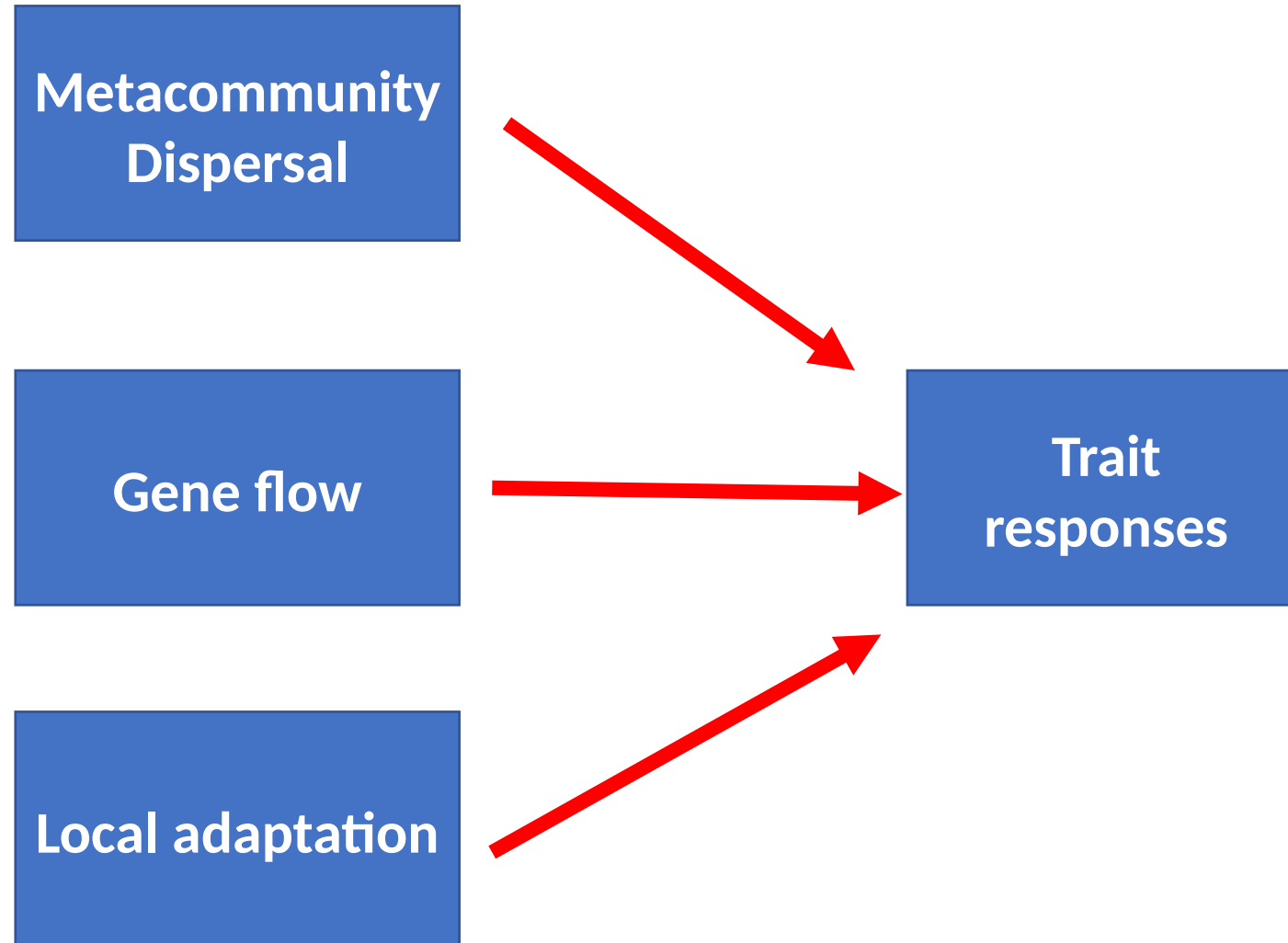
Manipulated the genetic composition of *Daphnia*

Establishment success of immigrants is affected by the source population of the resident *D. magna* and the presence/absence of macrophytes and the presence/absence of fish

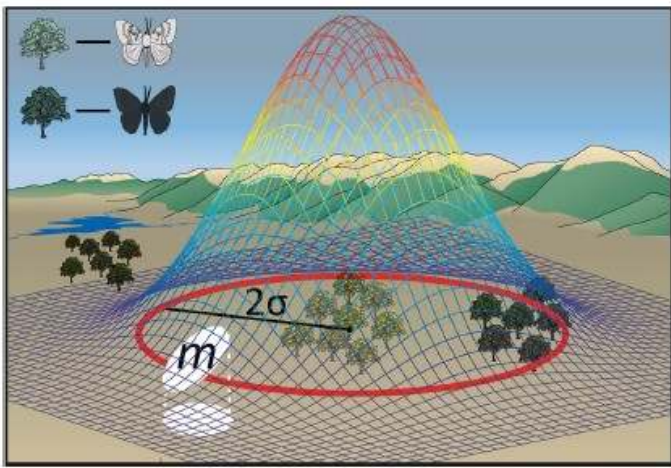


Evolving metacommunity perspective

Evolving Metacommunities



Evolving Metacommunities



TRENDS in Ecology & Evolution

Metacommunity
Dispersal

Gene flow

Local adaptation

Trait
responses

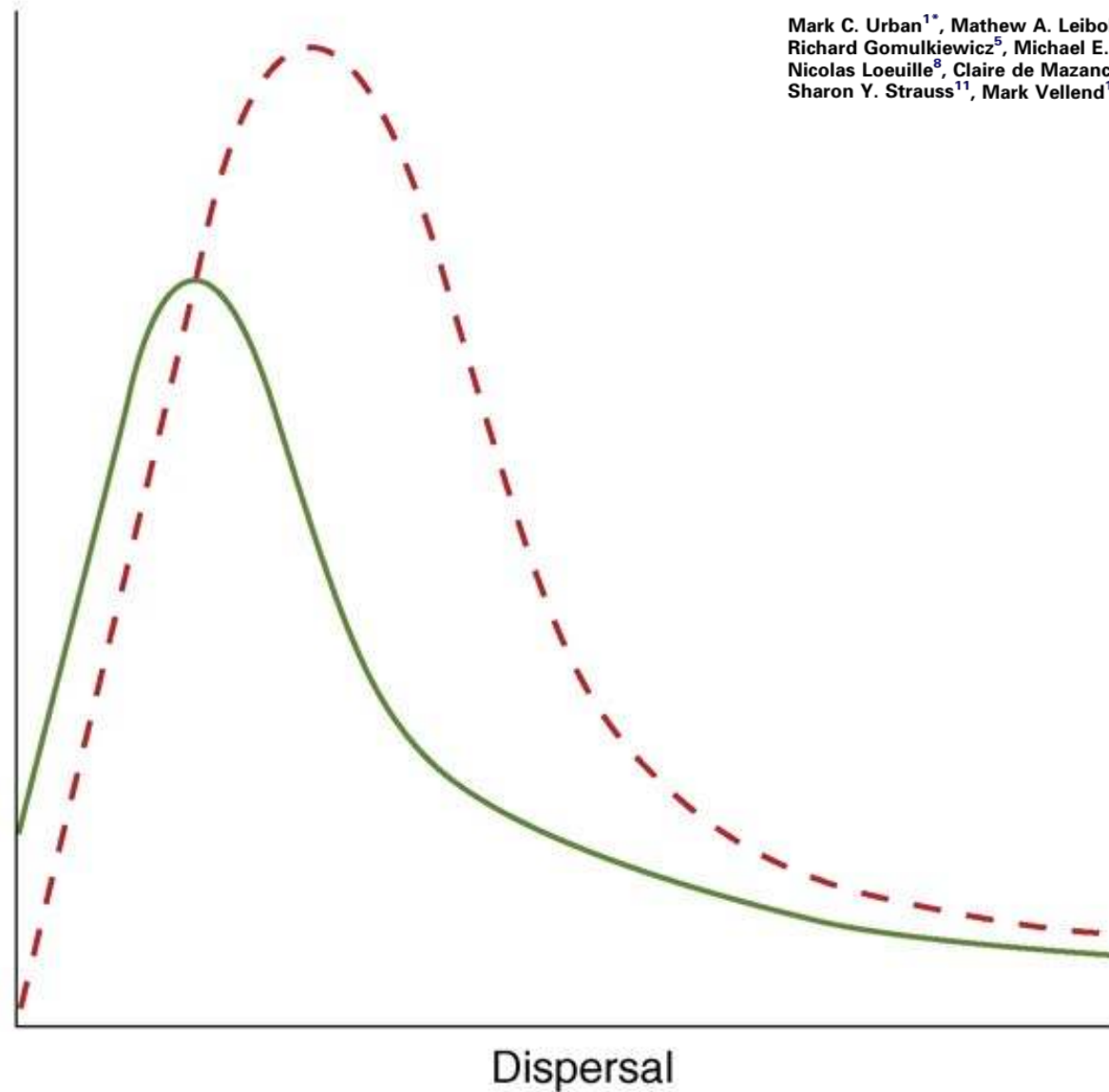
Microgeographic adaptation

Wrights => measure

EVOLVING METACOMMUNITIES: TOWARD AN EVOLUTIONARY PERSPECTIVE ON METACOMMUNITIES

MARK C. URBAN^{1,3} AND DAVID K. SKELLY^{1,2}

Adaptive and community trait
change due to selection

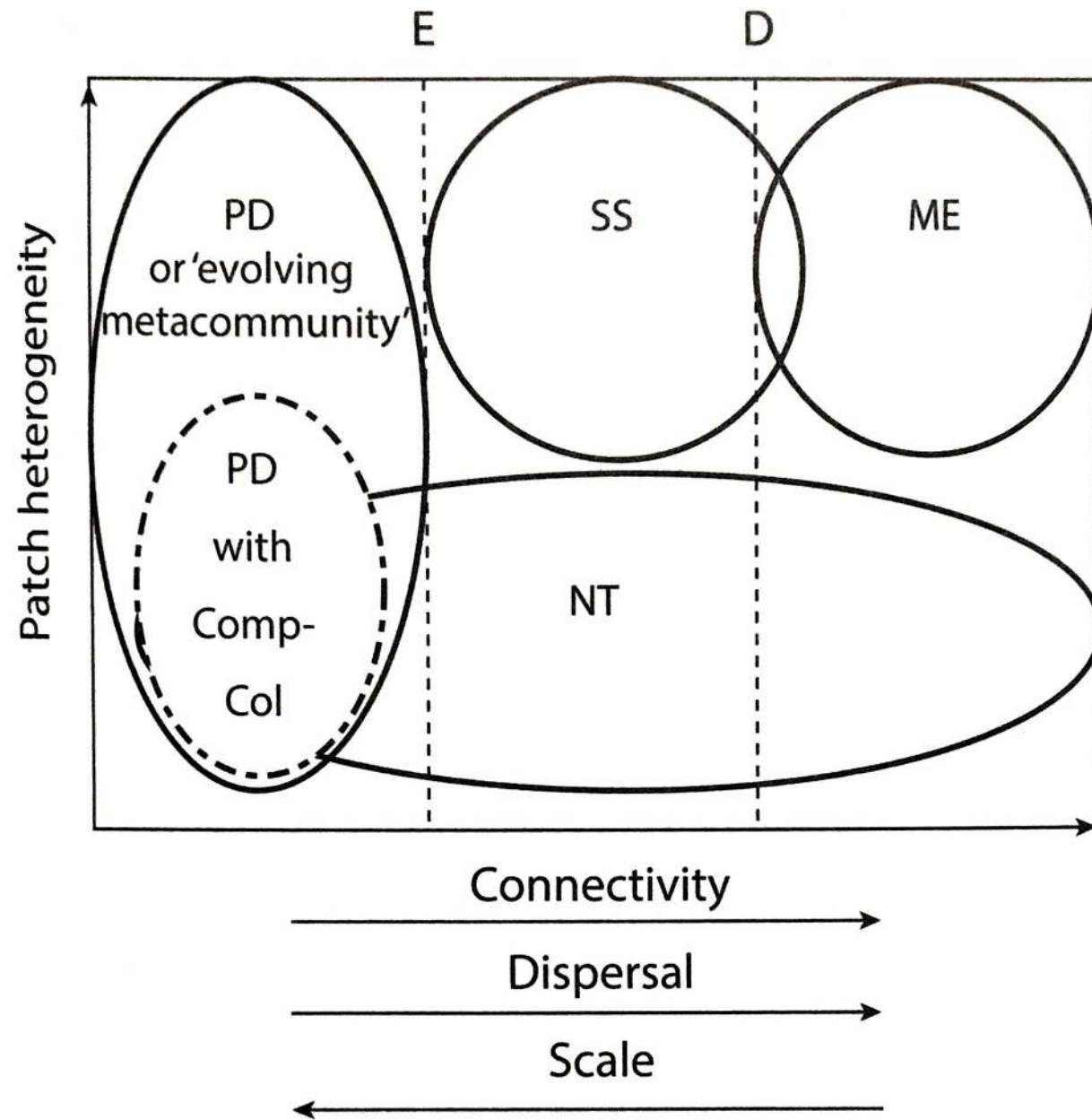


Dispersal

TRENDS in Ecology & Evolution

The evolutionary ecology of metacommunities

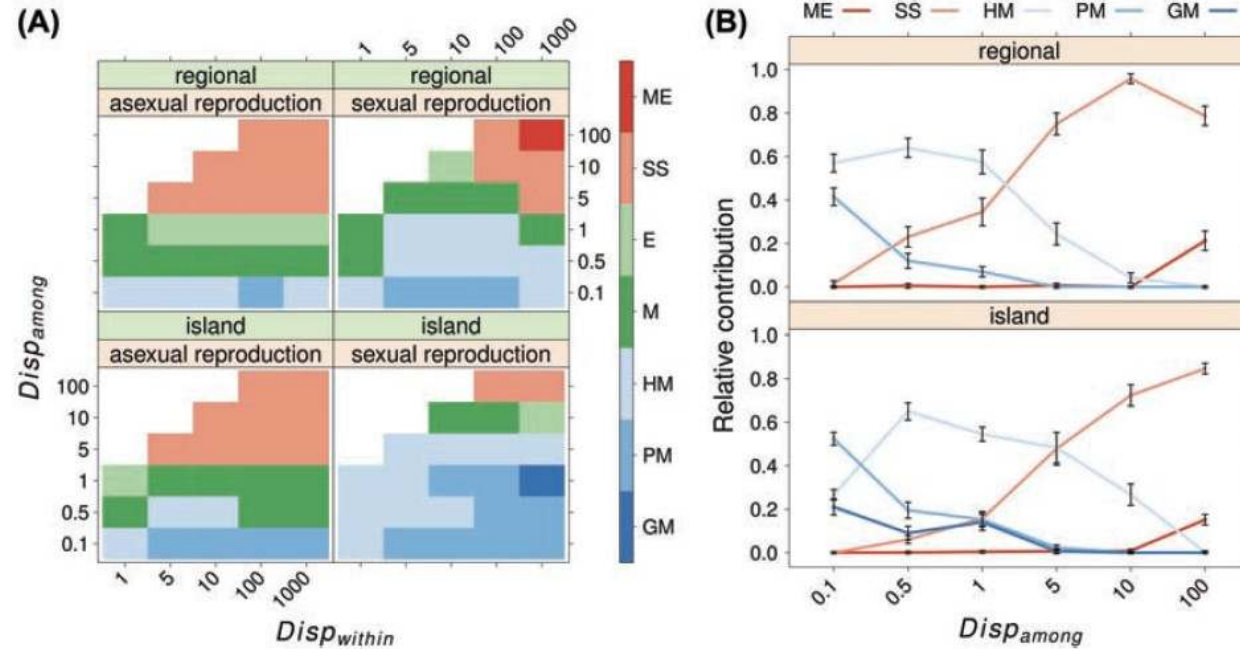
Mark C. Urban^{1*}, Mathew A. Leibold^{2*}, Priyanga Amarasekare³, Luc De Meester⁴, Richard Gomulkiewicz⁵, Michael E. Hochberg⁶, Christopher A. Klausmeier⁷, Nicolas Loeuille⁸, Claire de Mazancourt⁹, Jon Norberg¹⁰, Jelena H. Pantel², Sharon Y. Strauss¹¹, Mark Vellend¹² and Michael J. Wade¹³





Community assembly is a race between immigration and adaptation: eco-evolutionary interactions across spatial scales

Joost Vanoverbeke, Mark C. Urban and Luc De Meester



- The same set of ecological and evolutionary processes can determine community assembly at both regional and biogeographical scales.
- Metacommunity size and the proportion of empty spaces are also important
- Evolution often determines community assembly

Regional neutrality evolves through local adaptive niche evolution

Mathew A. Leibold^{a,1}, Mark C. Urban^b, Luc De Meester^c, Christopher A. Klausmeier^{d,e,f}, and Joost Vanoverbeke^c

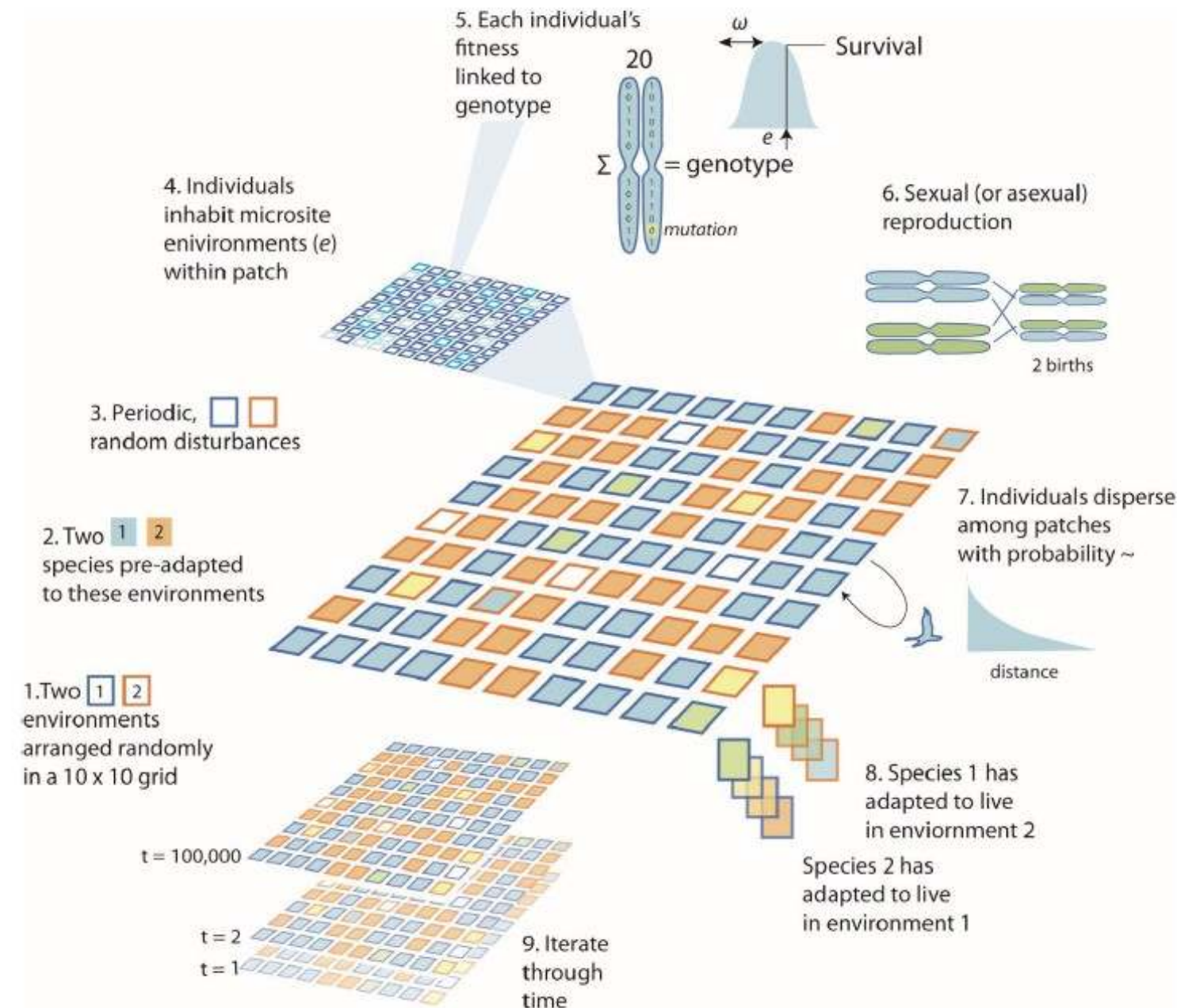
^aDepartment of Biology, University of Florida, Gainesville, FL 32611; ^bDepartment of Ecology and Evolutionary Biology, University of Connecticut, Storrs, CT 06269; ^cDepartment of Biology, Katholieke Universiteit Leuven, B-3000 Leuven, Belgium; ^dKellogg Biological Station, Michigan State University, Hickory Corners, MI 49060; ^eDepartment of Plant Biology, Michigan State University, Hickory Corners, MI 49060; and ^fProgram in Ecology, Evolutionary Biology & Behavior, Michigan State University, Hickory Corners, MI 49060

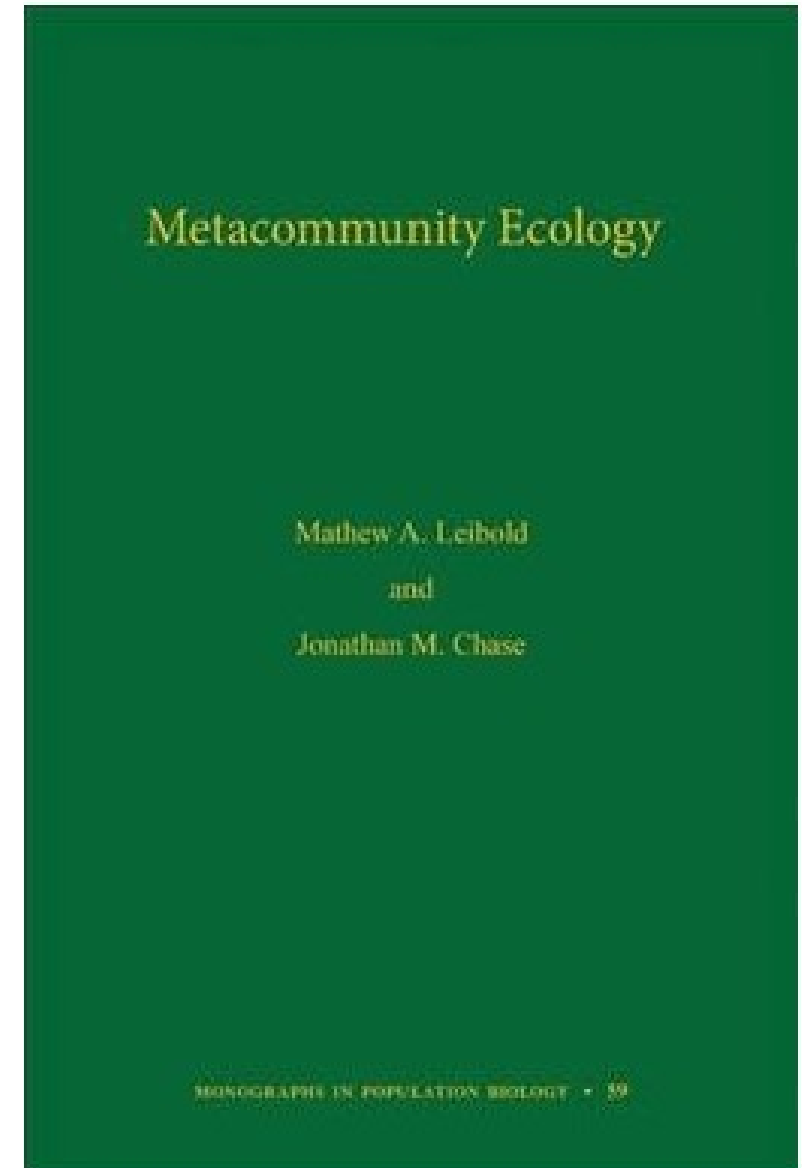
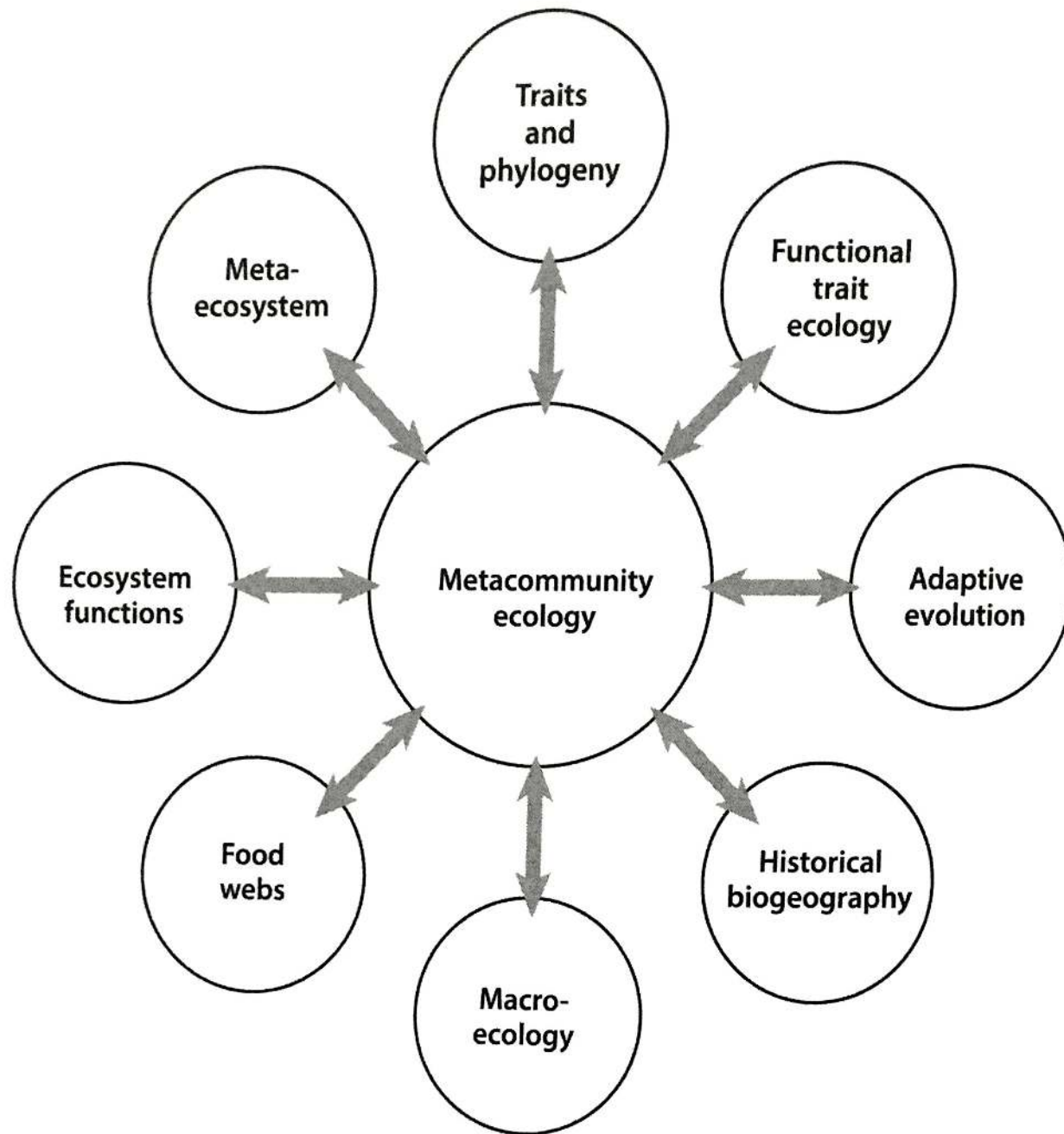
Leibold et al. 2019 PNAS

Incorporates
adaptation to
Chesson's MCT

Priority effects
and 'community
monopolization
hypothesis'

Neutrality emerge at regional
scale from local, niche-based
processes driven by local
adaptation





Is community ecology still a mess?

I can imagine a student reaching the end of this book and saying, “That’s all very interesting, really. I see how species diversity varies across space and time—fascinating stuff. And I understand how biodiversity can affect the functioning of communities and ecosystems and why the loss of biodiversity is such a critical concern. And I have learned a great deal about the nitty-gritty of species interactions and how these interactions impact species diversity. And I understand that species diversity in a community is a function of local interactions plus the movement of species between communities. But what I *don’t* get is how everything fits together. That’s what we are supposed to understand, isn’t it, if we want to use our ecological knowledge to help preserve the diversity of life in our world? So, just tell me—which of all the things that we studied is most important in determining the number and kinds of species I see around me?”

Do we still suffer from “physics envy”?

ecology is not just like doing them in other fields. Community ecology's uniqueness lies in the extraordinary diversity, variation, and heterogeneity of its objects of study, and in their historical legacy accumulated through evolution. Individual organisms, populations, species, and communities all are many orders of magnitude more variable internally than are the objects studied in physics, chemistry, or molecular biology, and the art in our science must reflect the difficulty of dealing with this variability. Differences among individuals such as age, sex, and nutritional condition are extremely important to how populations function, and likewise up through communities and ecosystems. Slight differences among species in trophic habit, ecological amplitude, and life history greatly affect the likelihood, intensity, and persistence of interactions among species. Most challenging, the stochasticity of the external environmental further varies population and community behavior. The same species in different en-

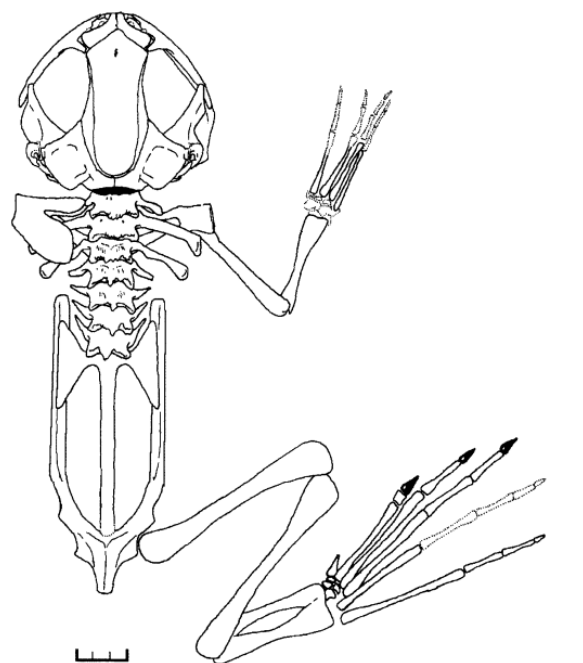
A cheering message

In Closing, I'd Like to Say...

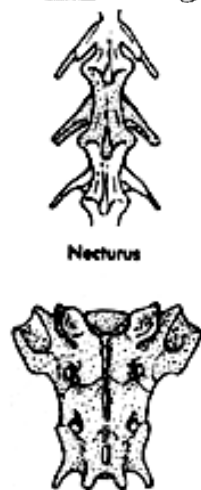
... it's an exciting time to be a community ecologist. We have the power to illuminate our understanding of the natural world as never before, to help conserve Earth's biodiversity, and to help provide the knowledge needed to develop a more sustainable world. I look forward to seeing what ecologists can accomplish in the coming decade, and I hope this book might contribute in some small way toward facilitating those accomplishments, perhaps by inspiring more young scientists to take up the challenge.



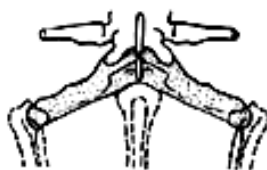
EvolCITY: How the urban environment changes the pace evolution of life



Frog's morphology is adapted for jumping, and relatively conserved throughout evolution



Necturus



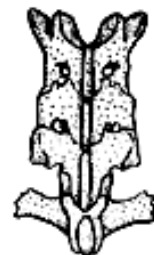
Frog



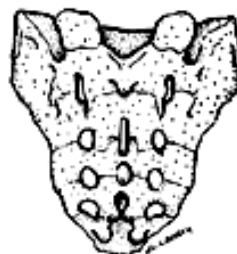
Alligator



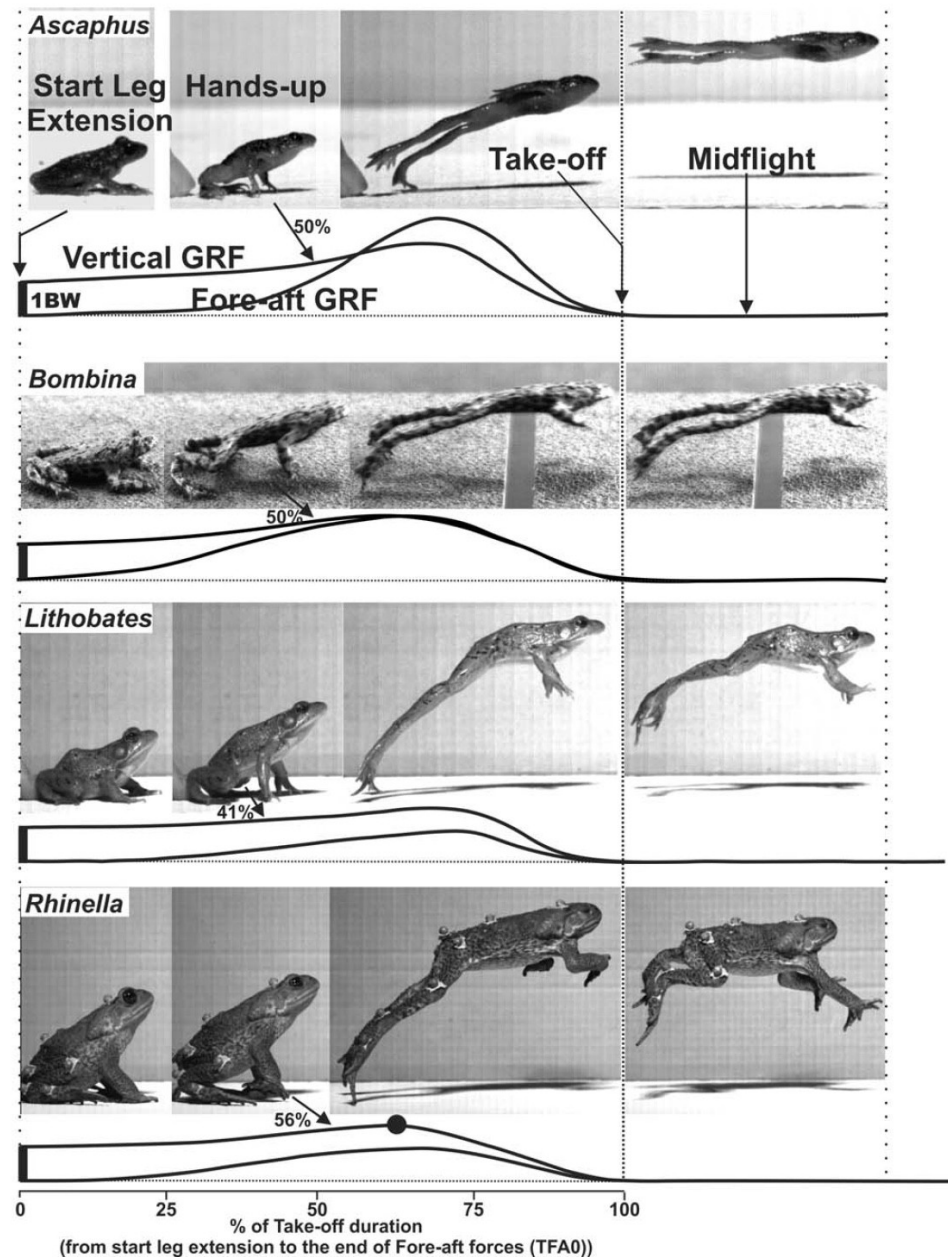
Cat



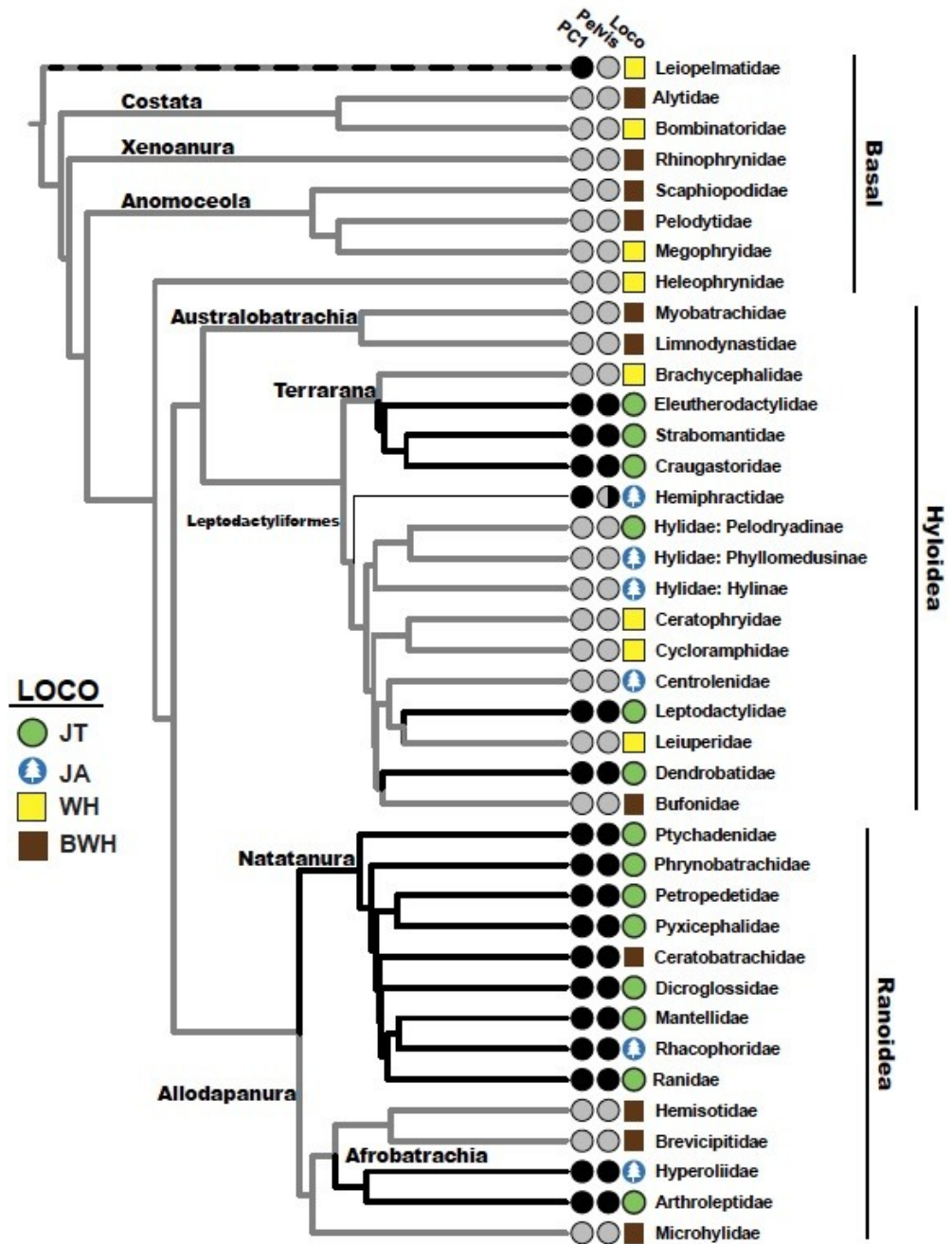
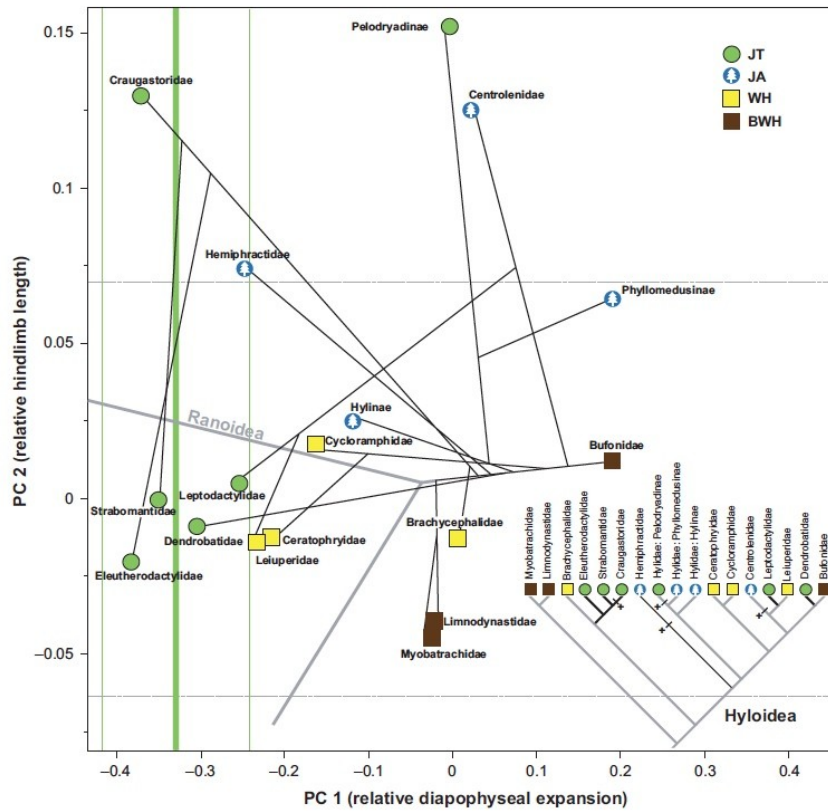
Hamster



Man



Pelvis morphology varies among groups depending on locomotion mode



- Neotropical frogs can be divided into 3 guilds:
 - 1) Generalists, eating preferably large prey
 - 2) Ant/termite specialist
 - 3) “avoids” ants/termites
- Most frogs are generalists, eating arthropods and other invertebrates

Head width and mandible length determine the size and type of prey (in specialist species)

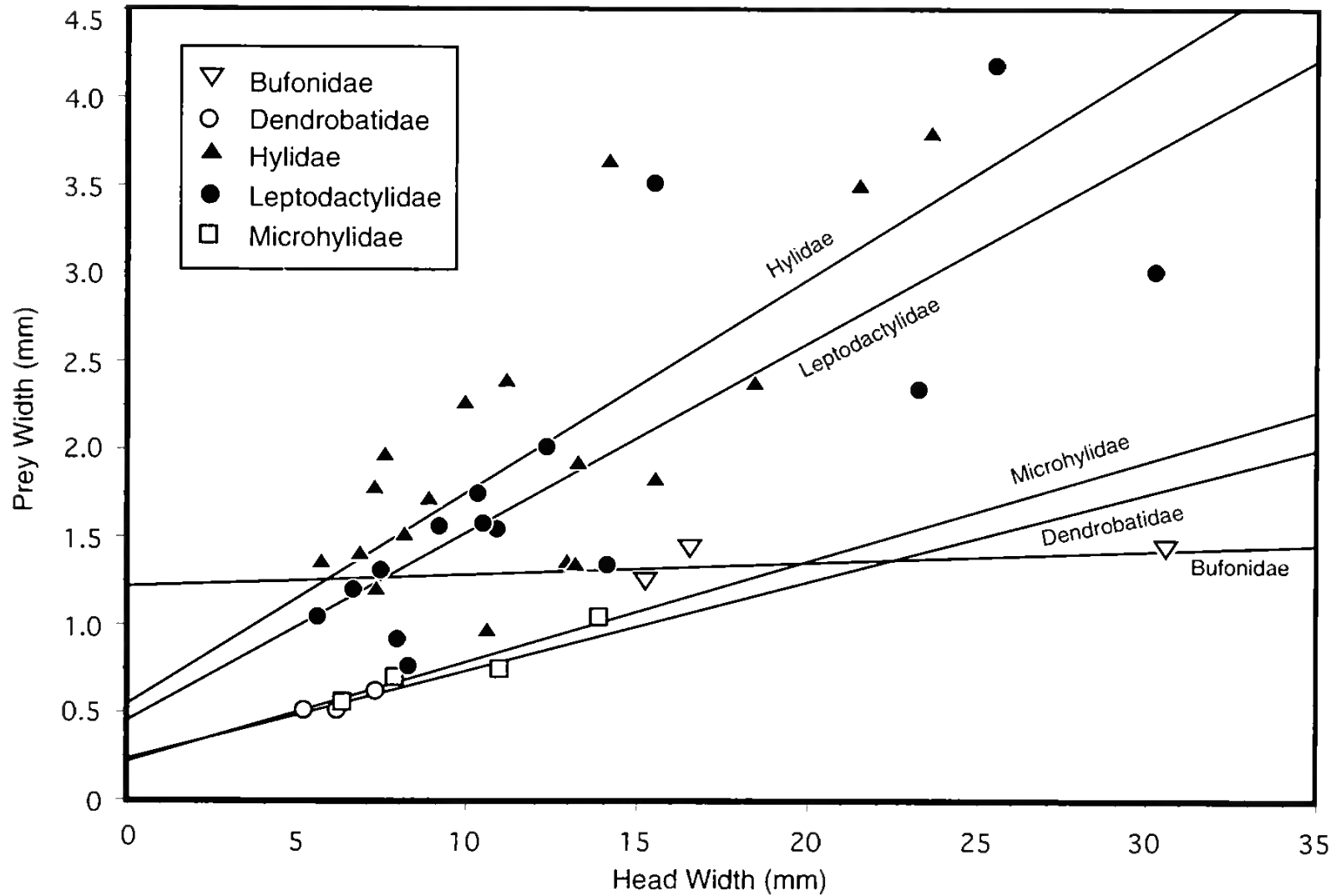
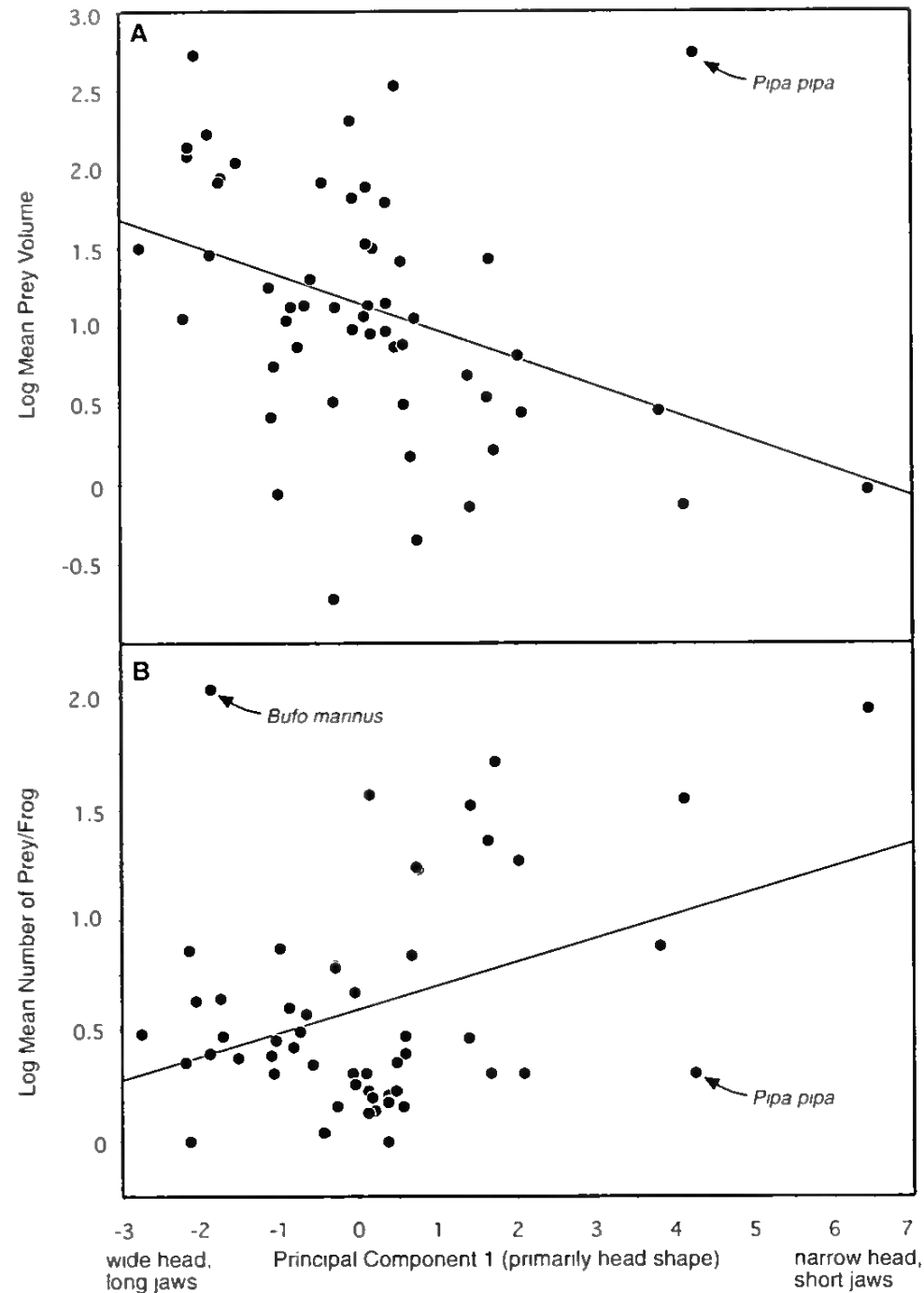
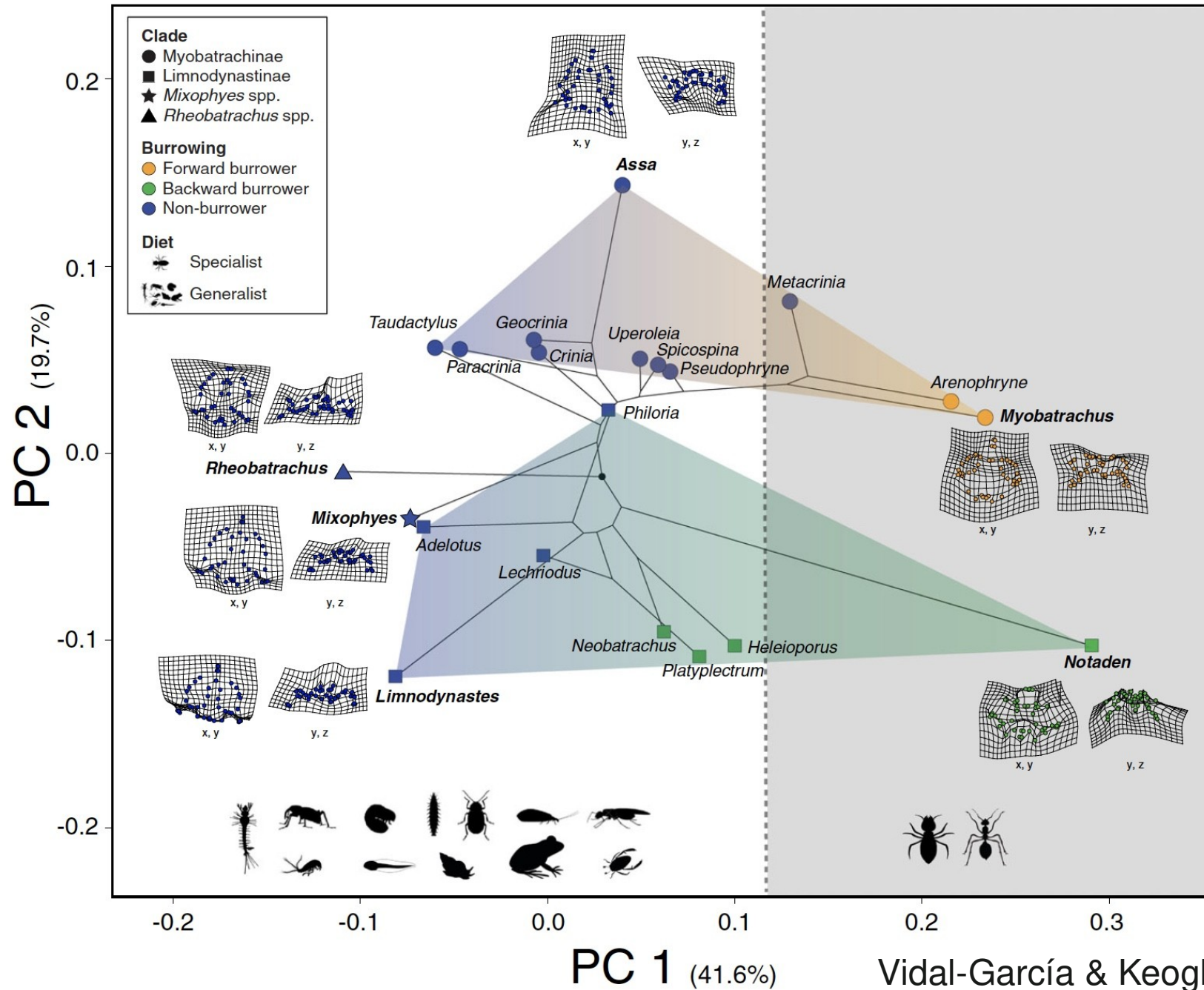


Fig. 12. Species means of prey width versus head width for five families of anurans.

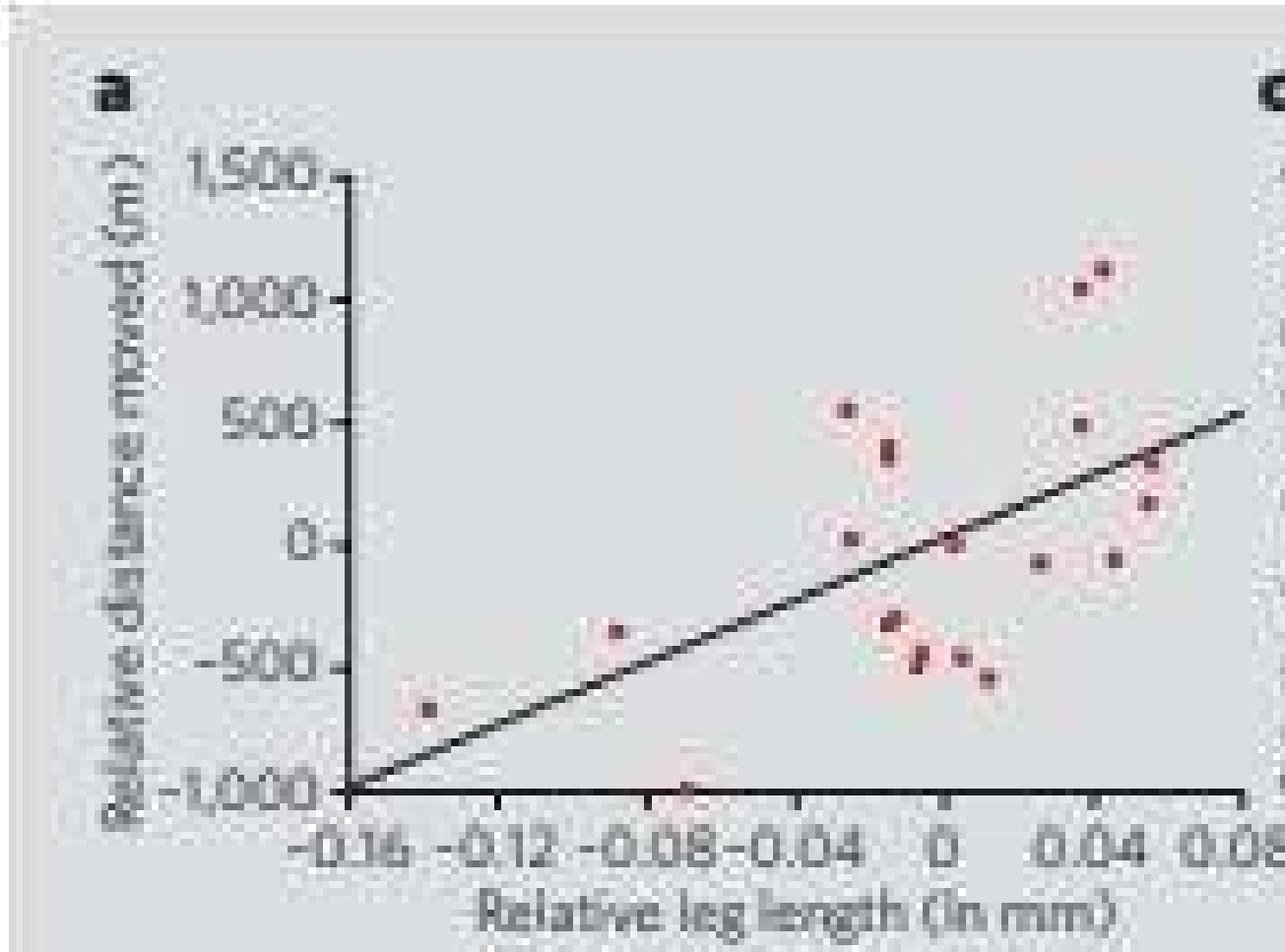


- Frogs that eat relatively **small, slow prey** have relatively **short jaws** and a **symmetrical feeding cycle** where equal amounts of time are spent catching and bringing prey into the mouth;
- Frogs that eat relatively **large, slow prey** have relatively **long jaws** and an **asymmetrical feeding cycle** where time to prey capture is less than the time to bring prey into the mouth; Emerson 1985

Myobatrachidae (Australo-Papuan)

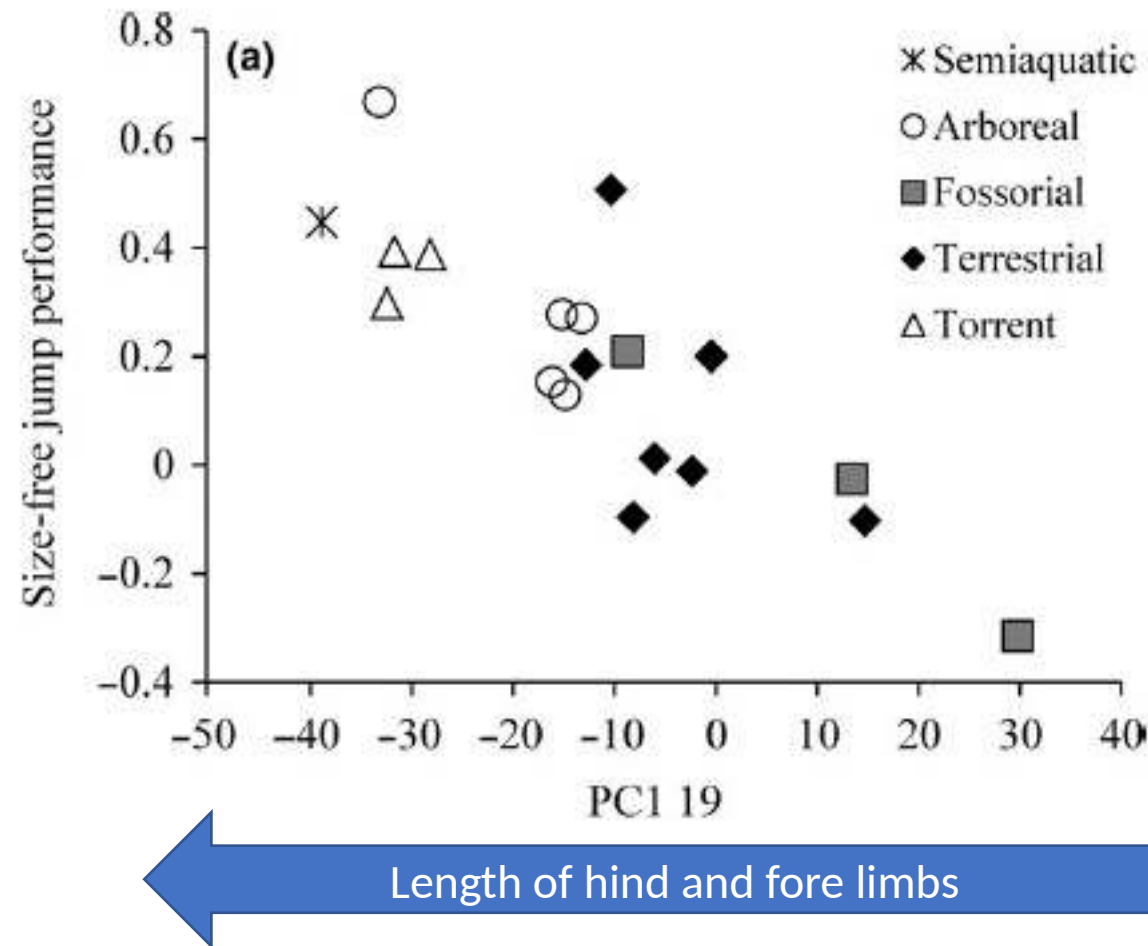


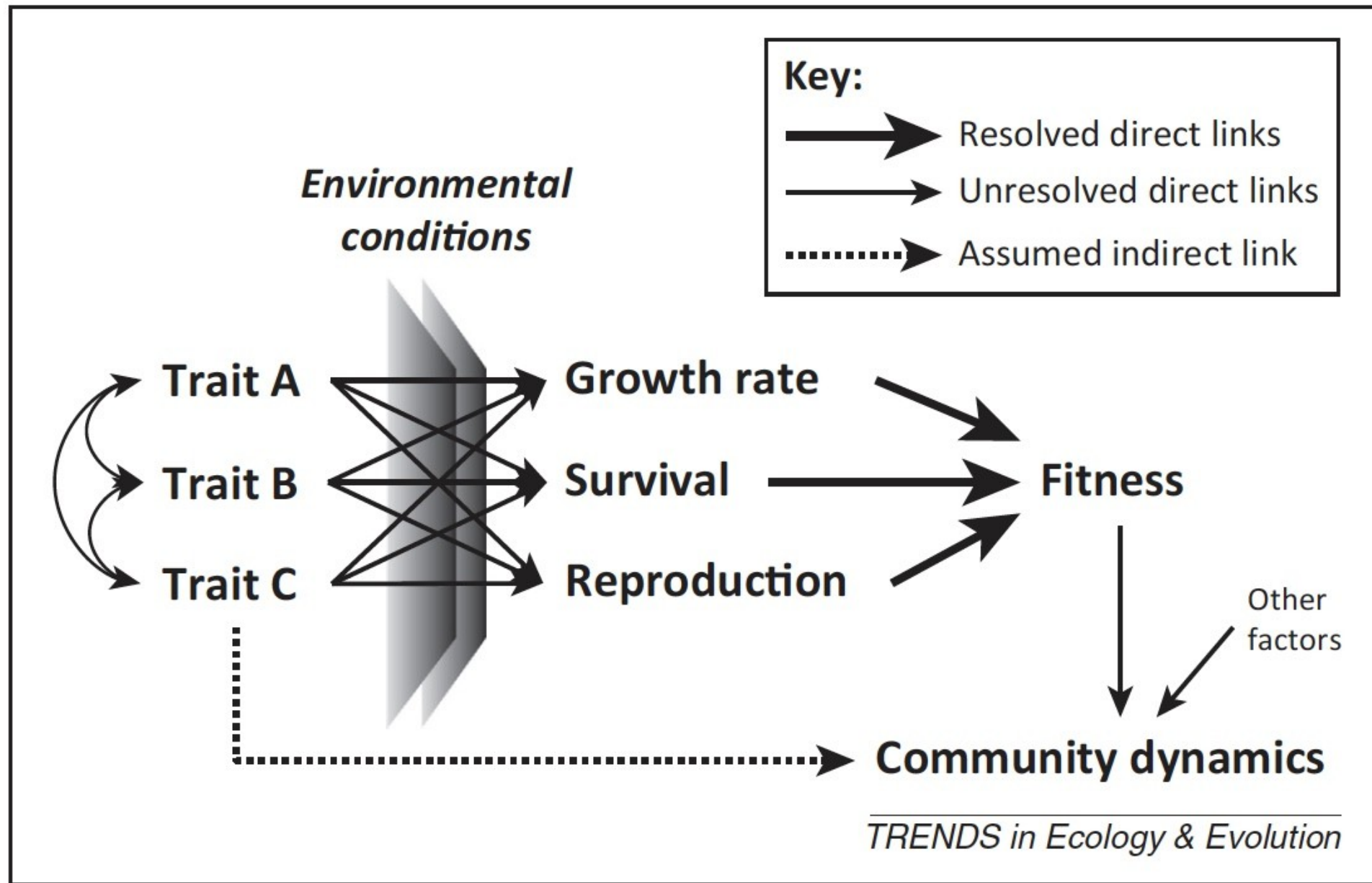
Relative leg length is positively related with dispersal distance

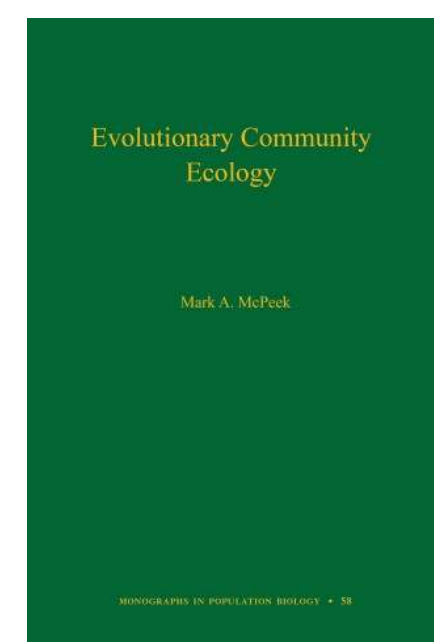


Rhinella marina (Australia)

Length of fore and hind limbs are positively related with jumping performance







Cap. 3 McPeck (2017)

Effect of a trait on fitness depends on the mechanism of interaction between individuals and populations

Bidirectional

Unidirectional

Reversion in the change direction in the fitness component with the trait change at different ranges of the trait

If trait change in one direction increases fitness in all range of the trait (e.g., escape speed)

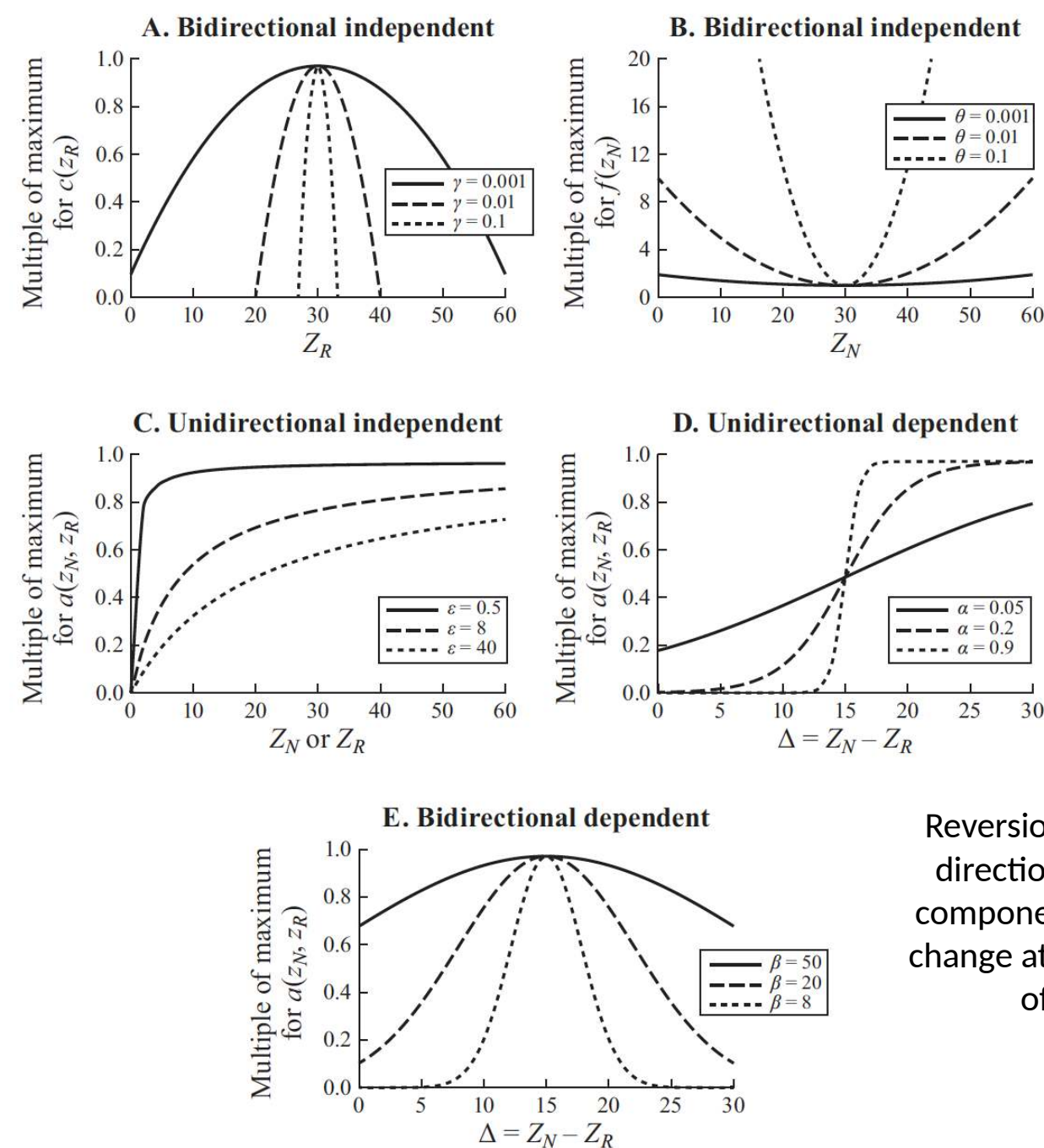
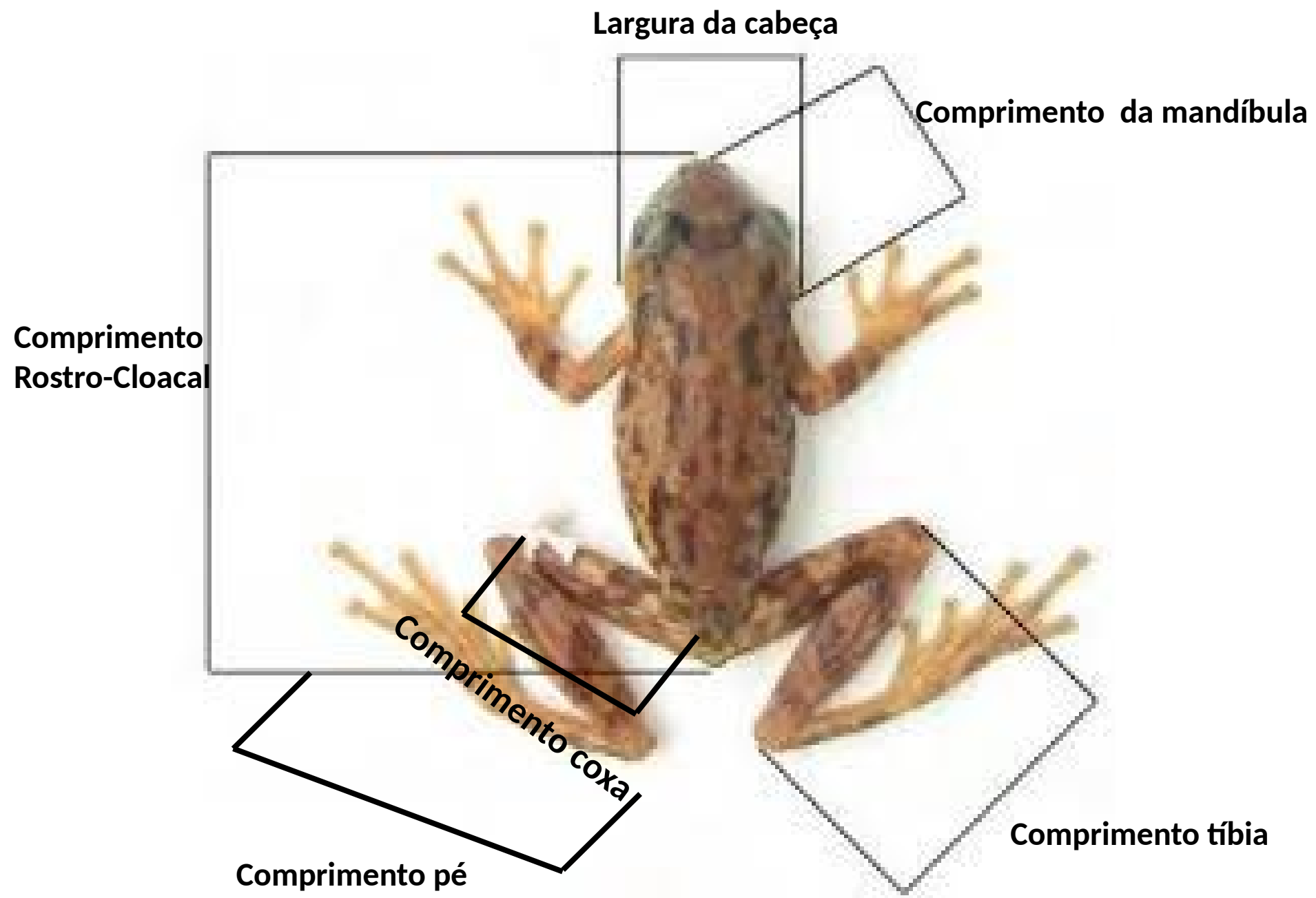


FIGURE 3.1. Functional forms used for the relationships between species trait values and param-

Main questions

- How does urbanization and species interaction affect phenotypic traits related to dispersal, feeding, and predator avoidance in frogs?
- Can time-since-urbanization change the fitness surface of key phenotypic traits?



Phenotypic traits

- Relative head width and mandible length: prey size ingested
- Leg length: jumping performance (locomotion)
- Body size: can represent competitive hierarchy, for resources and females (~ call frequency)

Capture-Mark-Recapture as a tool to infer demographic parameters in open populations

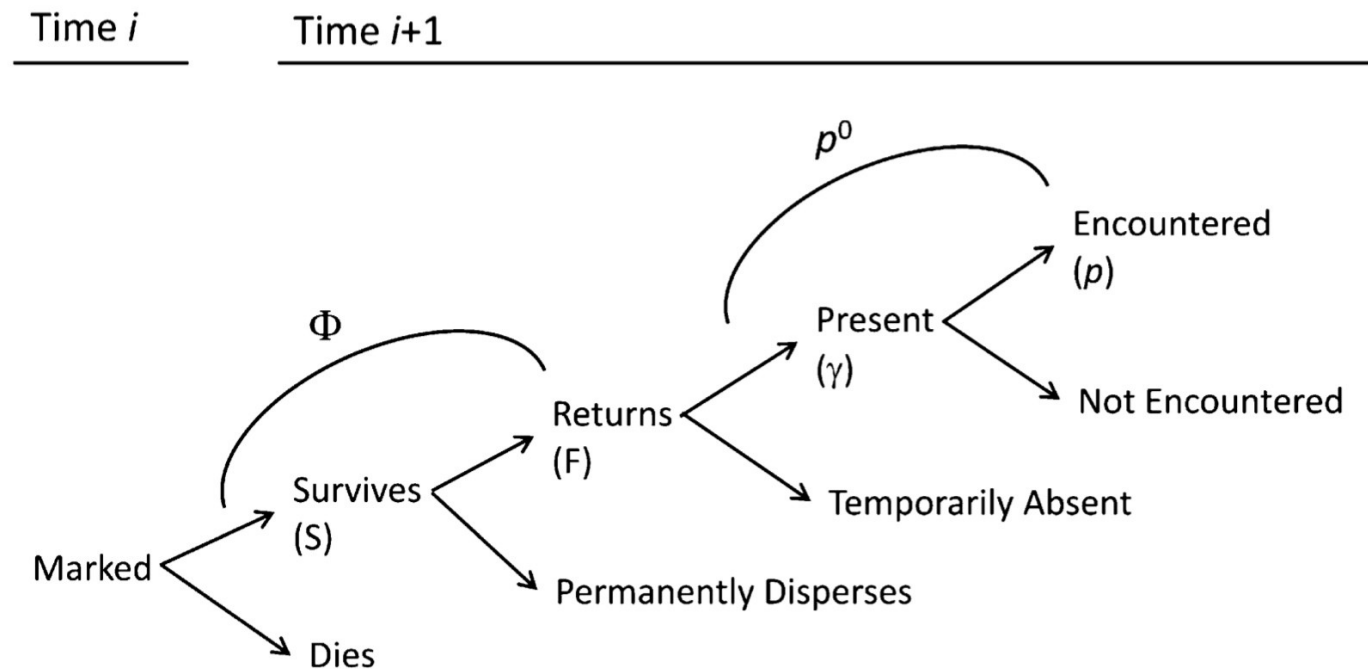




Foto: Leandro Drummond

Estimating dispersal in spatiotemporally variable environments using multievent capture–recapture modeling

HUGO CAYUELA,^{1,2,4} ROGER PRADEL,² PIERRE JOLY,¹ ERIC BONNAIRE,³ AND AURÉLIEN BESNARD²

True/Apparent survival (φ)

Dispersal between patches

ESTIMATING AND VISUALIZING FITNESS SURFACES USING MARK–RECAPTURE DATA

Olivier Gimenez,^{1,2} Arnaud Grégoire,^{1,3} and Thomas Lenormand^{1,4}

Oikos 000: 001–015, 2012

111/j.1600-0706.2012.20532.x

© 2012 THE AUTHORS. OIKOS © 2012 Nordic Society Oikos

Subject Editor: Matthew Symonds. Accepted 7 June 2012

Looking for a needle in a haystack: inference about individual fitness components in a heterogeneous population

Emmanuelle Cam, Olivier Gimenez, Russell Alpizar-Jara, Lise M. Aubry, Matthieu Authier, Evan G. Cooch, David N. Koons, William A. Link, Jean-Yves Monnat, James D. Nichols, Jay J. Rotella, Jeffrey A. Royle and Roger Pradel

Popul Ecol (2018) 60:101–109
DOI 10.1007/s10144-017-0598-x



SPECIAL FEATURE: ORIGINAL ARTICLE

Evolutionary demography: the dynamic and broad intersection of ecology and evolution

Estimating individual fitness in the wild using capture–recapture data

Olivier Gimenez¹ · Jean-Michel Gaillard²

The paradigm of body condition: a critical reappraisal of current methods based on mass and length

Jordi Peig^{1,*} and Andy J. Green²

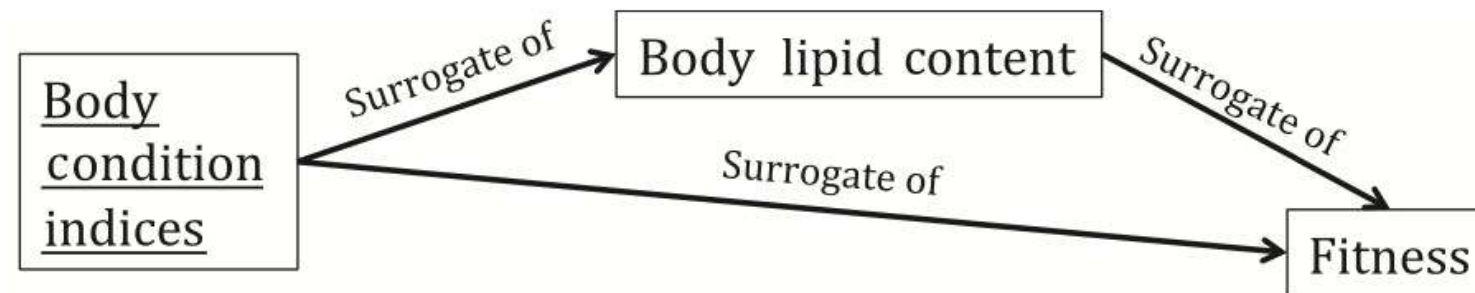


Fig. 1. Body condition indices are traditionally used either as a direct estimate of fitness or as an estimate of body lipid content, which is assumed to be related to fitness.

Critiques

Functional Ecology



Functional Ecology 2016, **30**, 108–115

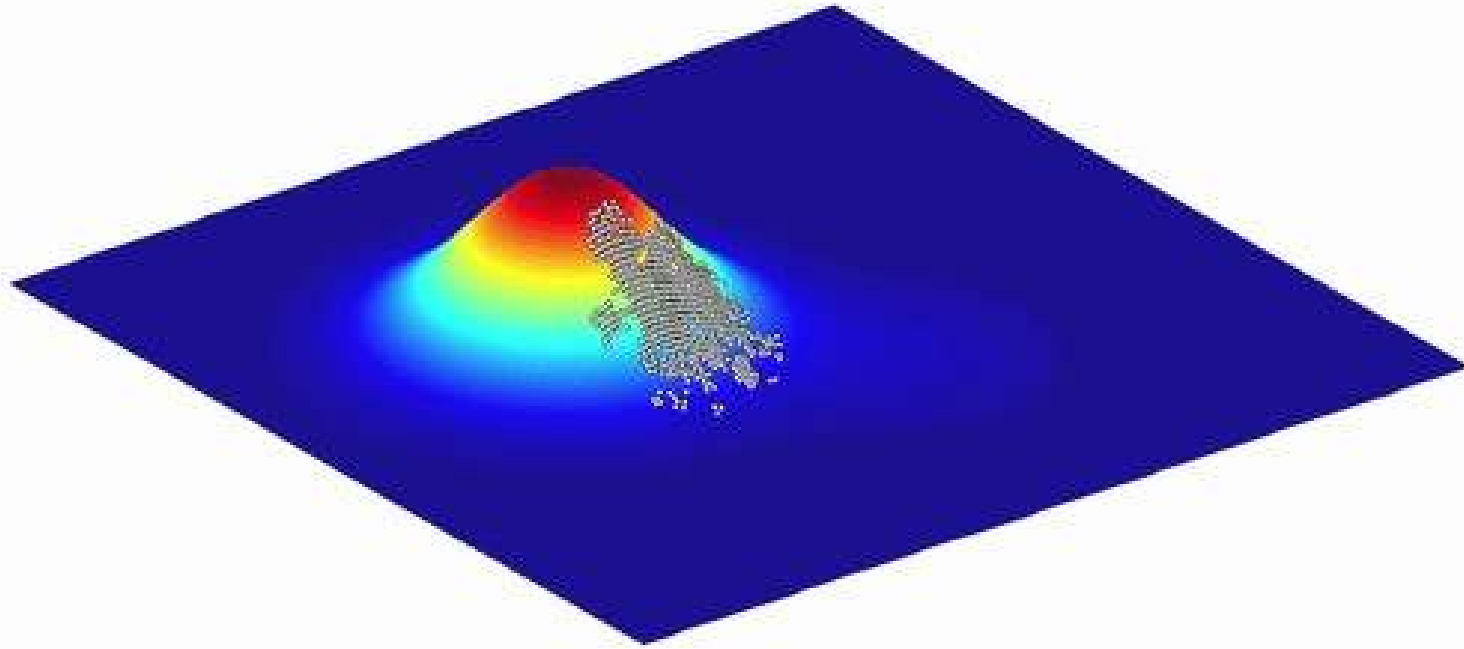
doi: 10.1111/1365-2435.12460

PERSPECTIVE

Moving beyond body condition indices as an estimate of fitness in ecological and evolutionary studies

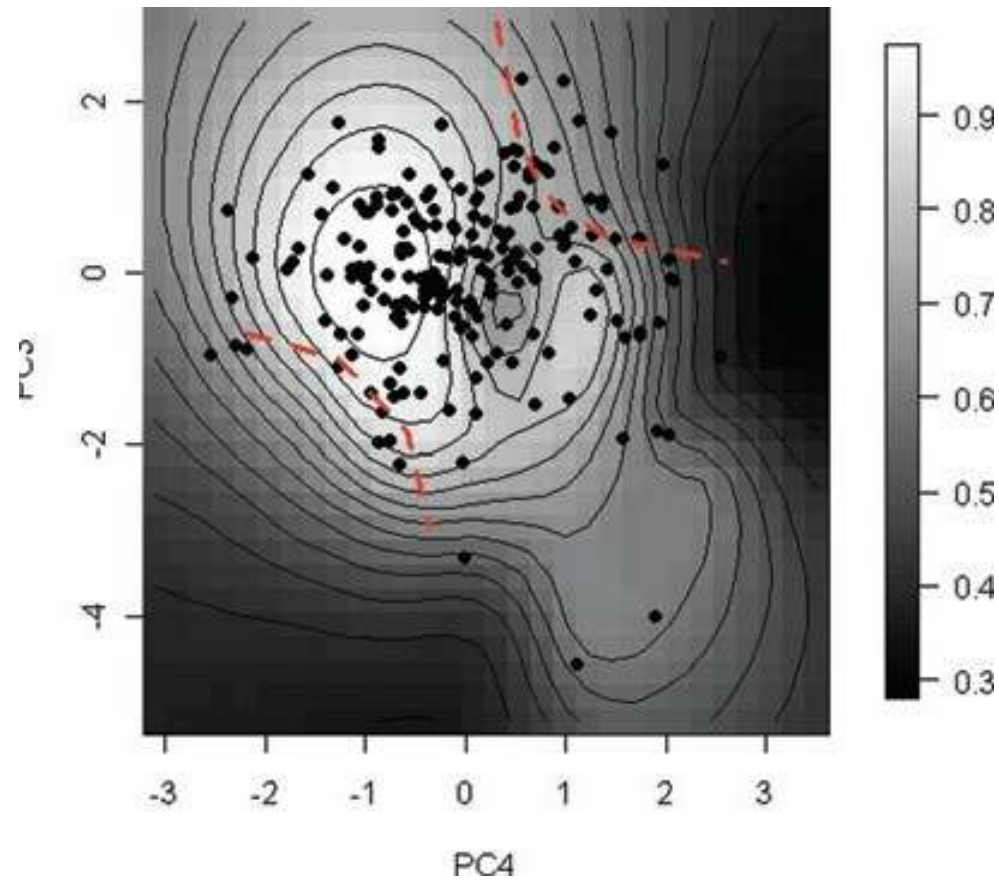
Shawn M. Wilder^{*,1,2}, David Raubenheimer^{1,3} and Stephen J. Simpson¹

Dynamic fitness landscape

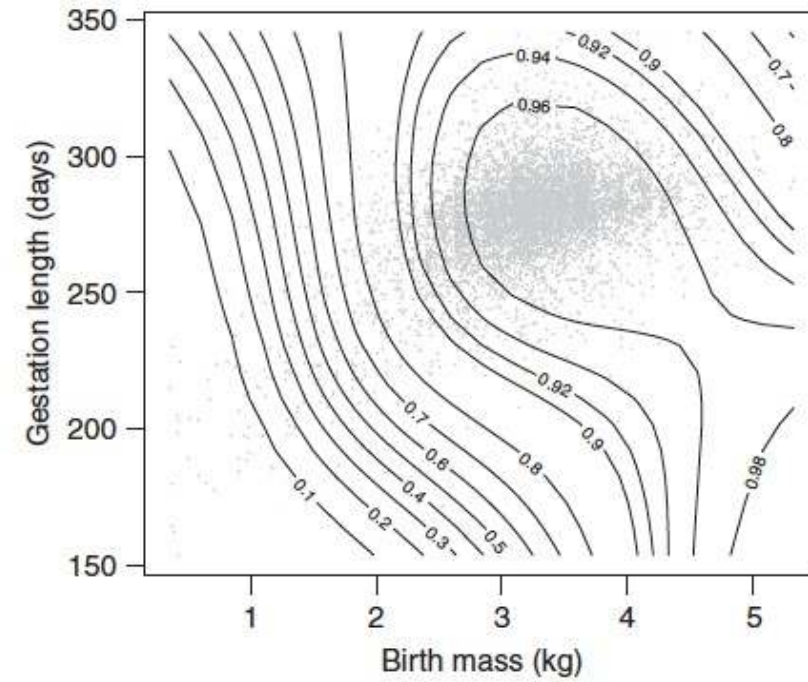


Population size, $N = 2,304$
Mutation rate, $\mu = 0.5$ per trait

© Randy Olson and Bjørn Østman

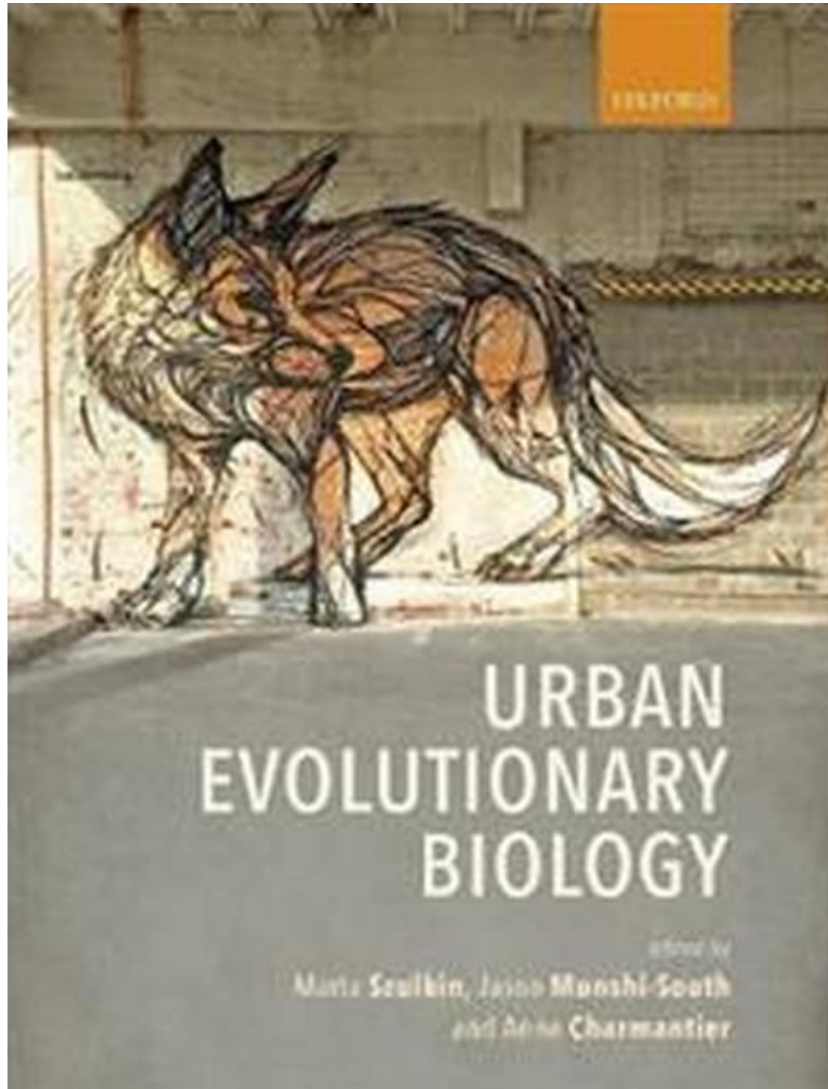


Gimenez et al. 2009



Morrissey & Sakrejda 2013

Hot topic



June 2020



2019

PERSPECTIVE

WILEY

Evolutionary Applications

Open Access

A roadmap for urban evolutionary ecology

L. Ruth Rivkin^{1,2,3} 

James S. Santangelo^{1,2,3} 

Marina Alberti⁴

Myla F. J. Aronson⁵

Charlotte W. de Keyzer^{1,3}

Sarah E. Diamond⁶

Marie-Josée Fortin^{1,3}

Lauren J. Frazee⁵

Amanda J. Gorton⁷

Andrew P. Hendry⁸

Yang Liu⁹ 

Jonathan B. Losos^{10,11}

J. Scott MacIvor^{1,3,12} 

Ryan A. Martin⁶

Mark J. McDonnell¹³

Lindsay S. Miles¹⁴

Jason Munshi-South¹⁵

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Mason R. Stothart¹⁶

Panagiotis Theodorou¹⁷

Ken A. Thompson¹⁸ 

Brian C. Verrelli^{19,20}

Andrew Whitehead²¹ 

Kristin M. Winchell^{11,22}

Marc T. J. Johnson^{2,3} 

The Complexity of Urban Eco-evolutionary Dynamics

MARINA ALBERTI, ERIC P. PALKOVACS, SIMONE DES ROCHES, LUC DE MEESTER,
LYNN GOVAERT, NANCY B. GRIMM, NYEEMA C. HARRIS, ANDREW P. HENDRY, CHRIS
MARTA SZULKIN, JASON MUNSHI-SOUTH, MARK C. URBAN, AND BRIAN C. VERRELLI

2020

Overview

Accepted: 29 June 2020

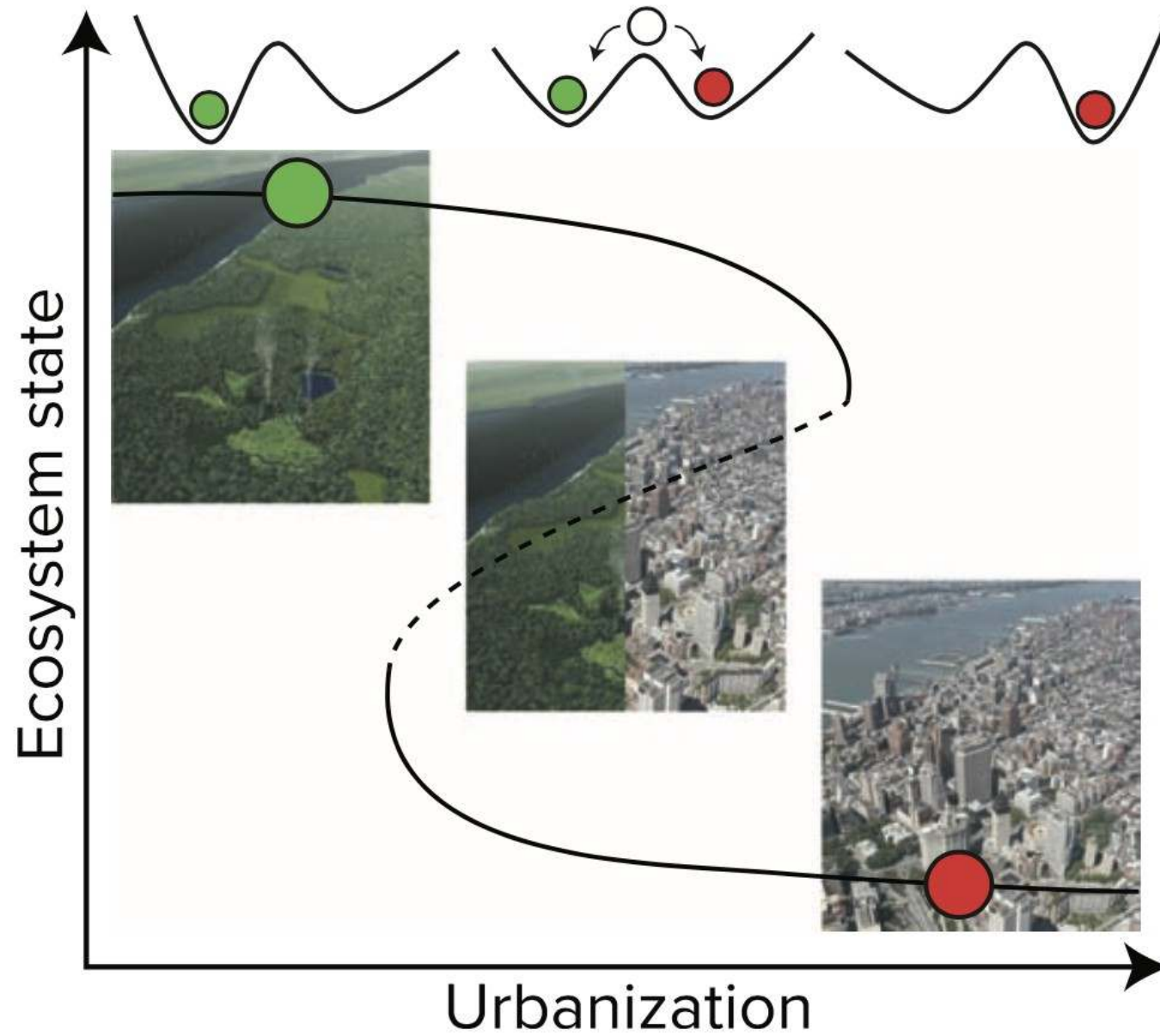
Received: 29 February 2020
DOI: 10.1111/eva.13065

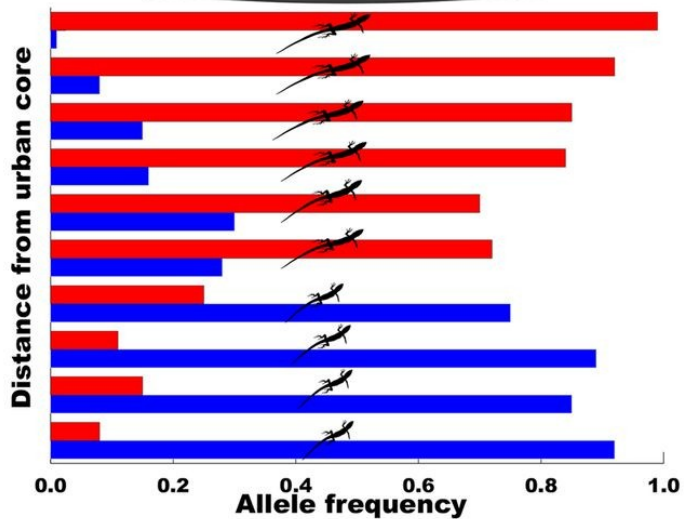
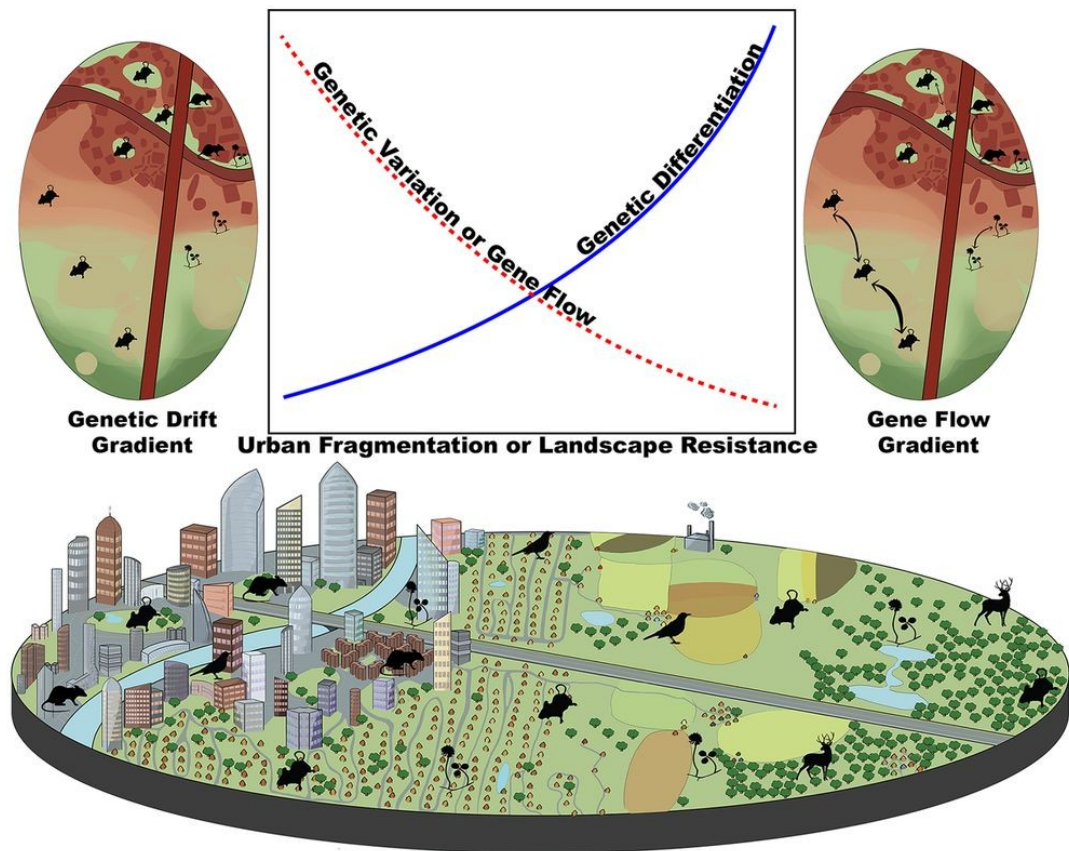
SPECIAL ISSUE PERSPECTIVE

Socio-eco-evolutionary dynamics in cities

Simone Des Roches¹ | Kristien I. Brans² | Max R. Lambert³ | L. Ruth Rivkin^{4,5,6} |
Amy Marie Savage⁷ | Christopher J. Schell⁸ | Cristian Correa^{9,10} | Nancy B. Grimm¹⁴ |
Luc De Meester^{2,11,12} | Sarah E. Diamond¹³ | Andrew P. Hendry¹⁸ | Eric P. Palkovacs²⁰ |
Nyeema C. Harris¹⁵ | Lynn Govaert^{16,17} | Jason Munshi-South¹⁹ | Brian C. Verrelli²³ |
Marc T. J. Johnson^{4,5,6} | Mark C. Urban²² | Marina Alberti¹

Evolutionary Applications
WILEY





REVIEW

EVOLUTION

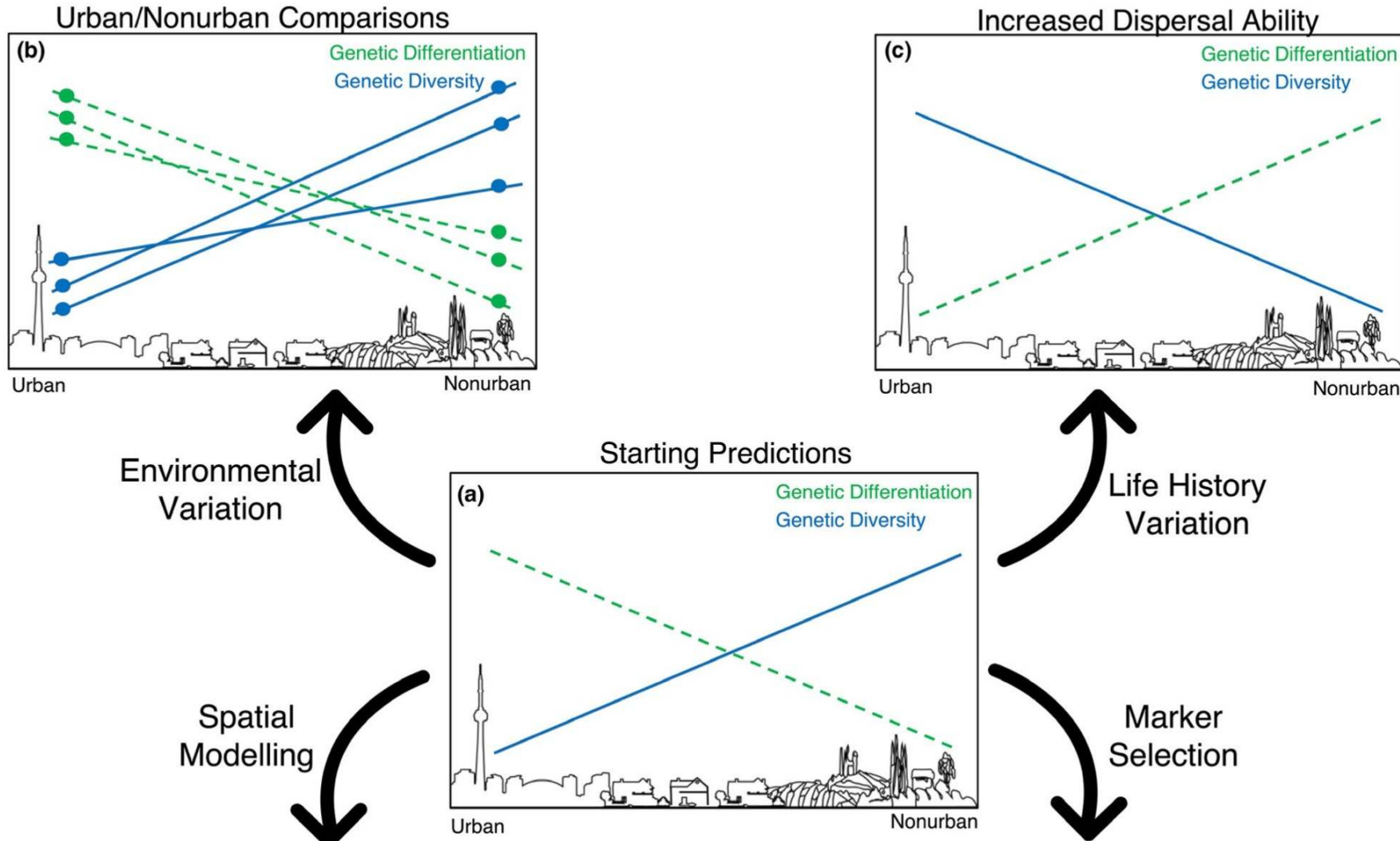
Evolution of life in urban environments

Marc T. J. Johnson^{1,2*} and Jason Munshi-South^{3*}

Johnson *et al.*, *Science* **358**, eaam8327 (2017) 3 November 2017

Gene flow and genetic drift in urban environments

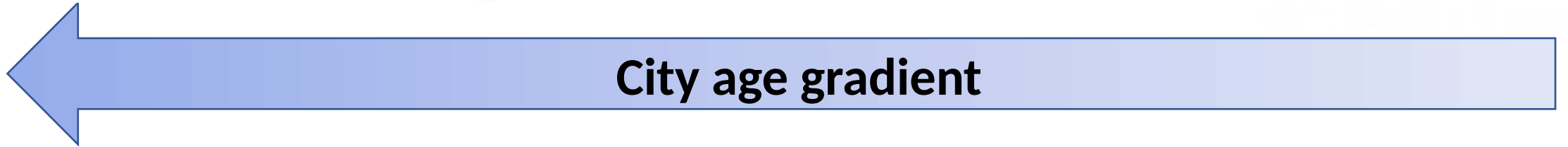
Lindsay S. Miles^{1,2}  | L. Ruth Rivkin^{2,3,4} | Marc T. J. Johnson^{2,3} |
Jason Munshi-South⁵  | Brian C. Verrelli⁶



Houston, we have a problem!

Urban (evo)ecology is (mostly)
based on single cities “case studies”

Need for a comparative approach



Does it affect adaptation?

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